Convex and Nonsmooth Optimization Notes

Vasco Villas-Boas

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1 Convex Sets

As a remark, we will restrict our focus to subsets of $\mathbb{R}^n$ in our analytical discussion. We can do a lot of interesting things by generalizing to a Hilbert space (an inner product space that's complete under the norm induced by the inner product), however for simplicity we will work in $\mathbb{R}^n$.

1.1 Affine and Convex Sets

In this section, we will introduce the concept of affine and convex sets and present some key examples of each.

Definition 1 (Affine Set)**.** A set C is *affine* if $\forall x_1, x_2 \in C$ and all $\theta \in \mathbb{R}$, we have that $\theta x_1 + (1-\theta)x_2 \in C$. For any $x_0 \in C$, we note that $V := C - x_0 := \{x - x_0 : x \in C\}$ is a subspace (see below) and we define $\dim(C) := \dim(V)$. Since the dimension of subspaces are invariant to translations, this dimension definition is sound.

Theorem 2 (Translated Affine Set is Subspace)**.** Let C be an affine set. Then, for any $x_0 \in C$, $V := C - x_0$ is a subspace.

Proof.

First, note that $\mathbf{0} \in V$ trivially as $x_0 \in C$. To see scalar multiplication, take $v \in V$ and consider λv for any $\lambda \in \mathbb{R}$, we have that

$$\begin{aligned}\lambda v &= \lambda(y - x_0), \text{ for some } y \in C \\ &= \underbrace{\lambda y + (1-\lambda)x_0}_{\in C, \text{ by affine-ness}} - x_0 \\ \implies \lambda v &\in V\end{aligned}$$

To see closed under addition, take $v_1, v_2 \in V$. We have that

$$\begin{aligned}v_1 + v_2 &= (y_1 - x_0) + (y_2 - x_0), \text{ for some } y_1, y_2 \in C \\ \implies \frac{1}{2}(y_1 - x_0) + \frac{1}{2}(y_2 - x_0) &\in C, \text{ by affine-ness}\end{aligned}$$

Then, we have that

$$\begin{aligned}\frac{1}{2}(y_1 - x_0) + \frac{1}{2}(y_2 - x_0) &= \underbrace{\frac{1}{2}y_1 + \frac{1}{2}y_2}_{\in C, \text{ by affine-ness}} - x_0 \\ \implies v_1 + v_2 &\in V\end{aligned}$$

□

Definition 3 (Affine Hull)**.** The *affine hull* of $S \subseteq \mathbb{R}^n$ is

$$\text{aff}(S) := \left\{ \sum_{k=1}^{K} \theta_k x_k : x_1, ..., x_K \in S, 1_K'\theta = 1, \theta \in \mathbb{R}^K, K \in \mathbb{N} \right\}$$

Example 4 (Affine Hull of Circle in $\mathbb{R}^2$)**.** The unit circle C in $\mathbb{R}^2$ is $\{x \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$. We have that $\text{aff}(C) = \mathbb{R}^2$.

Definition 5 (Relative Interior)**.** The *relative interior* of a set $S \subseteq \mathbb{R}^n$ is

$$\text{relint}(S) := \{x \in S : B(x, r) \cap \text{aff}(S) \subseteq S, \text{ for some } r > 0\}$$

where we use the notation $B(x, r)$ to denote a ball of radius r (with an implicit norm) around x. Recall that the interior of S is given by

$$\text{int}(S) := \{x \in S : \exists \epsilon > 0 \text{ s.t. } B(x, \epsilon) \subseteq S\}$$

The relative interior and interior are *not* equivalent concepts as we will see in Example 6.

Example 6 (Relative Interior and Interior of Line Segment)**.** Consider the line-segment given by $S := \{x \in \mathbb{R}^2 : x_1 \in [0, 1], x_2 = 0\}$. We have that $\text{int}(S) = \emptyset$ and $\text{relint}(S) = \{x \in \mathbb{R}^2 : x_1 \in (0, 1), x_2 = 0\}$.

Example 7 (Relative Interior and Interior of a Circle and Disk)**.** Consider again the unit circle in $\mathbb{R}^2$ given by $C := \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$. We have that $\text{int}(C) = \emptyset$ and $\text{relint}(S) = \emptyset$. Also consider the unit disk in $\mathbb{R}^2$ is given by $D := \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$. We have that $\text{int}(S) = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$ and $\text{relint}(S) = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$.

Definition 8 (Convex Set)**.** *A set C is convex if $\forall x_1, x_2 \in C$ and all $\theta \in [0,1]$, we have that $\theta x_1 + (1-\theta)x_2 \in C$.*

Definition 9 (Convex Hull)**.** The *convex hull* of a set $C \subseteq \mathbb{R}^n$ is

$$\operatorname{conv}(S) := \left\{ \sum_{k=1}^{K} \theta_k x_k : x_1, ..., x_K \in S, \theta \in \Delta^{K-1}, K \in \mathbb{N} \right\}$$

Lemma 10 (Convex Hull of Convex Set)**.** Suppose that $S \subseteq \mathbb{R}^m$ is convex. Then, we have that $S = \operatorname{conv}(S)$.

Proof.

The fact that $S \subseteq \operatorname{conv}(S)$ is obvious so we focus on the opposite direction. Take any $x \in \operatorname{conv}(S)$. Then, we have that $\exists x_1, ..., x_m$ such that $\sum_{i=1}^m \theta_i x_i = x$ and WLOG we can take all $\theta \gg 0$ with $\theta' 1 = 1$. If $m = 1$, we trivially have that $x \in S$. Suppose that we've satisifed the result for $m - 1 \in \mathbb{N}$ and now we consider for m. We have that

$$\begin{aligned} x &= \sum_{i=1}^{m} \theta_i x_i \\ &= \theta_m x_m + \sum_{i=1}^{m-1} \theta_i x_i \\ &= \theta_m x_m + \left(\sum_{j=1}^{m} \theta_j \right) \underbrace{\sum_{i=1}^{m-1} \frac{\theta_i}{\sum_{j=1}^{m=1} \theta_j} x_i}_{\in S \text{ by inductive hypothesis}} \\ &\in S, \text{ by convexity of } S \end{aligned}$$

$\square$

Definition 11 (Cone and Convex Cone)**.** A set $C \subseteq \mathbb{R}^n$ is a cone if $\forall x \in C$ and all $\theta \in \mathbb{R}_+$, we have that $\theta x \in C$. A *convex cone* is a cone that is convex.

Definition 12 (Hyperplane and Closed Halfspace)**.** A set $H \subseteq \mathbb{R}^n$ is a *hyperplane* if for some $a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and some $b \in \mathbb{R}$

$$H = \{x \in \mathbb{R}^n : a'x = b\}$$

We call a the normal for the hyperplane H. A set $C \subseteq \mathbb{R}^n$ is a *closed halfspace* if for some $a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and some $b \in \mathbb{R}$

$$H = \{x \in \mathbb{R}^n : a'x \leq b\}$$

Example 13 (Important Examples of Convex Cones)**.** $\mathbb{R}^n_+$ is a convex cone. It is very straightforward to see that this set is convex and a cone.

Consider the set

$$Q^{n+1}_+ := \{[x', t]' \in \mathbb{R}^{n+1} : ||x||_2 \leq t\}$$

We have that this set is a convex cone. To see that it's a cone, take any $[x', t]' \in Q^{n+1}_+$ and any $\lambda \in \mathbb{R}_+$. We have that

$$\begin{aligned} ||\lambda x||_2 &= \lambda ||x||_2 \\ &\leq \lambda t, \text{ since } [x', t]' \in Q_+^{n+1} \\ \implies \lambda [x', t]' &\in Q_+^{n+1} \end{aligned}$$

To see that it's convex, take any $[x_1', t_1]', [x_2', t_2]' \in Q_+^{n+1}$ and any $\lambda \in [0,1]$. We have that

$$\begin{aligned} ||\lambda x_1 + (1-\lambda)x_2||_2 &\leq \lambda ||x_1|| + (1-\lambda)||x_2||, \text{ by convexity of } ||\cdot|| \\ &\leq \lambda t_1 + (1-\lambda)t_2, \text{ since } [x_1', t_1]', [x_2', t_2]' \in Q_+^{n+1} \\ \implies \lambda [x_1', t_1]' + (1-\lambda)[x_2', t_2]' \in Q_+^{n+1} \end{aligned}$$

Let $S^n \subseteq \mathbb{R}^{n \times n}$ be the set of symmetric $n \times n$ matrices and define S_+^n as the set of Positive Semi-Definite symmetric $n \times n$ matrices

$$S_+^n := \{X \in S^n : x \succeq 0\}$$

We have that S_+^n is a convex cone. It is very straightforward to see that this set is a convex cone. As one more remark, we let S_{++}^n be the set of Positive Definite matrices.

1.2 Operations that Preserve Convexity

In this section, we will discuss various operations that preserve the convexity of a set.

Proposition 14 (Intersection Preserves Convexity)**.** If $S_1, S_2 \subseteq \mathbb{R}^n$ are convex, then $S_1 \cap S_2$ is convex. Note that this result generalizes to a countable or uncountable number of convex sets.

Proof.

Take any $x, y \in S_1 \cap S_2$ and any $\lambda \in [0,1]$. We know that $x, y \in S_1$ and $x, y \in S_2$. Thus, we have that $\lambda x + (1-\lambda)y \in S_1$ and $\lambda x + (1-\lambda)y \in S_2$ by the convexity of S_1 and S_2, respectively. Thus, we have that $\lambda x + (1-\lambda)y \in S_1 \cap S_2$ proving the convexity of $S_1 \cap S_2$. $\square$

Proposition 15 (Affine Function Preserves Convexity)**.** Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be an affine function (ie., $f(x) = Ax+b$). Then, the image of S under f, denoted $\mathrm{im}(f)$, is convex if S is convex. Also, the inverse image of S under f, denoted $f^{-1}(S) := \{x \in \mathbb{R}^n : f(x) \in S\}$, is convex if S is convex.

Proof.

[First claim]:

Suppose that S is convex. Take any $y_1, y_2 \in \mathrm{im}(S)$ and any $\lambda \in [0,1]$. We have that $y_1 = Ax_1 + b$ and $y_2 = Ax_2 + b$ for some $x_1, x_2 \in S$. By convexity of S, we know that $\lambda x_1 + (1-\lambda)x_2 \in S$. Thus, we have that

$$\begin{aligned} \mathrm{im}(S) \ni & A(\lambda x_1 + (1-\lambda)x_2) + b \\ &= \lambda(Ax_1 + b) + (1-\lambda)(Ax_2 + b) \\ &= \lambda y_1 + (1-\lambda)y_2 \end{aligned}$$

showing that $\mathrm{im}(S)$ is convex.

[Second claim]:

Suppose that S is convex. Take any $x_1, x_2 \in f^{-1}(S)$ and any $\lambda \in [0,1]$. We have that $Ax_1 + b = y_1$ and $Ax_2 + b = y_2$ for some $y_1, y_2 \in S$. By convexity of S, we have that

$$\begin{aligned} S &\ni \lambda y_1 + (1-\lambda)y_2 \\ &= \lambda(Ax_1 + b) + (1-\lambda)(Ax_2 + b) \\ &= A(\lambda x_1 + (1-\lambda)x_2) + b \\ \implies f^{-1}(S) &\ni \lambda x_1 + (1-\lambda)x_2 \end{aligned}$$

showing the convexity of $f^{-1}(S)$. □

Application 16 (Projection onto Some Coordinates Preserves Convexity)**.** Suppose that $f : \mathbb{R}^{n+m} \to \mathbb{R}^n$ is defined as

$$f(x) = [I_n, 0_{n\times m}]x$$

so that it projects its input onto its first n dimensions. Then, since f is an affine function, and $\mathbb{R}^{n+m}$ is convex, by Proposition 15, we have that $\operatorname{im}(f)$ is convex.

Proposition 17 (Cartesian Product Preserves Convexity)**.** Suppose that S_1 and S_2 are convex sets. Then $S_1 \times S_2$ is convex. Note that this result generalizes to a countable or uncountable number of convex sets.

Proof.

Take any $(x_1, x_2), (y_1, y_2) \in S_1 \times S_2$ and any $\lambda \in [0,1]$. By the convexity of S_1 we have that $\lambda x_1 + (1-\lambda)y_1 \in S_1$ and similarly, $\lambda x_2 + (1-\lambda)y_2 \in S_2$. Thus, we have that

$$\begin{aligned} S_1 \times S_2 &\ni (\lambda x_1 + (1-\lambda)y_1, \lambda x_2 + (1-\lambda)y_2) \\ &= \lambda(x_1, x_2) + (1-\lambda)(y_1, y_2) \end{aligned}$$

showing the convexity of $S_1 \times S_2$. □

Application 18 (Solution Set of Vector Inequality and Equality Preserves Convexity)**.** Consider the set $S := \{x \in \mathbb{R}^n : Ax \le b, Cx = d\}$ for $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times n}$, and $d \in \mathbb{R}^k$. We note that S is the inverse image of $\mathbb{R}^m_+ \times \{0\}^k$ under

$$f(x) = [b', d']' - \begin{bmatrix} A \\ C \end{bmatrix} x$$

We see that $\mathbb{R}^m_+ \times \{0\}^k$ is convex by Proposition 17 and f is affine so that by Proposition 15, we have that $S = f^{-1}(\mathbb{R}^m \times \{0\}^k)$ is convex.

Proposition 19 (Sum Preserves Convexity)**.** Suppose that S_1 and S_2 are convex. Then, we have that $S_1 + S_2 := \{x_1 + x_2 : x_1 \in S_1, x_2 \in S_2\}$ is convex.

Proof.

Take any $y_1, y_2 \in S_1 + S_2$ and any $\lambda \in [0,1]$. Then, we have that $y_1 = x_1^1 + x_2^1$ for some $(x_1^2, x_2^2) \in S_1 \times S_2$ and $y_2 = x_1^2 + x_2^2$ for some $(x_1^2, x_2^2) \in S_1 \times S_2$. By the convexity of S_1 we have $\lambda x_1^1 + (1-\lambda)x_1^2 \in S_1$ and similarly

$\lambda x_2^1 + (1-\lambda)x_2^2 \in S_2$. Thus, we we have that

$$\begin{aligned} S_1 + S_2 \ni \lambda x_1^1 + (1-\lambda)x_1^2 + \lambda x_2^1 + (1-\lambda)x_2^2 \\ = \lambda(x_1^1 + x_2^1) + (1-\lambda)(x_1^2 + x_2^2) \\ = \lambda y_1 + (1-\lambda)y_2 \end{aligned}$$

showing the convexity of $S_1 + S_2$. □

Application 20 (Solution Set of Linear Matrix Inequality (LMI) Preserves Convexity)**.** Consider the set $S := \{x \in \mathbb{R}^n : x_1A_1 + ... + x_nA_n \preceq B\}$ for $A_1, ..., A_n, B \in \mathbb{R}^{n\times n}$. We note that S is the inverse image of S_+^n under

$$f(x) := B - x_1A_1 + ... + x_nA_n$$

We know that S_+^n is convex by Example 13 so that by Proposition 15, with affine f, $S = f^{-1}(S_+^n)$ is convex.

Application 21 (Hyperbolic Cone is Convex)**.** Consider the set $S := \{x \in \mathbb{R}^n : x'Px \le (c'x)^2, c'x \ge 0\}$ for some $P \in S_+^n, c \in \mathbb{R}^n$. We note that S is the inverse image of Q_+^{n+1} under

$$f(x) = \begin{bmatrix} P^{1/2} \\ c' \end{bmatrix} x$$

We define $P^{1/2}$ as follows. P is symmetric and PSD so that we can use Singular Value Decomposition (SVD) to decompose it as $P = Q\Lambda Q'$ where $\Lambda := diag([\lambda_1, ..., \lambda_n])$ and $\{\lambda_1, ..., \lambda_n\}$ are the non-negative eigenvalues of P and $Q \in \mathbb{R}^{n\times n}$ has orthonormal eigenvectors of P as columns. We note that $(Q\Lambda^{1/2}Q')(Q\Lambda^{1/2}Q') = P$ so that we define $P^{1/2} := Q\Lambda^{1/2}Q'$. To see why S is the inverse image of Q_+^{n+1} under f, note that

$$\begin{aligned} ||P^{1/2}x||_2 &= \Big[(P^{1/2}x)'(P^{1/2}x)\Big]^{1/2} \\ &= [x'Px]^{1/2}, \text{ since } P^{1/2} \text{ is symmetric} \end{aligned}$$

which implies that

$$\big[x'Px \le (c'x)^2 \wedge c'x \ge 0\big] \iff ||P^{1/2}x||_2 \le c'x$$

Thus, since f is affine, we can apply Proposition 15 to say that $S = f^{-1}(Q_+^{n+1})$ is convex.

1.3 Separating and Supporting Hyperplanes

In this section, we will present the separating hyperplane theorem and some consequences of the result.

Lemma 22 (Convex Combination of Interior and Convex Set Belongs to Interior)**.** Let $S \subseteq \mathbb{R}^n$ be convex with $\operatorname{int}(S) \neq \emptyset$. Then, we have that $\lambda \operatorname{int}(S) + (1-\lambda)S \subseteq \operatorname{int}(S)$ for any $\lambda \in (0, 1]$.

Proof.

Take any $x \in \operatorname{int}(S)$ and any $y \in S$ and any $\lambda \in (0, 1]$. Then, $\exists\epsilon > 0$ with $B(x, \epsilon) \subseteq S$. We wish to show that $B(\lambda x + (1-\lambda), \lambda\epsilon) \subseteq S$ to conclude that $\lambda x + (1-\lambda) \in \operatorname{int}(S)$. To prove this, take any $z \in \mathbb{R}^n$ with $||\lambda x + (1-\lambda)y - z|| < \lambda\epsilon$. Dividing both sides of the inequality by λ reveals that $\frac{1}{\lambda}z - \left(\frac{1-\lambda}{\lambda}\right) y \in B(x, \epsilon) \subseteq S$. But then, we have that $z = \lambda\left(\frac{1}{\lambda}z - \left(\frac{1-\lambda}{\lambda}y\right)\right) + (1-\lambda)y \in S$, since S is convex. Thus, we have that $B(z, \lambda\epsilon) \subseteq S \implies z \in \operatorname{int}(S)$. □

Lemma 23 (Interior and Closure of Convex Set are Convex)**.** Let $S \subseteq \mathbb{R}^n$ be convex. Then, we have that both $\mathrm{int}(S)$ and $\mathrm{cl}(S)$ are convex sets.

Proof.

The claim of $\mathrm{int}(S)$ convex follows readily from the result of Lemma 22. For the second claim, take $x, y \in \mathrm{cl}(S)$ and $\lambda \in [0,1]$. Then, there are sequences (x_n) and (y_n) in S with $x_n \to x$ and $y_n \to y$. As S is convex $(\lambda x_n + (1-\lambda) y_n)$ is a sequence in S. Since this sequence converges to $\lambda x + (1-\lambda)y$ and $\mathrm{cl}(S)$ is a closed set, then we have that $\lambda x + (1-\lambda) y \in \mathrm{cl}(S)$. □

Theorem 24 (Separating Hyperplane Theorem)**.** Suppose that $C, D \subseteq \mathbb{R}^n$ are nonempty, disjoint, and convex sets. Then, $\exists a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$ such that

$$\begin{aligned} a'x &\leq b \; \forall x \in C \\ a'x &\geq b \; \forall x \in D \end{aligned}$$

Proof.

We will prove this result under the simplified assumption that $\mathrm{dist}(C, D) := \inf\{||u - v|| : u \in C, v \in D\} > 0$ and $\exists (c, d) \in C \times D$ with $||c - d|| = \mathrm{dist}(C, D)$. To prove the result more generally, we would need to introduce the topic of Weak Compactness and and look at Convex Analysis a little longer.

To prove this result, define $a := d - c$ and $b := \frac{||d||^2 - ||c||^2}{2}$. Let's also define the affine function $f(x) = (d - c)' \left(x - \frac{1}{2}(d + c)\right)$. We claim that f is nonnegative on D. Suppose not. Then there exists $u \in D$ such that $f(u) < 0$. For that u, we have that

$$\begin{aligned} f(u) &= (d - c)' \left(u - \frac{1}{2}(d + c)\right) \\ &= (d - c)'(u - d) + \frac{1}{2}||d - c||^2 \end{aligned}$$

Since $\frac{1}{2}||d - c||^2 \geq 0$ and $f(u) < 0$, that implies that $(d - c)'(u - d) < 0$. Let

$$\begin{aligned} g(t) &= ||(d - c) + t(u - d)||^2 \\ &= ||d - c||^2 + 2t(u - d)'(d - c) + O(t^2) \\ \implies \left.\frac{dg(t)}{dt}\right|_{t=0} &= 2(u - d)'(d - c) \\ &< 0, \text{ by above} \end{aligned}$$

Thus, by the definition of the derivative, for sufficiently small $t \in (0, 1)$, we have that

$$\begin{aligned} g(t) &< g(0) \\ &= ||d - c||^2 \\ \implies \sqrt{g(t)} &< ||d - c|| \end{aligned}$$

But then, we have that $||(d - c) + t(u - d)|| = ||(tu + (1 - t)d) - c|| < ||d - c||$. Since D is convex, $tu + (1 - t)d \in D$, which contradicts that $d \in D$ is the closest point to c. The argument to show that f is nonpositive on C is identical. □

Corollary 25 (Strict Separation of Point and Closed Convex Set)**.** Consider $C \subseteq \mathbb{R}^n$ closed and convex and $x_0 \in$

$\mathbb{R}^n \setminus C$. Then, we have that $\exists a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$ such that

$$a'x < b \,\forall x \in C$$
$$a'x_0 > b$$

Proof.

Since $x_0 \notin C$ and C is closed, $\exists \epsilon > 0$ such that $B(x_0, \epsilon) \cap C = \emptyset$. Also note that $B(x_0, \epsilon)$ is convex by a simple application of the triangle inequality and homogeneity of the norm. So by Theorem 24, $\exists a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $\tilde{b} \in \mathbb{R}$ such that

$$a'x \leq \tilde{b} \,\forall x \in C$$
$$a'x \geq \tilde{b} \,\forall x \in B(x_0, \epsilon)$$

Since $B(x_0, \epsilon) = \{x_0 + u : ||u|| \leq \epsilon\}$, we have that

$$a'(x_0 + u) \geq \beta \,\forall ||u|| \leq \epsilon$$

To minimize the LHS, we can pick $u^* = -\frac{a}{||a||}\epsilon$. Then, we have that

$$\begin{aligned} a'(x_0 + u) &\geq a'(x_0 + u^*) \\ &= a'x_0 - \epsilon||a|| \\ &\geq \tilde{b} \end{aligned}$$

Define a new hyperplane with the same a but set $b := \tilde{b} + \epsilon\frac{||a||}{2}$. We then have that

$$a'x \leq \tilde{b} < b \,\forall x \in C$$
$$a'x_0 \geq \tilde{b} > b$$

showing the desired result. □

Corollary 26 (Closed Convex Set and Half-Spaces)**.** Let C be a closed and convex set and let $S := \cap\{\text{half spaces containing } C\}$. We claim that $C = S$.

Proof.

Clearly, $x \in C \implies x \in S$. To see the reverse, for the sake of contradiction, assume that $x \in S$ and $x \notin C$. By Corollary 25, there exists a hyperplane containing C but not x. But then, this contradicting the fact that $x \in S$. □

Definition 27 (Supporting Hyperplane)**.** Consider any nonempty (not necessarily convex) $C \subseteq \mathbb{R}^n$ and take $x_0 \in \mathrm{bd}(C)$. If $a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ satisfies $a'x \leq a'x_0 \;\forall x \in C$, we call the set set $H := \{x \in \mathbb{R}^n : a'x = a'x_0\}$ a *supporting hyperplane* for C at x_0. Equivalently, C and $\{x_0\}$ are separated by H.

Theorem 28 (Existence of Supporting Hyperplane for Convex Set and Boundary Point)**.** For any convex $C \subseteq \mathbb{R}^n$ and any $x_0 \in \mathrm{bd}(C)$, there exists a supporting hyperplane to C at x_0.

Proof.

We proceed by cases on C. First, assume that $\mathrm{int}(C) \neq \emptyset$. By Lemma 23, we have that $\mathrm{int}(C)$ is convex so that we can apply Theorem 24 to separate $\mathrm{int}(C)$ and $\{x_0\}$ with a hyperplane. Alternatively, suppose that $\mathrm{int}(C) = \emptyset$.

Then, C and $\{x_0\}$ lie in an affine space with dimension strictly less than n.[a] As a result, there exists a hyperplane consisting of a vector $(a, b) \in \mathbb{R}^n \setminus \{\mathbf{0}\} \times \mathbb{R}$

$$a'x = b = a'x_0 \; \forall x \in C$$

To form the hyperplane, define $\tilde{C} = \text{aff}(C) - z$ for some $z \in C$ so that $\tilde{C}$ contains the origin. Then, we have that $\tilde{C}$ is a linear space by Theorem 2. This space has dimension less than n so that $\exists a \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that $a'x = 0 \; \forall x \in \tilde{C}$. We then note that $a'x = \underbrace{a'z}_{=:b} \; \forall x \in C \cup \{x_0\}$. □

[a] Suppose not. Then C has $n + 1$ affinely independent points. By convexity of C, it contains the convex hull of these $n + 1$ points which form a simplex and any point in the simplex with positive coefficients on all affinely independent points is in the interior of C, which contradicts $\text{int}(C) = \emptyset$. We note that this affine space must contain x_0 since $x_0 \in \text{cl}(C)$. In other words, x_0 is the limit point of a sequence in C and so x_0 can't reside off the same subspace where C lives.

1.4 Dual Cones and Generalized Inequalities

In this section, we will introduce the concept of a dual cones and discuss some applications. We will present sets that are self-dual.

Definition 29 (Dual Cone)**.** Let $K \subseteq \mathbb{R}^n$ be a cone. We define the *dual cone* of K, denoted by K^* as

$$K^* := \{y \in \mathbb{R}^n : x'y \geq 0 \; \forall x \in K\}$$

We observe that $y \in K^* \iff -y$ is the normal for a separating hyperplane that supports K at the origin.

Example 30 (Dual of $\mathbb{R}^n_+$)**.** If $K = \mathbb{R}^n_+$, we have that $K^* = \mathbb{R}^n_+$ so that $\mathbb{R}^n_+$ is self-dual.

Proof.

Take any $y \in K$. Then, we have that $x'y \geq 0 \; \forall x \in K$ so that $y \in K^*$. Now, take any $y \notin K$. Then, we have that $\exists i \in [n]$ such that $y_i < 0$. Take $e_i \in K$ as the ith canonical unit vector. We then have that $y'e_i < 0 \implies y \notin K^*$. □

Example 31 (Dual of S^n_+)**.** If we let $K = S^n_+$ we have that $K^* = S^n_+$ under the inner product $\langle \cdot, \cdot \rangle : S^n \times S^n \to \mathbb{R}$ given by $\langle X, Y \rangle = \text{Tr}(XY)$.

Proof.

Take any $Y \in K$ so that Y is PSD and we can write $Y = Q\Lambda Q' = \sum_{i=1}^n \lambda_i q_i q_i'$ by SVD where $\lambda_1, ..., \lambda_n \geq 0$ and $q_1, ..., q_n$ are orthonormal. Then, we have that for $X \in K$,

$$\begin{aligned}
\langle Y, X \rangle &= \text{Tr}\left(\left(\sum_{i=1}^n \lambda_i q_i q_i'\right) X\right) \\
&= \text{Tr}\left(\sum_{i=1}^n \lambda_i q_i' X q_i\right), \text{ by trace invariance to circular permutations} \\
&\geq 0 \;\text{ since } X \in K \text{ is PSD}
\end{aligned}$$

That implies that $Y \in K^*$. Now, suppose that $Y \notin K$. Then, $\exists v \in \mathbb{R}^n$ with $v'Yv < 0$. Let $X = vv'$. Then, we have that

$$\begin{aligned}\langle Y, X\rangle &= \mathrm{Tr}(Yvv') \\ &= \mathrm{Tr}(v'Yv) \\ &< 0\end{aligned}$$

which implies that $x \notin K^*$. □

Definition 32 (Norm Cone)**.** Let $||\cdot||$ be a norm on $\mathbb{R}^n$. We call $K = \{[x', t]' \in \mathbb{R}^{n+1} : ||x|| \leq t\}$ a *norm cone*. Note that in Example 13 we saw one example of such a cone Q_+^{n+1}, which was the $||\cdot||_2$-norm cone.

What we're calling a "norm cone" can be in fact shown to be a convex cone with an identical argument to that of Example 13 for Q_+^{n+1}.

Definition 33 (Dual Norm)**.** For a norm $||\cdot||$ on $\mathbb{R}^n$, we define the dual norm $||\cdot||_*$ by $||u||_* = \sup\{u'x : ||x|| \leq 1\}$. Since $||\cdot||_*$ is the pointwise supremum of a bunch of convex functions, it is convex by Example 49.

Proposition 34 (Dual of Norm Cone)**.** Consider the norm cone $K = \{[x', t]' \in \mathbb{R}^{n+1} : ||x|| \leq t\}$. The dual cone is $K^* = \{[u', v]' \in \mathbb{R}^{n+1} : ||u||_* \leq v\}$.

Proof.

To show the result, we must show that

$$\underbrace{[x', t][u', v]'}_{=u'x+vt} \geq 0 \text{ whenever } ||x|| \leq t \iff ||u||_* \leq v$$

[$\Longleftarrow$]:

Suppose that $||u||_* \leq v$ and $||x|| \leq t$ for some $t > 0$ (if $t = 0$, $x = \mathbf{0} \implies u'x + vt \geq 0$). Applying the definition of the dual norm and using the fact that $||-x/t|| \leq 1$, we have that

$$u'(-x/t) \leq ||u||_* \leq v$$

Therefore, we have that

$$\begin{aligned}&t(u'(-x/t)) \leq vt \\ \implies &u'x + vt \geq 0\end{aligned}$$

[$\Longrightarrow$]:

We will show this result by contraposition. Suppose that $||u||_* > v$. Then, by the definition of the dual norm, there exists an $x \in \mathbb{R}^n$ with $||x|| \leq 1$ and $x'u > v$. Taking $t = 1$, we have that

$$u'(-x) + v < 0$$

which means the LHS of the equivalence does not hold. □

2 Convex Functions

In the section, we will discuss convex functions and concepts that arise when pondering them.

2.1 Basic Properties and Extensions

In this section, we will define a convex function and discuss basic properties.

Definition 35 (Convex Function)**.** We say that a function $f : \mathbb{R}^n \to \mathbb{R}$ is *convex* if $\mathrm{dom}(f)$ is a convex set and if $\forall x, y \in \mathrm{dom}(f)$, and $\theta \in [0, 1]$, we have that

$$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta)f(y) \tag{1}$$

where $\mathrm{dom}(f)$ is the domain of f. Geometrically, this inequality means that the line segment between $(x, f(x))$ and $(y, f(y))$, which is the chord from x to y. We say that f is strictly convex if the domain of f is convex and the inequality in Equation (1) holds strictly $\forall x \neq y$ and all $\theta \in (0, 1)$.

Definition 36 (Concave Function)**.** We say that a function $f : \mathbb{R}^n \to \mathbb{R}$ is concave if $-f$ is convex.

Definition 37 (Convex Function Alternate Definition)**.** For an alternate definition of a convex function, for a function $f : \mathbb{R}^n \to \mathbb{R}$, we define its *extended-value extension* to be $\tilde{f} : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ by

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in \mathrm{dom}(f) \\ \infty & \text{o.w.} \end{cases}$$

By this definition, we say that f is convex if $\tilde{f}$ is convex.

The definitions of a convex function in Definitions 35 and 37 are equivalent. It's often nice to use the second definition as one doesn't need to include the domain qualification. In these notes in practice, we will use the first definition of a convex function given in Definition 35.

Theorem 38 (First Order Characterization of Convexity)**.** Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is such that it's domain is open and that ∇f exists at each point in $\mathrm{dom}(f)$. Then, we have that f is convex if and only if $\mathrm{dom}(f)$ is convex and

$$f(y) \ge f(x) + (\nabla f(x))' (y - x) \tag{2}$$

holds $\forall x, y \in \mathrm{dom}(f)$.[a]

Proof.

We will proceed by induction on n to show this result.

<u>$n = 1$</u>:

[$\implies$]:

Take any $x, y \in \mathrm{dom}(f)$. Since $\mathrm{dom}(f)$ is convex (ie., it's an interval), we conclude that $\forall t \in [0, 1]$, $x + t(y - x) \in \mathrm{dom}(f)$ and by the convexity of f,

$$\begin{aligned} (1-t)f(x) + tf(y) &\ge f(x + t(y-x)) \\ \implies f(y) &\ge f(x) + \frac{f(x + t(y-x)) - f(x)}{t} \\ &= f(x) + \left(\frac{f(x+h) - f(x)}{h}\right) \cdot (y - x)\,, \text{ defining } h := t(y-x) \\ &= f(x) + f'(x)(y-x), \text{ sending } h \to 0 \end{aligned}$$

[$\Longleftarrow$]:

Assume that f has a convex domain (ie., it's an interval) and it satisfies

$$f(y) \geq f(x) + f'(x) \cdot (y - x) \tag{3}$$

$\forall x, y \in \mathrm{dom}(f)$. Choose any $x \neq y \in \mathrm{dom}(f)$ and any $\theta \in [0, 1]$, letting $z := \theta x + (1 - \theta)y$. Applying the inequality in Equation (3), we get that

$$\begin{aligned} f(x) &\geq f(z) + f'(z)(x - z) \\ f(y) &\geq f(z) + f'(z)(y - z) \\ \Longrightarrow \theta f(x) + (1 - \theta)f(y) &\geq f(z) \end{aligned}$$

by multiplying the first equation by θ, the second by $1 - \theta$, and recalling the definition of z.

General case with arbitrary $n \in \mathbb{N}$:

Now, we wish to prove the general case with $f : \mathbb{R}^n \to \mathbb{R}$. Let $x, y \in \mathbb{R}^n$ and consider f restricted to the line passing through them (ie., consider the function defined by $g(t) = f(ty + (1 - t)x)$ for $ty + (1 - t)x \in \mathrm{dom}(f)$ so that $g'(t) = (\nabla f(ty + (1 - t)x))' (y - x)$).

[$\Longrightarrow$]:

First assume that f is convex, which in turn implies that g is convex. Thus, by the argument in the $n = 1$ case, we have that

$$\begin{aligned} & g(1) \geq g(0) + g'(0) \cdot (1 - 0) \\ \Longrightarrow \quad & f(y) \geq f(x) + (\nabla f(x))' (y - x) \end{aligned}$$

[$\Longleftarrow$]:

Now, assume that $\mathrm{dom}(f)$ is convex and that the inequality in Equation (2) holds for any $x, y \in \mathrm{dom}(f)$. Thus, we have that if $tx + (1 - t)y \in \mathrm{dom}(f)$ and $\tilde{t}x + (1 - \tilde{t})y \in \mathrm{dom}(f)$, we have that

$$\begin{aligned} & f(tx + (1 - t)y) \geq f(\tilde{t}x + (1 - \tilde{t})y) + \nabla f \left(\tilde{t}x + (1 - \tilde{t})y)\right)' (x - y)(t - \tilde{t}) \\ \Longrightarrow \quad & g(t) \geq g(\tilde{t}) + g'(\tilde{t})(t - \tilde{t}) \end{aligned}$$

which in turn implies that g is convex by the $n = 1$ case. In light of the arbitrariness of x, y in the construction of g, we have that f is also convex. □

[a] As a remark, this theorem states that from *local information* about a convex function (ie., its value and derivative at a point) we can derive *global information* (ie., a global underestimator of it).

Corollary 39 (First Order Condition is Optimal for Convex Function)**.** Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is such that its domain is open and that ∇f exists at each point in $\mathrm{dom}(f)$. Then, if f is convex and $\nabla f(x) = 0$, then $\forall y \in \mathrm{dom}(f)$, $f(y) \geq f(x)$.

Proof.

Apply Theorem 38 at x for any arbitrary $y \in \mathcal{X}$. □

Theorem 40 (Second Order Characterization of Convexity)**.** Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable. That is, $\nabla^2 f$, the *Hessian* of f, exists at each point in $\mathrm{dom}(f)$. Then, f is convex if and only if

$\mathrm{dom}(f)$ is convex and its *Hessian* is PSD $\forall x \in \mathrm{dom}(f)$.

Proof.

Omitted.

Example 41 (Examples of Convex Functions)**.** Here, we present some examples of typically encountered convex functions. We omit their proofs though one can look them up in the textbook by Boyd and Vandenberghe.

- $x \mapsto \exp(ax)$ is convex on $\mathbb{R}$ for any $a \in \mathbb{R}$.
- $x \mapsto x^a$ is convex on $\mathbb{R}_{++}$ when $a \geq 1$ or $a \leq 0$.
- $x \mapsto |x|^p$ is convex on $\mathbb{R}$ for $p \geq 1$.
- $x \mapsto -\log(x)$ is convex on $\mathbb{R}_{++}$.
- $x \mapsto x\log(x)$ is convex on $\mathbb{R}_{++}$.[a]
- $x \mapsto ||x||_p$ is convex on $\mathbb{R}^n$ $\forall p \in [1, \infty]$.
- $x \mapsto \log\left(\sum_{i=1}^n \exp(x_i)\right)$ is convex on $\mathbb{R}^n$.[b]
- $x \mapsto -\left(\Pi_{i=1}^n x_i\right)^{1/n}$ is convex for $x \in \mathbb{R}^n_{++}$
- $X \mapsto -\log\left(\det(X)\right)$ is convex on S^n_{++}.

[a] Technically, one can say that it's convex on $\mathbb{R}_+$ if one defines the output value as 0 at $x = 0$.
[b] This function can be interpreted as a differentiable approximation of the max function since $\max\{x_1, ..., x_n\} \leq f(x) \leq \max\{x_1, ..., x_n\} + \log(n) \forall x \in \mathbb{R}^n$. The second inequality is tight when all components of x are equal.

Definition 42 (Graph, Epigraph, and Hypograph of Functions)**.** The graph, epigraph, and hypograph of a function $f : \mathbb{R}^n \to \mathbb{R}$ are defined as

$$\begin{aligned} \mathrm{graph}(f) &:= \{(x, f(x)) \in \mathbb{R}^{n+1} : x \in \mathrm{dom}(f)\} \\ \mathrm{epi}(f) &:= \{[x', t]' \in \mathbb{R}^{n+1} : x \in \mathrm{dom}(f), f(x) \leq t\} \\ \mathrm{hypo}(f) &:= \{[x', t]' \in \mathbb{R}^{n+1} : x \in \mathrm{dom}(f), f(x) \geq t\} \end{aligned}$$

respectively.

Theorem 43 (Epigraph Characterization of Convex Function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if $\mathrm{epi}(f)$ is convex.

Proof.

[$\implies$]:

Take any $[x', t]', [y', u]' \in \mathrm{epi}(f)$ and any $\theta \in [0, 1]$. We know that $f(x) \leq t$ and $f(y) \leq u$. By convexity of f, we know that $\theta x + (1-\theta)y \in \mathrm{dom}(f)$ and that

$$\begin{aligned} f(\theta x + (1-\theta)y) &\le \theta f(x) + (1-\theta)f(y) \\ &\le \theta t + (1-\theta)u \\ \implies \mathrm{epi}(f) &\ni \begin{bmatrix} \theta x + (1-\theta)y \\ \theta t + (1-\theta)u \end{bmatrix} \\ &= \theta \begin{bmatrix} x \\ t \end{bmatrix} + (1-\theta) \begin{bmatrix} y \\ u \end{bmatrix} \end{aligned}$$

showing the convexity of $\mathrm{epi}(f)$.

[$\Longleftarrow$]:

Take any $x, y \in \mathrm{dom}(f)$ and any $\theta \in [0,1]$. By the convexity of $\mathrm{epi}(f)$, we know that

$$\begin{aligned} \theta \begin{bmatrix} x \\ f(x) \end{bmatrix} + (1-\theta) \begin{bmatrix} y \\ f(y) \end{bmatrix} &= \begin{bmatrix} \theta x + (1-\theta)y \\ \theta f(x) + (1-\theta)f(y) \end{bmatrix} \\ &\in \mathrm{epi}(f) \end{aligned}$$

That implies that

$$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta)f(y)$$

showing the convexity of f. $\square$

Lemma 44 (Convexity Implies Locally Bounded)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex with $\mathrm{dom}(f) = C$. Then, we have that for any $x_0 \in \mathrm{int}(\mathrm{dom}(f))$, $\exists \epsilon > 0$ such that $\forall x \in B[x_0, \epsilon]$, $f(x) \le M(x_0, \epsilon)$.

Proof.

Since $x_0 \in \mathrm{int}(\mathrm{dom}(f))$, we have that $\exists \epsilon > 0$ such that $\{x_0 \pm \epsilon e_i : i \in [n]\} \subseteq \mathrm{int}(\mathrm{dom}(f))$. Let $H[x_0, \epsilon] = \{x \in \mathbb{R}^n : ||x - x_0||_\infty \le \epsilon\}$. It's obvious that $B[x_0, \epsilon] \subseteq H[X_0, \epsilon]$ so that we have that

$$\begin{aligned} \max_{x \in B[x_0,\epsilon]} f(x) &\le \max_{x \in H[x_0,\epsilon]} f(x) \\ &= \max_{i \in [n]} \max(f(x_0 + \epsilon e_i), f(x_0 - \epsilon e_i)), \text{ by convexity of } f \\ &=: M(x_0, \epsilon) \end{aligned}$$

$\square$

Definition 45 (Lipschitz and Locally Lipschitz)**.** $f : \mathbb{R}^m \to \mathbb{R}^n$ is *Lipschitz* with constant $L \in \mathbb{R}_{++}$ on a set $S \subseteq \mathbb{R}^m$ if

$$||f(x) - f(y)|| \le L||x - y|| \; \forall x, y \in S$$

If f is Lipschitz on a neighborhood of $z \in \mathbb{R}^m$, then we say that f is *locally Lipschitz* at $z \in \mathbb{R}^m$.

Lemma 46 (Convexity Implies Locally Lipschitz)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex with $\mathrm{dom}(f) = C$. Then, we have that $\forall x_0 \in \mathrm{int}(\mathrm{dom}(f))$, $\exists \epsilon > 0$ and $N(x_0, \epsilon) > 0$ such that

$$|f(y) - f(x)| \le N(x_0, \epsilon)||x - y|| \; \forall x, y \in B[x_0, \epsilon]$$

Proof.

Assume that $y \neq x_0$ (as otherwise the result is trivial). Let $\alpha := \frac{||y-x_0||}{\epsilon}$. We then observe that we can write

$$y = (1-\alpha)x_0 + \alpha z$$
$$x_0 = \frac{1}{1+\alpha}y + \frac{\alpha}{1+\alpha}u$$

for some $z, u \in \mathrm{bd}(B[x_0, \epsilon])$. We then have by convexity of f that

$$\begin{aligned} & f(y) \leq (1-\alpha)f(x_0) + \alpha f(z) \\ \implies & f(y) - f(x_0) \leq \alpha[f(z) - f(x_0)] \\ \implies & f(y) - f(x_0) \leq \frac{f(z) - f(x_0)}{\epsilon}||y - x_0||, \text{ by definition of } \alpha \\ \implies & f(y) - f(x_0) \leq \frac{M(x_0,\epsilon) - f(x_0)}{\epsilon}||y - x_0||, \text{ by Lemma 44} \end{aligned}$$

We then also have by convexity of f that

$$\begin{aligned} & f(x_0) \leq \frac{1}{1+\alpha}f(y) + \frac{\alpha}{1+\alpha}f(u) \\ \implies & (1+\alpha)f(x_0) \leq f(y) + \alpha f(u) \\ \implies & f(x_0) - f(y) \leq \alpha[f(u) - f(x_0)] \\ \implies & f(x_0) - f(y) \leq \frac{M(x_0,\epsilon) - f(x_0)}{\epsilon}||y - x_0||, \text{ using an argument like above} \end{aligned}$$

Thus, after repeating the exercise with $x \neq x_0$ and leveraging the Triangle inequality, we get the claim holds with $N := 2\left(\frac{M(x_0,\epsilon)-f(x_0)}{\epsilon}\right)$. □

2.2 Operations that Preserve Convexity

In this section, we will discuss operations on functions that preserve convexity.

Example 47 (Nonnegative Weighted Sums and Integrals)**.** Suppose that $f_1, ..., f_n : \mathbb{R}^n \to \mathbb{R}$ are convex for $n \in \mathbb{N}$ and $w \in \mathbb{R}^n_+$. Then, we have that $g(x) = w'f(x)$ is convex. One can show this result by direct verification. Firstly, one restricts $\mathrm{dom}(g) = \cap_{i=1}^n \mathrm{dom}(f_i)$, which is convex by Proposition 14. Then, take any $x, y \in \mathrm{dom}(g)$ and any $\theta \in [0, 1]$. Then, we have that

$$\begin{aligned} g(\theta x + (1-\theta)y) &= \sum_{i=1}^n w_i f_i(\theta x + (1-\theta)y) \\ &\geq \sum_{i=1}^n w_i \theta f_i(x) + (1-\theta)f_i(y), \text{ since } w_i \geq 0 \text{ and } f_i \text{ is convex for } i \in [n] \\ &= \theta w'f(x) + (1-\theta)w'f(y) \\ &= \theta g(x) + (1-\theta)g(y) \end{aligned}$$

In fact, it's easy to extend this result to infinite sums and integrals using the same argument.[a] For the integral case, suppose that $f : \mathbb{R}^n \times A \to \mathbb{R}$ is convex in the first argument for each element of the second argument. Then, we have that

$$g(x) = \int_A w(y)f(x, y)dy$$

[a] We assume whatever normality conditions are necessary so that the infinite sum and integral exist.

Example 48 (Composition with Affine Mapping)**.** Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex. Then, we also have that $g(x) := f(Ax + b)$ is convex for $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$.

Proof.

Restrict the domain of g to be $D := \{x \in \mathbb{R}^n : Ax + b \in \mathrm{dom}(f)\}$. D is convex by Proposition 15. Take any $x, y \in D$ and $\theta \in [0, 1]$. We have that

$$\begin{aligned} g(\theta x + (1-\theta)y) &= f\left(A(\theta x + (1-\theta)y) + b\right) \\ &= f(\theta(Ax + b) + (1-\theta)(Ay + b)) \\ &\geq \theta f(Ax + b) + (1-\theta)f(Ay + b), \text{ by convexity of } f \\ &= \theta g(x) + (1-\theta)g(y) \end{aligned}$$

□

Example 49 (Pointwise Maximum and Supremum)**.** If $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ are convex functions, there pointwise maximum $f(x) := \max\{f_1(x), f_2(x)\}$ with $\mathrm{dom}(f) = \mathrm{dom}(f_1) \cap \mathrm{dom}(f_2)$ is also convex. This result can be easily shown by taking $x, y \in \mathrm{dom}(f)$ and any $\theta \in [0, 1]$. We have that

$$\begin{aligned} f(\theta x + (1-\theta)y) &= \max\{f_1(\theta x + (1-\theta)y), f_2(\theta x + (1-\theta)y)\} \\ &\leq \max\{\theta f_1(x) + (1-\theta)f_1(y), \theta f_2(x) + (1-\theta)f_2(y)\}, \text{ since } f_1, f_2 \text{ are convex} \\ &\leq \theta \max\{f_1(x), f_2(x)\} + (1-\theta)\max\{f_1(y), f_2(y)\} \\ &= \theta f(x) + (1-\theta)f(y) \end{aligned}$$

This result extends to a countable pointwise maximum of functions and even a supremum over a collection of functions. For this argument, suppose that $f : \mathbb{R}^n \times A \to \mathbb{R}$ is convex in the first argument for all entries in the second argument. Then, we have that $g(x) = \sup_{y\in A} f(x, y)$ is convex in x where $\mathrm{dom}(g) = \{x \in \mathbb{R}^n : (x, y) \in \mathrm{dom}(x \mapsto f(x, y))\ \forall y \in A, \sup_{y\in A} f(x, y) < \infty\}$. This result can be shown by taking the intersection of the epigraphs for the collections of functions given by $\{x \mapsto f(x, y) : y \in A\}$ where we use the property that the intersection of convex sets is convex. These results can be shown to prove the following functions are convex

- For $x \in \mathbb{R}^n$, the sum of the $r \leq n$ largest entries of x. In particular, is is the maximum of the sum over $\binom{n}{r}$ choices for selecting r coordinates from x.

- For $C \subseteq \mathbb{R}^n$, $C \neq \emptyset$, not necessarily convex, we define the *support function* S_C associated with C as $S_C(c) = \sup_{y\in C}\{x'y\}$ where $\mathrm{dom}(S_C) = \{x \in \mathbb{R}^n : S_C(x) < \infty\}$. It's the supremum of a bunch of linear functions and its domain is convex so that it's also convex.

- For $C \subseteq \mathbb{R}^n$. The distance in any norm $||\cdot||$ to the farthest point in C, $f(x) = \sup_{y\in C} ||x - y||$ is convex on $\mathbb{R}^n$. To see that, note that we're taking the pointwise supremum of a collection of convex functions of the form $||x - y||$.

Example 50 (Maximum Eigenvalue of Symmetric Matrix)**.** The function $f : S^n \to \mathbb{R}$ where $f(X) = \lambda_{\max}(X)$ is convex.

Proof.

Clearly, the set of symmetric matrices is also convex. Next, observe that $f(X) = \lambda_{\max}(X) = \sup_{||y||_2=1} y'Xy$. Thus, we have that f is the supremum of a collection of functions linear in X so that f is convex.[a] □

[a] To see why $\lambda_{\max}(X) = \sup_{||y||_2=1} y'Xy$, note that since X is symmetric, we can diagonalize it as $Q\Lambda Q'$ for Q with orthonormal

columns (and rows) and Λ a diagonal matrix of eigenvalues of X. Then, observe that for any $y \in \mathbb{R}^n$, $||Q'y||_2^2 = (Q'y)'Q'y = y' \underbrace{QQ'}_{=I_n} y = ||y||_2^2 \implies ||Q'y||_2 = ||y||_2$. Also note that since Q has orthonormal columns, $\text{col}(X) = \mathbb{R}^n$. Thus, we have that $\sup_{||y||_2=1} y'Xy = \sup_{||y||_2=1} y'Q\Lambda Q'y = \sup_{||z||_2=1} z'\Lambda z = \lambda_{\max}(X)$.

Example 51 (Spectral Norm of Matrix)**.** Consider $f : \mathbb{R}^{m\times n} \to \mathbb{R}$ where $f(X) = ||X||_2 := \sup_{y\in\mathbb{R}^n\setminus\{\mathbf{0}\}} \frac{||Xy||_2}{||y||_2} = \sigma_{\max(X)}$. We have that f is convex.

Proof.

Clearly, $\text{dom}(f) = \mathbb{R}^{m\times n}$ is convex. Note that $\sup_{y\in\mathbb{R}^n\setminus\{\mathbf{0}\}} \frac{||Xy||_2}{||y||_2} = \sup_{||y||_2=1} ||Xy||_2 = \sigma_{\max}(X)$ where the first equality follows since the ratio is homogeneous in y with degree 0. Then, we get that f is the supremum of a bunch of convex functions so that it's convex as well. □

2.3 The Conjugate Function

In this section, we introduce the concept of a conjugate function and present some examples.

Definition 52 (Conjugate Function)**.** For a function $f : \mathcal{X} \to \mathbb{R}$, not necessarily convex, where $\mathcal{X}$ is finite dimensional and has on it the inner product $\langle\cdot,\cdot\rangle$, we define the conjugate of f, denoted by $f^* : \mathcal{X} \to \mathbb{R}$ as

$$f^*(y) = \sup_{x\in\text{dom}(f)} \langle y, x\rangle - f(x)$$

where the domain of f^* consists of all $y \in \mathcal{X}$ such that the supremum is finite. We note that f^* is convex (even though f is not necessarily convex) as it is the pointwise supremum of a bunch of linear (and therefore convex) functions.

Example 53 (Conjugate of Univariate Affine Function)**.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ is defined as $f(x) = ax + b$. Then, we have that the domain of f^* is $\{a\}$ and $f^*(a) = -b$. Elsewhere, the supremum in Definition 52 is infinity.

Example 54 (Conjugate of Negative Logarithm)**.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ is defined as

$$f(x) = \begin{cases} -\log(x), & \text{if } x > 0 \\ \infty, & \text{o.w.} \end{cases}$$

The convex conjugate of f is given by

$$f^*(y) = \sup_{x\in\mathbb{R}_{++}} yx + \log(x)$$

We note that if $y \geq 0$, then $f^*(y) = \infty$. Suppose now that $y < 0$. Then, we have that $x \mapsto yx + \log(x)$ is a smooth concave function by Theorem 40. Defining $g(x) = yx + \log(x)$, we can in turn calculate the minimum by Theorem 39. We get that $0 = g'(x^*) = y + \frac{1}{x^*} \implies x^* = -\frac{1}{y}$. Plugging into g, we get that $g(x^*) = -1 - \log(-y)$. Thus, we can write the conjugate of f as

$$f^*(y) = \begin{cases} \infty, & \text{if } y \geq 0 \\ -1 - \log(-y), & \text{if } y < 0 \end{cases}$$

Example 55 (Convex Conjugate of Log Determinant of Matrix)**.** Suppose that $f : S^m \to \mathbb{R}$ is defined as

$$f(X) = \begin{cases} -\log(\det(X)), & \text{if } X \succ 0 \\ \infty, & \text{o.w.} \end{cases}$$

We wish to compute $f^* : S^n \to \mathbb{R}$ defined by $f^*(Y) = \sup_{X \succ 0} \operatorname{Tr}(YX) + \log(\det(X))$. To that end, we will first show that $X \mapsto \operatorname{Tr}(YX) + \log(\det(X))$ is unbounded above unless $Y \prec 0$. To that end, if $Y \not\prec 0$, then we know that Y has an eigenvalue $\lambda \geq 0$ with corresponding eigenvector v WLOG letting $||v||_2 = 1$. Then, we consider $X = I_n + tvv'$ for $t > -1$. We find that

$$\begin{aligned} \operatorname{Tr}(YX) + \log(\det(X)) &= \operatorname{Tr}(Y) + t\lambda + \log\left(\det\left(I_n + tvv'\right)\right) \\ &= \operatorname{Tr}(Y) + t\lambda + \log(1 + t), \\ &\quad \text{since } \operatorname{eig}(I_n + tvv') = [\underbrace{1, ..., 1}_{n-1 \text{ times}}, t+1] \text{ and } \det(I_n + tvv') = \prod_{\lambda \in \operatorname{eig}(I_n + tvv')} \lambda \end{aligned}$$

so that this quantity can be sent to infinity by sending $t \to \infty$.

Next, suppose that $Y \prec 0$. Let's search for the maximizing X of $g(X) = \operatorname{Tr}(YX) + \log(\det(X))$ by looking at the first order condition. We note that this function is concave since the first term is linear in X and the second term is concave in X by Example 41. Thus, we can apply Theorem 39 to find the X that maximizes $g(X)$ when $Y \prec 0$. We compute

$$\begin{aligned} 0 &= \nabla_X g(X^*) \\ &= Y' + (X^*)^{-1}, \text{ using matrix cookbook identities} \\ &= Y + (X^*)^{-1}, \text{ since } Y \text{ is symmetric} \\ \implies X^* &= -Y^{-1} \end{aligned}$$

Plugging in X^* to g, we get that $g(X^*) = -n + \log\left(\det\left(-Y^{-1}\right)\right)$. Thus, we have that

$$f^*(Y) = \begin{cases} -n + \log\left(\det\left(-Y^{-1}\right)\right), & \text{if } Y \prec 0 \\ \infty, & \text{o.w.} \end{cases}$$

3 Optimization Programs and Duality

In this section, we define and discuss general (constrained) optimization programs, convex optimization programs, linear programs, and duality.

Definition 56 (Optimization Program, Convex Optimization Program, and Linear Program)**.** We define an *optimization program* as

$$\begin{aligned} \inf_{x \in \mathbb{R}^m} \ & f_0(x), && \text{“objective function”} \\ \text{s.t. } & f_i(x) \leq 0 \ \forall i \in [n], && \text{“inequality constraints”} \\ & h_i(x) = 0 \ \forall i \in [p], && \text{“equality constraints”} \end{aligned}$$

We call this program the *primal* program and denote its optimal value by p^*. The domain of the optimization problem is given by $\mathcal{D} := (\cap_{i=0}^n \operatorname{dom}(f_i)) \cap (\cap_{i=1}^p \operatorname{dom}(h_i))$. This optimization program is a *convex optimization program* if f_i is convex for $i \in [0:n]$ and h_i is affine for $i \in [p]$. If in addition, f_i is affine for $i \in [0:n]$, then this optimization program is a *linear program*. The *standard form* of a linear program is

$$\begin{aligned} &\inf_x c'x \\ &\quad \text{s.t. } Ax = b \\ &\quad x \geq 0 \end{aligned}$$

Definition 57 (The Lagrangian and Lagrangian Dual)**.** Take the first optimization problem from Definition 56 and call that program (P), which stands for the primal problem. We define the *Lagrangian function* for (P) as $L : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$

$$L(x, \lambda, \nu) = f_0(x) + \lambda' f(x) + \nu' h(x)$$

We call the *Lagrangian dual function* $g : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ with

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\ &\geq -\infty \end{aligned}$$

We write the domain of g as

$$\text{dom}(g) := \{(\lambda', \nu')' \in \mathbb{R}^{n \times p} : g(\lambda, \nu) > -\infty\}$$

and say that all $(\lambda', \nu')' \in \text{dom}(g)$ are dual feasible.

Now, we make two remarks:

- Let p^* be the optimal value of (P). Then, we have that $\forall \lambda \in \mathbb{R}^n_+$, all $\nu \in \mathbb{R}^p$, and all $\tilde{x} \in \mathcal{D}$ such that $f(\tilde{x}) \leq 0$ and $h(\tilde{x}) = 0$,

$$\begin{aligned} L(\tilde{x}, \lambda, \nu) &= f_0(\tilde{x}) + \lambda' f(\tilde{x}) + \nu' h(\tilde{x}) \\ &\leq f_0(\tilde{x}) \\ \implies g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\ &\leq L(\tilde{x}, \lambda, \nu) \\ &\leq f_0(\tilde{x}) \end{aligned}$$

  Since this series of inequalities holds $\forall \tilde{x} \in \mathcal{D}$, we have that $g(\lambda, \nu) \leq p^*$.

- Since g is the pointwise infimum of a bunch of affine functions, it is concave.

Definition 58 (Lagrangian Dual Problem)**.** We define the dual problem to the first problem in Definition 56 with the Lagrangian dual function g defined in Definition 57 as

$$\sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$$

which is a convex optimization problem since g is concave, by the second remark. We denote this problem by (D) and its optimal value by d^*.

3.1 Primal and Dual Problems in Linear Programs

Recall the linear program in standard form in Definition 56. That is

$$\begin{aligned}\inf_{x\in\mathbb{R}^m}\ & c'x \\ \text{s.t. } & Ax = b \\ & x \geq 0\end{aligned}$$

The Lagrangian function of this problem (after turning it into a convex optimization program) is

$$\begin{aligned}L(x,\lambda,\nu) &= c'x - \lambda'x + \nu'(Ax-b) \\ &= (c-\lambda+A'\nu)'x - b'\nu\end{aligned}$$

The Lagrangian dual function is

$$\begin{aligned}g(\lambda,\nu) &= \inf_{x\in\mathbb{R}^m} L(x,\lambda,\nu) \\ &= \begin{cases}-\infty, & \text{if } c-\lambda+A'\nu \neq 0 \\ -b'\nu, & \text{o.w.}\end{cases}\end{aligned}$$

The Lagrangian dual problem is

$$\begin{aligned}\sup_{\lambda\geq 0,\nu} g(\lambda,\nu) = \sup_{\lambda\geq 0,\nu}\ & -b'\nu \\ \text{s.t. } & c-\lambda+A'\nu = 0 \\ = \sup_{\nu}\ & -b'\nu \\ \text{s.t. } & c + A'\nu \geq 0\end{aligned}$$

where the first equality follows from the characterization of g and the second by substituting in λ to the first constraint. We note that the Lagrangian dual problem can be written as another linear program in "standard form".

Let's now consider what happens if we take the dual of the dual? The dual problem can be written as (defining $y := -\nu$)

$$\begin{aligned}\inf_{y\in\mathbb{R}^p}\ & -b'y \\ \text{s.t. } & A'y - c \leq 0\end{aligned}$$

The Lagrangian function is

$$\begin{aligned}\tilde{L}(y,\pi) &= -b'y + \pi'(A'y-c) \\ &= (-b+A\pi)'y - c'\pi\end{aligned}$$

The Lagrangian dual function is

$$\begin{aligned}\tilde{g}(\pi) &= \inf_{y\in\mathbb{R}^p} \tilde{L}(y,\pi) \\ &= \begin{cases}-c'\pi, & \text{if } A\pi = b \\ -\infty, & \text{if } A\pi \neq b\end{cases}\end{aligned}$$

The Lagrangian dual problem is

$$\begin{aligned}\sup_{\pi\geq 0} \tilde{g}(\pi) = \min_{\pi\in\mathbb{R}^n}\ & c'\pi \\ \text{s.t. } & A\pi = b, \pi \geq 0\end{aligned}$$

which is precisely the original primal problem.

3.2 Weak Duality

Let (P) be a primal problem and (D) be a dual problem and let p^* be the optimal value attained by the primal problem and d^* the optimal value attained by the dual problem. From the first remark in Section 3, we had that $g(\lambda, \nu) \leq p^* \; \forall \lambda \geq 0, \forall \nu$. By definition of the dual problem in Definition 58, we have that

$$d^* \leq p^*$$

which we call *weak duality*.

3.3 Strong Duality

In this section, we will discuss conditions that guarantee strong duality (ie., $p^* = d^*$).

Definition 59 (Slater's Condition)**.** Let (P) be a convex optimization problem. If the domain $\mathcal{D}$ of the problem satisfies $\exists \tilde{x} \in \operatorname{relint}(\mathcal{D})$ such that $f_i(\tilde{x}) < 0 \; \forall i \in [n]$ and $h_i(\tilde{x}) = 0 \; \forall i \in [p]$, then we say that (P) satisfies *Slater's condition*.

If (P) is a linear program, then if the feasible region is nonempty, then we say that (P) satisfies Slater's condition.

Before showing the statement and proof of the strong duality theorem, we present some notation to set the stage.

Let $\mathcal{G} := \{(f_1(x), ..., f_n(x), h_1(x), ..., h_p(x), f_0(x)) : x \in \mathcal{D}\} \subseteq \mathbb{R}^n \times \mathbb{R}^p \times \mathbb{R}$. It's clear that

$$p^* = \inf_{(u',v',t)' \in \mathcal{G}} \{t : u \leq 0, v = 0\}$$

and that

$$g(\lambda, \nu) = \inf_{(u',v',t)' \in \mathcal{G}} \{(\lambda', \nu', 1)(u', v', t)'\}$$

Then, for any $(\lambda', \nu', 1)'$, we have that

$$(\lambda', \nu', 1)(u', v', t)' \geq g(\lambda, \nu) \; \forall (u', v', t)' \in \mathcal{G}$$

In other words, we have that $(\lambda', \nu', 1)'$ is a supporting hyperplane for $\mathcal{G}$. We have $t \geq (\lambda', \nu', 1)(u', v', t)'$ if $\lambda \geq 0$, $u \leq 0$, and $v = 0$. Then by the characterization of p^* above, for any $(\lambda', \nu', 1)'$ with $\lambda \geq 0$, we have that

$$\begin{aligned}
p^* &= \inf_{(u',v',t)' \in \mathcal{G}} \{t : u \leq 0, v = 0\} \\
&\geq \inf_{(u',v',t)' \in \mathcal{G}} \{(\lambda', \nu', 1)(u', v', t)' : u \leq 0, v = 0\}, \text{ since } \lambda \geq 0 \\
&\geq \inf_{(u'v',t)' \in \mathcal{G}} \{(\lambda', \nu', 1)(u', v', t)'\} \\
&= g(\lambda, \nu)
\end{aligned}$$

Since $d^* = \sup_{\lambda \geq 0} g(\lambda, \nu)$ by above, we show weak duality $d^* \leq p^*$.

Now define the following set.

$$\begin{aligned}
\mathcal{A} &= \mathcal{G} + \left(\mathbb{R}^n_+ \times \{\mathbf{0}\} \times \mathbb{R}_+\right) \subseteq \mathbb{R}^{n+p+1} \\
&= \{(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^p \times \mathbb{R} : \exists x \in \mathcal{D} \text{ s.t. } f_i(x) \leq u_i \; \forall i \in [n], h_i(x) = v_i \; \forall i \in [p], f_0(x) \leq t\}
\end{aligned}$$

In general $\mathcal{G}$ is not convex but $\mathcal{A}$ is convex if the optimization problem is convex.[1]

[1] See example 60 for an example of when $\mathcal{G}$ is not convex.

Example 60 (Example when $\mathcal{G}$ is not Convex)**.** Consider the following optimization problem

$$\begin{aligned}&\min_{x\in\mathbb{R}} x^2\\ &\text{s.t.}(x-2)^2-1\leq 0\end{aligned}$$

We note that this is a convex optimization problem since we're minimizing a convex function subject to a convex constraint. We note that $(-1,4),(3,16)\in\mathcal{G}$ but $\frac{5}{12}(3,16)+\frac{7}{12}(-1,4)=\left(\frac{1}{3},9\right)\notin\mathcal{G}$.

Lemma 61 (Convexity of $\mathcal{A}$)**.** Take the context in Section 3.3. We have that if (P) is a convex optimization program with domain $\mathcal{D}$, then $\mathcal{A}$ is a convex set.

Proof.

Take any $(u^{(1)},v^{(1)},t^{(1)}),(u^{(2)},v^{(2)},t^{(2)})\in\mathcal{A}$ and any $\theta\in[0,1]$. By inclusion of those points in $\mathcal{A}$, for $j\in\{1,2\}$, we have that $\exists x_j\in\mathcal{D}$ such that $f_i(x_j)\leq u_i^{(j)}$ $\forall i\in[n]$, $h_i(x_j)=v_i^{(j)}$ $\forall i\in[p]$, and $f_0(x_j)\leq t^{(j)}$. Since each of f_i $\forall i\in[0:n]$ are convex (and all h_i $\forall i\in[p]$ are affine) and the intersection of their convex domains is also a convex set by Proposition 14, we have that $\mathcal{D}$ is convex so that $\theta x_1+(1-\theta)x_2\in\mathcal{D}$. Then, we have that for any $i\in[n]$,

$$\begin{aligned}&\theta f_i(x_1)+(1-\theta)f_i(x_2)\geq f_i(\theta x_1+(1-\theta)x_2), \text{ by convexity of } f_i\\ \implies&\theta u_i^{(1)}+(1-\theta)u_i^{(2)}\geq f_i(\theta x_1+(1-\theta)x_2), \text{ by transitivity}\end{aligned}$$

Following an identical argument, by the convexity of f_0, we have that

$$\begin{aligned}\theta t^{(1)}+(1-\theta)t^{(2)}&\geq\theta f_0(x_1)+(1-\theta)f_0(x_2)\\ &\geq f_0(\theta x_1+(1-\theta)x_2), \text{ by the convexity of } f_0\end{aligned}$$

Since h_i $\forall i\in[p]$ are affine functions, we have that for any $i\in[p]$,

$$\begin{aligned}h_i(\theta x_1+(1-\theta)x_2)&=\theta h_i(x_1)+(1-\theta)h_i(x_2), \text{ by affineness of } h_i\\ &=\theta v_i^{(1)}+(1-\theta)v_i^{(2)}\end{aligned}$$

Combining, we have that $\theta x_1+(1-\theta)x_2\in\mathcal{D}$ is such that

$$\begin{aligned}f_i(\theta x_1+(1-\theta)x_2)&\leq\theta u_i^{(1)}+(1-\theta)u_i^{(2)}\ \forall i\in[n]\\ f_0(\theta x_1+(1-\theta)x_2)&\leq\theta t^{(1)}+(1-\theta)t^{(2)}\\ h_i(\theta x_1+(1-\theta)x_2)&=\theta v_i^{(1)}+(1-\theta)v_i^{(2)}\ \forall i\in[p]\end{aligned}$$

which implies that

$$\begin{aligned}\theta(u^{(1)},v^{(1)},t^{(1)})+(1-\theta)(u^{(2)},v^{(2)},t^{(2)})&=\left(\theta u^{(1)}+(1-\theta)u^{(2)},\theta v^{(1)}+(1-\theta)v^{(2)},\theta t^{(1)}+(1-\theta)t^{(2)}\right)\\ &\in\mathcal{A}\end{aligned}$$

proving the convexity of $\mathcal{A}$. $\square$

Theorem 62 (Strong Duality Theorem)**.** If (P) is a convex program (with domain $\mathcal{D}$) and it satisfies Slater's condition, then we have that $p^*=d^*$ where d^* is the optimal value from the dual program (D).[a]

Proof.

We will prove this theorem under the assumptions that $\exists \tilde{x} \in \operatorname{int}(\mathcal{D})$ such that $f_i(\tilde{x}) < 0 \, \forall i \in [n]$ and $h_i(\tilde{x}) = 0 \, \forall i \in [p]$ and that A, the relevant matrix describing the affine equality constraints has full row rank. We also let b be the the vector so that $Ax = b$ in the affine equality constraint. $\mathcal{A}$ is a convex set by Lemma 61. Let us define the convex set

$$B := \{(0, 0, s) \in \mathbb{R}^{n+p+1} : s < p^*\}$$

We claim that $\mathcal{A} \cap B = \emptyset$. If not, $\exists (u, v, t) \in \mathcal{A} \cap B$. Since $(u, v, t) \in A$, $\exists x \in \mathcal{D}$ with $f_i(x) \leq 0 \, \forall i \in [n], Ax = b$, and $f_0(x) \leq t < p^*$, which is impossible since p^* is assumed to be the optimal value of the primal problem.

Now, we apply Theorem 24. With this theorem, $\exists (\tilde{\lambda}', \tilde{\nu}', \tilde{\mu})' \neq \mathbf{0}$ and $\alpha \in \mathbb{R}$ such that

$$\begin{aligned} &(u', v', t)' \in A \implies \tilde{\lambda}' u + \tilde{\nu}' v + \tilde{\mu} t \geq \alpha \\ &(0, 0, t) \in B \implies \tilde{\mu} t \leq \alpha \end{aligned}$$

From the first implication, we see that $\tilde{\lambda} \geq 0$ and $\tilde{\mu} \geq 0$. Otherwise, $\tilde{\lambda}' u + \tilde{\mu} t$ is unbounded below over $\mathcal{A}$. From the second implication, we see that $\tilde{\mu} t \leq \alpha \, \forall t < p^*$. Jointly, both implications tell us that $\forall x \in \mathcal{D}$ by taking $(u = f(x), v = Ax - b, t = f_0(x)) \in \mathcal{A}$:

$$\begin{aligned} \tilde{\lambda}' f(x) + \tilde{\nu}'(Ax - b) + \tilde{\mu} f_0(x) &\geq \alpha \\ &\geq \tilde{\mu} p^* \end{aligned} \tag{4}$$

We now proceed by cases.

$[\tilde{\mu} > 0]$:

Divide the previous equation by $\tilde{\mu}$ to get that

$$L\left(x, \underbrace{\frac{\tilde{\lambda}}{\tilde{\mu}}}_{=:\hat{\lambda}}, \underbrace{\frac{\tilde{\nu}}{\tilde{\mu}}}_{=:\hat{\nu}}\right) \geq p^* \; \forall x \in \mathcal{D}$$

and minimizing over x we get that $g\left(\hat{\lambda}, \hat{\nu}\right) \geq p^*$. By weak duality, we have that $g(\hat{\lambda}, \hat{\nu}) \leq p^*$. Then, since $d^* := \sup_{\lambda \geq 0} g(\lambda, \nu)$ by definition, we get that $d^* = p^*$.

$[\tilde{\mu} = 0]$:

By Equation (4), we have that

$$\tilde{\lambda}' f(x) + \tilde{\nu}'(Ax - b) \geq 0 \, \forall x \in \mathcal{D}$$

Applying to the Slater point $\tilde{x}$,

$$\begin{gathered} f(\tilde{x}) < 0, A\tilde{x} - b = 0 \\ \implies \tilde{\lambda}' f(\tilde{x}) \leq 0 \, \wedge \tilde{\lambda}' f(\tilde{x}) \geq 0 \end{gathered}$$

which implies that $\tilde{\lambda} = \mathbf{0}$. Since $(\tilde{\lambda}, \tilde{\nu}, \tilde{\mu}) \neq \mathbf{0}$, that implies that $\tilde{\nu} \neq \mathbf{0}$. In other words, we have that

$$x'(A'\tilde{\nu}) - b'\tilde{\nu} \geq 0 \, \forall x \in \mathcal{D}$$

But then $\tilde{x}'(A'\tilde{\nu}) - b'\tilde{\nu} = 0$. Since $\tilde{x} \in \text{int}(\mathcal{D})$, we can perturb $\tilde{x}$ to $\hat{x} \in \text{int}(\mathcal{D})$, such that

$$\hat{x}'(A'\tilde{\nu}) - b'\tilde{\nu} < 0$$

which is only not possible if $A'\tilde{\nu} = 0$. Since A' has linearly independent columns, that implies that $\tilde{\nu} = 0$, which is a contradiction. □

[a] When the dual of the dual is the primal, then Slater's condition just needs to hold for one of the primal or dual for strong duality to hold.

Corollary 63 (Strong Duality Linear Program Refinement)**.** If (P) is a linear program, then $p^* = d^*$ if there exists a feasible point in the primal problem.

Proof.

Omitted.

3.3.1 Comment on Strong Duality for Convex Programs

In this section we introduce the KKT conditions for optimality in convex programs.

Proposition 64 (Kuhn-Karush-Tucker Conditions (KKT Conditions))**.** Take a convex optimization problem as in Definition 56. Additionally, suppose that $f_0, f_1, ..., f_n$ are all differentiable. Suppose that $\tilde{x}$ are primal variables and $\tilde{\lambda}, \tilde{\nu}$ dual variables that satisfy the following set of conditions called the Kuhn-Karush-Tucker (KKT Conditions):

$$\begin{aligned}
&0 \geq f_i(\tilde{x}) \ \forall i \in [n] \\
&0 = h_i(\tilde{x}) \ \forall i \in [p] \\
&0 \leq \lambda_i \ \forall i \in [n] \\
&0 = \tilde{\lambda}_i f_i(\tilde{x}) \ \forall i \in [n] \\
&0 = \nabla f_0(\tilde{x}) + \sum_{i=1}^{n} \tilde{\lambda}_i \nabla f_i(\tilde{x}) + \sum_{i=1}^{p} \tilde{\nu}_i \nabla h_i(\tilde{x})
\end{aligned}$$

Then, $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ are primal and dual optimal with zero duality gap. The converse is also true.[a]

Proof.

[$\implies$]:

Note that the first two conditions state that $\tilde{x}$ is primal feasible. Since $\tilde{\lambda}_i \geq 0$, then $L(x, \tilde{\lambda}, \tilde{\nu})$ is convex in x so that the last KKT condition states that its gradient vanishes with respect to x at $x = \tilde{x}$ so that $\tilde{x}$ minimizes $L(x, \tilde{\lambda}, \tilde{\nu})$ over $x \in \mathcal{D}$. From this, we conclude that

$$\begin{aligned}
g(\tilde{\lambda}, \tilde{\nu}) &= \inf_{x \in \mathcal{D}} L(x, \tilde{\lambda}, \tilde{\nu}) \\
&= f_0(\tilde{x}) + \tilde{\lambda}' f(\tilde{x}) + \tilde{\nu}' h(\tilde{x}) \\
&= f_0(\tilde{x}), \text{ since } h(\tilde{x}) = 0 \text{ and } \tilde{\lambda}_i f_i(\tilde{x}) = 0 \ \forall i \in [n]
\end{aligned}$$

showing that $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ have zero duality gap and therefore are primal and dual optimal by weak duality.

[$\impliedby$]:

If x^* and (λ^*, ν^*) are primal and dual optimal then they must be feasible so the top three KKT conditions hold.

Since strong duality holds, we have that

$$\begin{aligned}
f_0(x^*) &= g(\lambda^*, \nu^*) \\
&= \inf_{x \in \mathcal{D}} f_0(x) + (\lambda^*)' f(x) + (\nu^*)' h(x) \qquad (5) \\
&\leq f_0(x^*) + \underbrace{(\lambda^*)' f(x^*)}_{\leq 0} + (\nu^*)'(\underbrace{h(x^*)}_{=0}) \\
&\leq f_0(x^*) \\
\implies& (\lambda^*)' f(x^*) = 0 \\
\implies& \lambda_i = 0 \ \vee \ f(x^*) = 0 \, \forall i \in [n]
\end{aligned}$$

solidifying the fourth KKT condition. Finally, since $\lambda \geq 0$, the problem in Equation 5 is convex so that the first order condition equaling 0 is necessary for optimality solidifying the fifth optimality condition. ☐

[a] The condition $\tilde{\lambda}_i f_i(\tilde{x}) = 0 \, \forall i \in [n]$ is known as *complementary slackness*. It says that when the constraint is not binding, the corresponding Lagrangian multiplier must be 0 and when the constraint is binding, the Lagrangian multiplier can be nonzero.

3.3.2 *Comment on Strong Duality for Linear Programs*

Take the context from Section 3.1. Suppose that $d^* = p^* < \infty$. Then, $\exists x^*, y^*$ such that $b'y^* = c'x^*$. Let $\lambda^* = c - A'y^*$. Then, we have that

$$\begin{aligned}
(x^*)'\lambda^* &= (x^*)'c - (x^*)'A'y^* \\
&= c'x^* - b'y^* = 0 \\
\implies& \ x_i^* = 0 \ \vee y_i^* = 0 \, \forall i \in [m]
\end{aligned}$$

since $x^*, \lambda^* \geq 0$. We view $x_i = 0$ is an *active primal constraint* and $\lambda_i = 0$ is an active dual constraint.

The KKT conditions for the linear program are

$$\begin{aligned}
&0 \geq -x^* \\
&0 = Ax^* - b \\
&0 \leq y^* \\
&x_i^* = 0 \ \vee \lambda_i^* = 0 \, \forall i \in [m] \\
&0 = c - \lambda^* - A'y^*
\end{aligned}$$

3.4 Semidefinite Programming

In the section, we define semidefinite programs and then look at their duality properties.

Definition 65 (Semidefinite Program)**.** We define a *semidefinite program* as

$$\begin{aligned}
&\inf_{X \in S^m} \langle C, X \rangle \\
&\text{s.t. } \langle A_i, X \rangle = b_i \, \forall i \in [p], \ X \succeq 0
\end{aligned}$$

This problem is a linear program if C, X are all diagonal. We define $\langle C, X \rangle := \mathrm{Tr}(CX) = \sum_{i,j} C_{ij} X_{ij}$.

The Lagrangian function for a semidefinite program (see Definition 65) is given by

$$\begin{aligned}
L(X, \Lambda, \nu) &= \langle C, X \rangle - \langle \Lambda, X \rangle + \sum_{i=1}^{p} \nu_i \left(\langle A_i, X \rangle - b_i \right) \\
&= \langle X, C - \Lambda + \sum_{i=1}^{p} \nu_i A_i \rangle - b'\nu
\end{aligned}$$

The Lagrangian dual function is[2]

$$\begin{aligned}
g(\Lambda,\nu) &= \inf_{X\in S^m} L(X,\Lambda,\nu) \\
&= \begin{cases} -b'\nu, & \text{if } \Lambda - \sum_{i=1}^p \nu_i A_i = C \\ -\infty, & \text{if } \Lambda - \sum_{i=1}^p \nu_i A_i \neq C \end{cases}
\end{aligned}$$

For any feasible $\tilde{X} \in S^m$, (ie., $\langle A_i, \tilde{X}\rangle = b_i \ \forall i \in [p],\ \tilde{X} \succeq 0$), assuming $\Lambda \succeq 0$, we claim that

$$L(\tilde{X},\Lambda,\nu) \leq \langle C, \tilde{X}\rangle$$

because $\langle \Lambda, \tilde{X}\rangle \geq 0$. To see that, since Λ and $\tilde{X}$ are PSD, they have factorizations $\Lambda = G'G$ and $\tilde{X} = H'H$ (where the factors G and H are found as is Application 21). Thus, we have that

$$\begin{aligned}
\langle \Lambda, \tilde{X}\rangle &= \mathrm{Tr}(G'GH'H) \\
&= \mathrm{Tr}(GH'HG') \\
&= \mathrm{Tr}((HG')'(HG')) \\
&= ||HG'||_F^2 \\
&\geq 0
\end{aligned}$$

So, we have that $g(\Lambda,\nu) = \inf_{X\in S^m} L(X,\Lambda,\nu) \leq p^*$ (assuming $\Lambda \succeq 0$), proving weak duality.

The Lagrangian dual problem is given by (defining $y := -\nu$)

$$\begin{aligned}
\sup_{\Lambda\succeq 0, \nu\in\mathbb{R}^p} g(\Lambda,\nu) = & \sup_{\Lambda, y\in\mathbb{R}^p} b'y \\
& \quad \text{s.t. } \Lambda + \sum_{i=1}^p y_i A_i = C,\ \Lambda \succeq 0 \\
= & \inf_{y\in\mathbb{R}^p} -b'y \\
& \quad \text{s.t. } C - \sum_{i=1}^p y_i A_i \succeq 0
\end{aligned}$$

Let's now consider what happens if we take the dual of the dual? The Lagrangian of the dual can be written as

$$\tilde{L}(y,\pi) = -b'y + \langle \pi, \left(\sum_{i=1}^p y_i A_i\right) - C\rangle$$

The Lagrangian dual function is

$$\begin{aligned}
\tilde{g}(\pi) &= \inf_{y\in\mathbb{R}^p} \tilde{L}(y,\pi) \\
&= \begin{cases} -\langle C,\pi\rangle, & \text{if } \langle A_i,\pi\rangle = b_i \ \forall i \in [p] \\ -\infty, & \text{o.w.} \end{cases}
\end{aligned}$$

Thus, the Lagrangian dual problem is

$$\begin{aligned}
\sup_{\pi\succeq 0} \tilde{g}(\pi) = & \inf_{\pi\succeq 0} \langle C,\pi\rangle \\
& \quad \text{s.t. } \langle A_i,\pi\rangle = b_i \ \forall i \in [p]
\end{aligned}$$

which precisely matches the primal problem.

[2] To argue that we require $\Lambda - \sum_{i=1}^p \nu_i A_i = C$ for the problem to have a finite solution, we can ponder what happens if that equality doesn't hold meaning that $C - \Lambda + \sum_{i=1}^p \nu_i A_i \neq 0$. In this case, this matrix has a nonzero eigenvalue and we can construct $X = tvv'$ where v is the corresponding eigenvector. By sending t to ∞ or $-\infty$, we can send the value of the Lagrangian dual function to $-\infty$.

3.4.1 Comment on Strong Duality in Semidefinite Programming

Suppose now that $d^* = p^*$ with attainment by $\tilde{X}, \Lambda \succeq 0$ and $y \in \mathbb{R}^p$. Then, $\tilde{X}$ and Λ have factorizations $\Lambda = G'G$ and $\tilde{X} = H'H$ (where the factors are found as in Application 21 and Section 3.4). We have shown that $\langle \tilde{X}, \Lambda \rangle \geq 0$ since $\tilde{X}, \Lambda \succeq 0$. In fact, $\langle \tilde{X}, \Lambda \rangle$ is the duality gap. To see why, note that

$$\begin{aligned}
\langle \tilde{X}, C \rangle &= \langle \tilde{X}, \Lambda + \sum_{i=1}^{p} y_i A_i \rangle, \text{ by dual feasibility} \\
&= \langle \tilde{X}, \Lambda \rangle + \sum_{i=1}^{p} y_i \langle \tilde{X}, A_i \rangle \\
&= \langle \tilde{X}, \Lambda \rangle + y'b, \text{ by primal feasibility}
\end{aligned}$$

When we recognize $\langle \tilde{X}, C \rangle$ as the objective of the primal problem and $y'b$ of the dual problem, we see that $\langle \tilde{X}, \Lambda \rangle$ is the duality gap. When there's no duality gap, we thus see that $\langle \tilde{X}, \Lambda \rangle = 0$. By the previous section, that in turn implies that $||HG'||_F^2 = 0$, which implies that

$$\begin{aligned}
\tilde{X}\Lambda &= H' \underbrace{HG'}_{=0} G \\
&= 0
\end{aligned}$$

and that

$$\begin{aligned}
\Lambda \tilde{X} &= G'GH'H \\
&= (G'GH'H)' \\
&= 0, \text{ by above}
\end{aligned}$$

So $\exists Q \in \mathbb{R}^{n \times n}$ with $Q'Q = I_m$ such that $\tilde{X} = QD_{\tilde{X}}Q'$ and $\tilde{\Lambda} = QD_\Lambda Q'$ where $D_{\tilde{X}} := \text{diag}(\xi)$ and $D_\Lambda := \text{diag}(\lambda)$ contain the eigenvalues of $\tilde{X}$ and Λ, respectively.[3] We have that

$$\begin{aligned}
0 &= \tilde{X}\Lambda \\
&= QD_{\tilde{X}} \overset{I_m}{\cancel{Q'Q}} D_\Lambda Q' \\
&= QD_{\tilde{X}} D_\Lambda Q' \\
\implies 0 &= Q'QD_{\tilde{X}} D_\Lambda Q'Q \\
\implies 0 &= D_{\tilde{X}} D_\Lambda \\
\implies \xi_i &= 0 \ \vee \ \lambda_i = 0 \ \forall i \in [m]
\end{aligned}$$

This last implication: $\xi_i = 0 \ \vee \ \lambda_i = 0 \ \forall i \in [m]$, is known as complementarity for positive semidefinite programs.

Definition 66 (Slater's Condition for Semidefinite Programming)**.** Slater's condition for Semidefinite programming holds given the problem in Definition 65 if $\exists X \succ 0$ such that $\langle A_i, X \rangle = b_i \ \forall i \in [p]$.

Theorem 67 (Strong Duality for Semidefinite Programming)**.** Take the context of Definition 65 and the start of this section. Then, if Slater's condition holds for the primal problem *or* the dual problem $p^* = d^*$.

Proof.

[3] The question here is why can we diagonalize $\tilde{X}$ and Λ using the same orthonormal eigenvectors? Well, since $\tilde{X}$ is symmetric PSD, we can diagonalize it with orthonormal eigenvectors. Some of those eigenvectors correspond to the null space of $\tilde{X}$ and others to the range of $\tilde{X}$ and its easy to see that those two spaces are orthogonal (eigenvectors corresponding to different eigenvalues are orthogonal for symmetric matrices). We also note that the eigenvectors of $\tilde{X}$ span $\mathbb{R}^m$. Finally, we note that $\text{col}(\Lambda) \subseteq \text{null}(\tilde{X})$ and that $\text{col}(\tilde{X}) \subseteq \text{null}(\Lambda)$ since $\tilde{X}\Lambda = 0$ and $\Lambda\tilde{X} = 0$, respectively. Thus, we can form Q by concatenating orthonormal vectors spanning both null spaces and that will space $\mathbb{R}^m$ since $\text{span}(\{\text{null}(\tilde{X}), \text{col}(\tilde{X})\}) \subseteq \text{span}(\{\text{null}(\tilde{X}), \text{null}(\Lambda)\})$ and the former spans $\mathbb{R}^m$.

Omitted.

3.5 Zero-Sum Game Interpretation

Here, we present strong duality from the perspective of a zero-sum game.

Theorem 68 (Max of Min is Weakly Less than Min of Max)**.** We have that

$$\sup_{z \in Z} \inf_{w \in W} h(w, z) \leq \inf_{w \in W} \sup_{z \in Z} h(w, z)$$

for any $h : Z \times W \to \mathbb{R}$ and any nonempty $W \subseteq \mathbb{R}^n$, $Z \in \mathbb{R}^m$.

Proof.

Let $H(z) := \inf_{w \in W} h(w, z)$ for any $z \in Z$. We have that $\forall (w, z) \in W \times Z$, $h(w, z) \geq H(z)$. That means that $\forall w \in W$

$$\sup_{z \in Z} h(w, z) \geq \sup_{z \in Z} H(z)$$

That means that

$$\inf_{w \in W} \sup_{z \in Z} h(w, z) \geq \sup_{z \in Z} \inf_{w \in W} h(w, z)$$

□

Example 69 (Game Interpretation of Theorem 68)**.** Suppose that 2 players are playing a zero-sum game where the payoff is given by $h(w, z)$. Player 1 wishes to send $h(w, z)$ down to $-\infty$ by controlling w and player 2 wishes to send $h(w, z)$ to ∞ by controlling z. Generally, player 1 wishes for player 2 to go first and player 2 wants the opposite as can be seen by Theorem 68. But if strong duality holds, then then it doesn't matter who goes first.

Suppose $W = Z = \mathbb{R}$. If $h(w, z) = -w^2 + z^2$, then $\sup_{z \in Z} \inf_{w \in W} h(w, z) = -\infty$ and $\inf_{w \in W} \sup_{z \in Z} h(w, z) = \infty$. If $h(w, z) = w^2 - z^2$, then $\sup_{z \in Z} \inf_{w \in W} h(w, z) = 0 = \inf_{w \in W} \sup_{z \in Z} h(w, z)$ so that strong duality holds.

4 Descent Methods

In this section, we will develop a context to think about descent methods and gradient descent.

Definition 70 (Strongly Convex)**.** We say that $f : \mathbb{R}^n \to \mathbb{R}$ is *strongly convex* if $\exists m > 0$ such that $\nabla^2 f(x) \succeq mI_n \; \forall x \in \mathrm{dom}(f)$. We say that f is strongly convex on a set S if $\exists m > 0$ such that $\nabla^2 f(x) \succeq mI_n \; \forall x \in S$.

The function $f(x) = \exp(x)$ is strictly convex but not strongly convex. It is strictly convex since for any $x, y \in \mathbb{R}$ and any $\theta \in (0, 1)$, we have that $\exp(\theta x + (1 - \theta)y) < \theta \exp(x) + (1 - \theta) \exp(y)$. It is not strongly convex since $f''(x) = \exp(x)$ and thus $\nexists m \in \mathbb{N}$ such that $f''(x) > m \; \forall x \in \mathbb{R}$.

Definition 71 (Closed Function)**.** We say that $f : \mathbb{R}^n \to \mathbb{R}$ is *closed* if all of its sublevel sets are closed. That is, all sets of the form

$$S_{x_0} = \{x \in \mathrm{dom}(f) : f(x) \leq f(x_0)\}$$

are closed sets.

We consider the following problem with $f : \mathbb{R}^n \to \mathbb{R}$, $f \in C^2$,[4] and f closed. Moreover, we assume that x_0 forms a sublevel set S on which f is strongly convex with parameter $m > 0$. is The problem is

$$\min_{x \in \mathrm{dom}(f)} f(x)$$

and we assume that the minimum is attained with x^* and the value is p^*. By Taylor's Theorem in one variable, we can write $\forall x, y \in S$, there exists $z \in S$ between x and y such that

$$f(y) = f(x) + (\nabla f(x))'(y-x) + \frac{1}{2}(y-x)'\nabla^2 f(z)(y-x)$$

so that by strong convexity of f

$$f(y) \geq f(x) + (\nabla f(x))'(y-x) + \frac{m}{2}(y-x)'(y-x) \tag{6}$$

We note that when $m = 0$, we recover the characterization of convex differentiable functions given in Theorem 38.

The RHS of Equation (6) is a convex function of y for fixed x so that the first order condition characterizes the minimum. The first order condition is

$$\begin{aligned} 0 &= \nabla f(x) + m(\tilde{y} - x) \\ \implies \tilde{y} &= x - \frac{1}{m}\nabla f(x) \end{aligned}$$

So, $\forall x, y \in S$, we have that

$$\begin{aligned} f(y) &\geq f(x) + (\nabla f(x))'(\tilde{y}-x) + \frac{m}{2}(\tilde{y}-x)'(\tilde{y}-x) \\ &= f(x) + (\nabla f(x))'\left(-\frac{1}{m}\nabla f(x)\right) + \frac{m}{2}\frac{1}{m^2}(\nabla f(x))'(\nabla f(x)) \\ &= f(x) - \frac{1}{2m}||\nabla f(x)||_2^2 \\ \implies p^* &\geq f(x) - \frac{1}{2m}||\nabla f(x)||_2^2 \end{aligned} \tag{7}$$

This last result suggests that if the gradient is small then the point is near optimal.

Note that Equation (6) tell us that for all $x, y \in S$

$$\begin{aligned} f(y) &\geq f(x) + (\nabla f(x))'(y-x) + \frac{m}{2}(y-x)'(y-x) \\ \implies \frac{m}{2}||y-x||_2^2 &\leq f(y) - f(x) + (\nabla f(x))'(y-x) \\ \implies ||y-x||_2^2 &\leq \frac{2}{m}\left(f(x_0) - f(x^*)\right) < \infty, \text{ taking } x \leftarrow x^*, y \leftarrow x_0 \end{aligned}$$

so that S is bounded. Since $f \in C^2$ and S is also closed (and hence compact by boundedness), we have that $x \mapsto \lambda_{max}(\nabla^2 f(x))$, which is a continuous function attains its maximum in S. That implies, $\exists M > 0$ such that

$$MI_n \succeq \nabla^2 f(x) \succeq mI_n$$

for all $x \in S$.

Similarly to Equation (6) we have that for all $x, y \in S$

$$f(y) \leq f(x) + (\nabla f(x))'(y-x) + \frac{M}{2}(y-x)'(y-x) \tag{8}$$

Minimizing both sides via $y \in S$ like above leads to

$$p^* \leq f(x) - \frac{1}{2M}||\nabla f(x)||_2^2 \tag{9}$$

Technically, one must be careful here. When minimizing the RHS, it's possible that the point which satisfies the first order condition does not lie in S. I believe this point is addressed by the analysis that conlcudes in Remark 74.

[4] In other words, f is twice continuously differentiable.

Definition 72 (Condition Number of Function). Let $f \in C^2$ be a closed strongly convex function on S. Let m, M be the smallest and largest eigenvalues of $\nabla^2 f(x)$ for $x \in S$ where the eigenvalues may be attained at different points. We call $\kappa = \frac{M}{m}$ the condition number of f.

4.1 Descent Algorithm

We consider the following general descent method:

Algorithm 1 General Descent Method

Input: : x_0, f convex, Stopping Criterion
Output: : x^*, p^*
Start Algorithm:
$x \leftarrow x_0$
while Stopping Criterion is not satisfied **do**
  $\Delta x \leftarrow$ descent direction
  $t \leftarrow$ line search result
  $x \leftarrow x + t\Delta x$
end while
return $x, f(x)$

We make some comments on Algorithm 1:

- From convexity of f, we know that $(\nabla f(x))'(y-x) \geq 0 \implies f(y) \geq f(x)$ so that for $y-x$ to be a step direction we need that $(\nabla f(x))'(y-x) < 0$.

- The second step in the while loop is called a *line search*. We pick which t on the line $\{x + t\Delta x : t \in \mathbb{R}_+\}$ to move.

- The stopping criterion is generally of the form $||\nabla f(x)||_2 \leq \eta$ for some hyperparameter η.

4.2 Exact Line Search

One line search method used is called *exact line search*. In this method, we pick t so as to minimize

$$t = \arg\min_{s>0} f(x + s\Delta x)$$

This method is often used when the cost of computing the optimal point along the ray is much lower than the cost of finding a good search direction. In some special cases, the point along the ray can be found analytically.

4.3 Gradient Descent

In this section, we define the gradient descent algorithm and analyze its convergence guarantees in the problem described in Section 4.

Definition 73 (Gradient Descent). Given a problem like the one defined in Section 4, the *gradient descent algorithm* is Algorithm 1 with $\Delta x \leftarrow -\nabla f(x)$.

4.3.1 Convergence Analysis

Assume we know m, M as defined in Section 4 to be lower and upper bounds on the eigenvalues of the hessian of f in S. From Equation (8), starting with $x = x_0$ and restricting $t > 0$ such that $x^+ := x - t\nabla f(x) \in S$, we have that

$$\begin{aligned} f(x - t\nabla f(x)) &\leq f(x) - t||\nabla f(x)||^2 + \frac{M}{2}t^2||\nabla f(x)||^2 \\ &= f(x) + \left(\frac{Mt^2}{2} - t\right)||\nabla f(x)||^2 \end{aligned}$$

The RHS is minimized by picking $t = \frac{1}{M}$. Consider a fixed step method with $t = \frac{1}{M}$ (instead of the exact line search or the backtracking line search). Then, we have that

$$f\left(x - \frac{1}{M}\nabla f(x)\right) \leq f(x) - \frac{1}{2M}||\nabla f(x)||^2$$

Remark 74 (Soundness of Choosing $t = \frac{1}{M}$)**.** Note that by moving from x to $x - t\nabla f(x)$, we're reducing the objective by the previous line so that we're staying in S, by definition.

Now, We also know that $||\nabla f(x)||^2 \geq 2m(f(x) - p^*)$ by a simple rearrangement of Equation (7). As a result, defining $x^+ := x - \frac{1}{M}\nabla f(x)$, we have that

$$\begin{aligned} f(x^+) &\leq f(x) - \frac{1}{2M}\left(2m(f(x) - p^*)\right) \\ \implies f(x^+) - p^* &\leq \left(1 - \frac{m}{M}\right)(f(x) - p^*) \end{aligned} \tag{10}$$

Applying this result recursively l times, we get that

$$f(x^{(l)}) - p^* \leq \left(1 - \frac{m}{M}\right)^l (f(x) - p^*) \tag{11}$$

which is geometric convergence to the optimum. Let $c := 1 - \frac{m}{M} < 1$ and $\epsilon_0 := f(x) - p^*$. Then rewriting the above we have that

$$f(x^{(l)}) - p^* \leq c^l (f(x) - p^*)$$

If we want the LHS less than ϵ, we need to go l iterations such that $c^l \leq \frac{\epsilon}{\epsilon_0}$. Taking logs, we get that

$$\begin{aligned} l\log(c) &\leq \log(\epsilon/\epsilon_0) \\ \implies l &\geq \frac{\log(\epsilon_0/\epsilon)}{\log(1/c)} \\ &= O(\log(1/\epsilon)) \end{aligned}$$

4.4 Backtracking Line Search

In this section we define the backtracking line search and look at its convergence properties. Most line searches used in practice are inexact. The step is chosen to approximately minimize f along the ray $\{x + t\Delta x : t \in \mathbb{R}_{++}\}$, or even just to reduce f "enough". One commonly used and effective inexact line search is called *backtracking line search* and is explained in Algorithm 2.

Algorithm 2 Backtracking Line Search

Input: : Δx, f convex, x, $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$
Output: : t
Start Algorithm:
$t \leftarrow 1$
while $f(x + t\Delta x) > f(x) + \alpha t(\nabla f(x))'\Delta x$ **do**
  $t \leftarrow \beta t$
end while
return t

Since Δx is a descent direction, we have that $(\nabla f(x))'\Delta x < 0$ so that for t small enough, by continuity, we will have that

$$\begin{aligned} f(x + t\Delta x) &< f(x) + t(\nabla f(x))'\Delta x \\ &< f(x) + \alpha t(\nabla f(x))'\Delta x \end{aligned} \tag{12}$$

The parameter $\alpha \in (0, 1/2)$ can be interpreted as the fraction of the decrease in f predicted by linear extrapolation that we will accept. Finding t so that Equation (12) holds is called solving the *Armijo condition*.

4.4.1 Convergence Analysis

Assume that we know m, M as defined in Section 4 to be lower and upper bounds on the eigenvalues of the hessian of f in S. Suppose we set the descent direction $\Delta x \leftarrow -\nabla f(x)$. Then, setting $x^+ := x - t\nabla f(x)$, the loop termination condition in Algorithm 2 becomes:

$$f(x^+) \leq f(x) - \alpha t||\nabla f(x)||_2^2$$

Recall from Equation (8) with $\Delta x := -\nabla f(x)$ we have that

$$\begin{aligned} f(x^+) &\leq f(x) - t||\nabla f(x)||^2 + \frac{M}{2}t^2||\nabla f(x)||^2 \\ &= f(x) + \left(\frac{Mt^2}{2} - t\right)||\nabla f(x)||^2 \\ &\leq f(x) - \frac{t}{2}||\nabla f(x)||^2, \text{ for } t \in [0, 1/M] \end{aligned}$$

since for $t \in [0, 1/M]$ we know that $-t + \frac{Mt^2}{2} \leq -\frac{t}{2}$. Since $\alpha \in (0, 1/2)$ in back-tracking line search, we see that

$$\begin{aligned} & f(x^+) \leq f(x) - \frac{t}{2}||\nabla f(x)||^2 \\ \implies & f(x^+) \leq f(x) - \alpha t||\nabla f(x)||_2^2 \end{aligned}$$

Therefore, we find that the back-tracking line search while loop terminates with either $t = 1$ or $t \geq \frac{\beta}{M}$. We in turn have that in each iteration:

$$f(x^+) \leq f(x) - \min\left(\alpha, \frac{\beta\alpha}{M}\right)||\nabla f(x)||^2$$

Now, we can proceed as in Section 4.3.1 to get that

$$f(x^+) - p^* \leq \left(1 - \min\left(2m\alpha, \frac{2\beta\alpha m}{M}\right)\right)(f(x) - p^*)$$

Applying this recursively l times:

$$f(x^{(l)}) - p^* \leq \left(1 - \min\left(2m\alpha, \frac{2\beta\alpha m}{M}\right)\right)^l (f(x) - p^*)$$

which is geometric convergence to the optimum. Further analysis is again identical to Section 4.3.1.

4.5 Newton's Method

In this section, we define Newton's Method and look at its convergence properties.

Definition 75 (Newton's Method)**.** Given a problem like the one defined in Section 4, the *Newton's Method* is Algorithm 1 with Δx solving $\nabla^2 f(x)(\Delta x) = -\nabla f(x)$. The line search is solved by backtracking line search as in Algorithm 2. See Algorithm 3.

Algorithm 3 Newton's Method

Input: : f convex, x, $\epsilon > 0$
Output: : x^*
Start Algorithm:
while True **do**
    $\Delta x \leftarrow -(\nabla^2 f(x))^{-1} \nabla f(x)$
    $\lambda^2 \leftarrow (\nabla f(x))' \left(\nabla^2 f(x)\right)^{-1} \nabla f(x)$ **Quit if**: $\frac{\lambda^2}{2} \leq \epsilon$
    $t \leftarrow$ backtracking line search result
    $x \leftarrow x + t\Delta x$
end while
return x

To motivate Newton's Method, from a point x, consider a second order Taylor expansion of the objective f at x:

$$\phi(v) := f(x) + (\nabla f(x))'(v-x) + \frac{1}{2}(v-x)'\nabla^2 f(x)(v-x) + o(||v-x||^2)$$

Minimizing ϕ gives $v - x =: \Delta x$ solving $\nabla^2 f(x)(\Delta x) = -\nabla f(x)$.

Take the optimization problem in Section 4. In Newton's method, we have the search direction Δx is found by the solution to:

$$\nabla^2 f(x)(\Delta x) = -\nabla f(x)$$

To solve this equation, we usually use the Cholesky factorization:

$$\nabla^2 f(x) = LL'$$

where L is a lower-triangular matrix. We solve this by:

(1) Forward solving $L'y = -\nabla f(x)$ for y.

(2) Backward solving $L'\Delta x$ = y for Δx.

4.5.1 Convergence Analysis

Assume that we know m, M as defined in Section 4 to be lower and upper bounds on the eigenvalues of the hessian of f in S. We additionally assume that the hessian of f is Lipschitz continuous on S with constant L, ie.,

$$||\nabla^2 f(x) - \nabla^2 f(y)||_2 \leq L||x-y||_2, \ \forall x, y \in S$$

Lemma 76 (Newton's Method Convergence Lemma)**.** There exist numbers $\eta \in (0, m^2/L]$ and $\gamma > 0$ such that (i) if $||\nabla f(x^{(k)})||_2 \geq \eta$, then

$$f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$$

and (ii) if $||\nabla f(x^{(k)})||_2 < \eta$, then the backtracking line search selects $t^{(k)} = 1$ and

$$\frac{L}{2m^2}||\nabla f(x^{(k+1)})||_2 \leq \left(\frac{L}{2m^2}||\nabla f(x^{(k)})||_2\right)^2$$

Proof.

[(i)]:

Omitted. See BV pg. 489-490.

[(ii)]:

We omit the proof that $t = 1$ is selected by the line search, see BV pg., 490-491.

To see the second part, with a step $t = 1$, observe that

$$\begin{aligned}
||\nabla f(x^{+})||_2 &= ||\nabla f(x + \Delta x_{nt}) - \nabla f(x) - \nabla^2 f(x)\Delta x_{nt}||_2, \text{ by the definition of } \Delta x_{nt} \text{ (nt = Newton)} \\
&= \left|\left|\int_0^1 \left(\nabla^2 f(x + t\Delta x_{nt}) - \nabla^2 f(x)\right) \Delta x_{nt} dt\right|\right|_2 \\
&\leq \left|\left|\int_0^1 \left(\nabla^2 f(x + t\Delta x_{nt}) - \nabla^2 f(x)\right) dt\right|\right|_2 ||\Delta x_{nt}||_2, \text{ by Cauchy-Schwarz} \\
&\leq \int_0^1 \left|\left|\left(\nabla^2 f(x + t\Delta x_{nt}) - \nabla^2 f(x)\right)\right|\right| dt ||\Delta x_{nt}||_2, \text{ by Jensen's inequality} \\
&\leq \frac{L}{2}||\Delta x_{nt}||_2^2, \text{ by the Lipschitz condition and integrating} \\
&= \frac{L}{2}|| \left(\nabla^2 f(x)\right)^{-1} \nabla f(x)||_2^2, \text{ by definition} \\
&\leq \frac{L}{2}||(\nabla^2 f(x))^{-1}||_2^2||\nabla f(x)||_2^2, \text{ by Cauchy-Schwarz} \\
&\leq \frac{L}{2}\lambda_{max}\left(\left[(\nabla^2 f(x))^{-1}\right]' \left[(\nabla^2 f(x))^{-1}\right]\right) ||\nabla f(x)||_2^2 \\
&= \frac{L}{2m^2}||\nabla f(x)||_2^2
\end{aligned}$$

□

Remark 77 (Quadratic Convergence in (ii) of Lemma 76)**.** Suppose that $||\nabla f(x^{(k)})|| < \eta$. Then, we have that

$$\begin{aligned}
\frac{L}{2m^2}||\nabla f(x^{(k+1)})||_2 &\leq \left(\frac{L}{2m^2}||\nabla f(x^{(k)})||_2\right)^2 \\
\implies ||\nabla f(x^{(k+1)})||_2 &\leq \frac{L}{2m^2}||\nabla f(x^{(k)})||_2^2 \\
&\leq \frac{L}{2m^2}\eta^2, \text{ since } ||\nabla f(x^{(k)})||_2 < \eta \\
&\leq \frac{\eta}{2}, \text{ since } \eta \leq \frac{m^2}{L}
\end{aligned}$$

Anyways, applying this inequality recursively, we find that for any $l \geq k$, we have that $||\nabla f(x^{(l)})|| < \eta$ so that the algorithm takes a full Newton step with $t^{(l)} = 1$. Also applying the inequality recursively, we have that for $l \geq k$,

$$\begin{aligned}
\frac{L}{2m^2}||\nabla f(x^{(l)})||_2 &\leq \left(\frac{L}{2m^2}||\nabla f(x^{(k)})||_2\right)^{2^{l-k}} \\
&\leq \left(\frac{1}{2}\right)^{2^{l-k}} \qquad (13)
\end{aligned}$$

and hence

$$\begin{aligned}
f(x^{(l)}) - p^* &\leq \frac{1}{2m}||\nabla f(x^{(l)})||_2^2, \text{ by a simple rearrangement of Equation (7)} \\
&\leq \frac{2m^3}{L^2}\left(\frac{1}{2}\right)^{2^{l-k+1}}, \text{ applying a rearranged Equation (13)}
\end{aligned}$$

This last inequality shows that convergence to the optimum is very very fast once condition (ii) in Lemma 76 is hit. This phenomenon is called *quadratic convergence*.

In light of Lemma 76 and Remark 77, we wish to answer the question of how many iterations are needed till convergence in Newton's Method up to a tolerance ϵ? By (i) of Lemma 76, when $||\nabla f(x)||_2 \geq \eta$, we know that the value of the objective decreases by at least γ. That means that we can say that there are at most $\frac{f(x)-p^*}{\gamma}$ steps taken in this initial phase.

In the quadratically convergence phase covered in Remark 77 with $||\nabla f(x)||_2 < \eta$, suppose we want $f(x^{(l)}) - p^* \leq \epsilon$, we can force this by making $\frac{2m^3}{L^2}\left(\frac{1}{2}\right)^{2^{l-k+1}} < \epsilon$ by Equation (13) where k is the first step where we reach this phase and $l-k+1$ steps are taken in this phase. That means that

$$l - k + 1 \geq \log_2\left(\log_2\left(\frac{\epsilon_0}{\epsilon}\right)\right)$$

where $\epsilon_0 := \frac{2m^3}{L^2}$. In total, we get convergence to a degree of tolerance ϵ in at most

$$\frac{f(x) - p^*}{\gamma} + \log_2\left(\log_2\left(\frac{\epsilon_0}{\epsilon}\right)\right) \text{ steps}$$

4.6 Nesterov's Lower Complexity Bound on Gradient Like Methods

In this section, we will remark on Nesterov's upper bound for convergence in gradient descent as described in Section 4.3.1. After, we will show a lower bound on convergence speed for any gradient based method.

Remark 78 (Nesterov's Upper Bound on Convergence)**.** Take any optimization problem like that in Section 4 with objective f where we assume we know m, M to be lower and upper bounds on the eigenvalues of $\nabla^2 f$ in some sublevel set S given by start point x. Recall we showed in Equation (11) that

$$f(x^{(l)}) - p^* \leq \left(1 - \frac{m}{M}\right)^l (f(x) - p^*)$$

and in Definition 72 we defined the condition number of f to be $\kappa := \frac{M}{m}$. Nesterov showed another bound:

$$f(x^{(l)}) - p^* \leq \kappa \left(\frac{1 - \frac{1}{\kappa}}{1 + \frac{1}{\kappa}}\right)^{2l} (f(x) - p^*)$$

for exact line search with $t = \frac{2}{m+M}$. Let's consider the factor $c' := \left(\frac{1-\frac{1}{\kappa}}{1+\frac{1}{\kappa}}\right)^2$ and compare it to the factor $c := \left(1 - \frac{1}{\kappa}\right)$ we derived. One can show $c' \leq c$. Defining $a = \frac{1}{\kappa}$, we have that

$$\begin{aligned}
&\left(\frac{1 - \frac{1}{\kappa}}{1 + \frac{1}{\kappa}}\right)^2 < \left(1 - \frac{1}{\kappa}\right) \\
\iff &\left(\frac{1-a}{1+a}\right)^2 < (1 - a) \\
\iff &1 - a < (1 + a)^2, \text{ since } \frac{m}{M} = a < 1 \\
\iff &0 < a(3 + a)
\end{aligned}$$

which is true since $\frac{m}{M} = a > 0$ so we conclude the original inequality is true. The Nesterov factor leads with a κ, which can make the Nesterov bound worse for small l.

Assumption 79 (Assumption A)**.** Assume a descent algorithm from x on f that generates $x^{(1)}, x^{(2)}, ...$ for which an "oracle" or a "black box" computes $f(x)$, $\nabla f(x)$, $f(x^{(l)})$, and $\nabla f(x^{(l)})$ for each $l \in \mathbb{N}$. We will additionally assume that

$$x^{(l)} - x \in \text{span}(\{\nabla f(x), \nabla f(x^{(1)}), ..., \nabla f(x^{(l-1)})\})$$

Assumption 80 (Assumption B)**.** Consider the set of all optimization problems $P(x, m, M)$ starting from x of the form:

$$\min_{y \in \text{dom}(f)} f(y)$$

with $f \in \mathcal{F}$. The class of functions $\mathcal{F}$ consists of twice differentiable strongly convex functions with member f satisfying

$$\begin{aligned} \text{dom}(f) &= l_2 \\ &= \{x \in \mathbb{R}^\infty : ||x||^2 = \sum_{i=1}^{\infty} x_i^2 < \infty\} \end{aligned}$$

and $MI_\infty \succeq \nabla^2 f(y) \succeq mI_\infty \; \forall y \in S$ where $S := \{y \in \text{dom}(f) : f(y) \leq f(x)\}$.

Theorem 81 (Nesterov's Main Theorem)**.** For any $x \in l_2$ and any $M > m > 0$, there exists an optimization problem in the set $P(x, m, M)$ discussed in Assumption 80 with objective F such that for all methods satisfying Assumption 79,

$$||x^{(l)} - x^*||^2 \geq \left(\frac{1 - \frac{1}{\sqrt{\kappa}}}{1 + \frac{1}{\sqrt{\kappa}}} \right)^{2l} ||x - x^*||^2$$

and

$$F(x^{(l)}) - p^* \geq \frac{m}{2} \left(\frac{1 - \frac{1}{\sqrt{\kappa}}}{1 + \frac{1}{\sqrt{\kappa}}} \right)^{2l} ||x - x^*||^2$$

Proof.

Define $F(x) := \frac{M-m}{8} \left(x_1^2 + \sum_{i=1}^{\infty} (x_i - x_{i+1})^2 - 2x_1 \right) + \frac{m}{2} ||x||^2$, which is twice continuous differentiable. We have that

$$\begin{aligned} \frac{\partial F(x)}{\partial x_1} &= \frac{M-m}{8} (2x_1 + 2(x_1 - x_2) - 2) + mx_1 \\ &= \frac{M-m}{8} (4x_1 - 2x_2 - 2) + mx_1 \end{aligned}$$

And for $j \in \mathbb{N} \setminus \{1\}$, we have that

$$\frac{\partial F(x)}{\partial x_j} = \frac{M-m}{8} (4x_j - 2x_{j+1} - 2x_{j-1}) + mx_j$$

In vector form, that is:

$$\nabla F(x) = \left(\frac{M-m}{4} T + mI_\infty \right) x - \frac{M-m}{4} e_1$$

where e_1 is the first canonical basis vector and T is the *tridiagonal matrix*:

$$T := \begin{bmatrix} 2 & -1 & 0 & 0 & \cdots \\ -1 & 2 & -1 & 0 & \cdots \\ 0 & -1 & 2 & -1 & \cdots \\ 0 & 0 & -1 & 2 & \ddots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix}$$

In addition, we have that

$$\nabla^2 F = \frac{M-m}{4} T + m I_\infty$$

We have that $m I_\infty \preceq \nabla^2 F \preceq M I_\infty$ so that in fact F is an objective in $P(x, m, M)$ as described in Assumption 80. Since F is convex, the solution x^* to the problem is defined by $\nabla F(x^*) = 0$. That is for $0 = (\nabla F(x^*))_1$

$$\begin{aligned} \frac{M-m}{4}(2x_1^* - x_2^*) + m x_1^* &= \frac{M-m}{4} \\ \iff 2x_1^* - x_2^* + \frac{4m x_1^*}{M-m} &= 1 \\ \iff \left(\frac{2(M-m)+4m}{M-m}\right) x_1^* &= x_2^* + 1 \\ \iff 0 = x_2^* - 2x_1^* \left(\frac{M+m}{M-m}\right) + 1 \end{aligned}$$

and for $j \in \mathbb{N} \setminus \{1\}$ $0 = (\nabla F(x^*))_j$

$$\begin{aligned} 0 &= \frac{M-m}{4}\left(-x_{j-1}^* + 2x_j^* - x_{j+1}^*\right) + m x_j^* \\ \iff 0 &= x_{j+1}^* - 2\left(\frac{M+m}{M-m}\right) x_j^* + x_{j-1}^* \end{aligned}$$

which is a *3-term recurrence*. We solve this 3-term recurrence by substituting $x_j^* := q^j$ to get that

$$\begin{aligned} 0 &= q^{j+1} - 2\left(\frac{M+m}{M-m}\right) q^j + q^{j-1} \\ \implies 0 &= q^2 - 2q\left(\frac{M+m}{M-m}\right) + 1, \text{ if } q \neq 0 \end{aligned}$$

The roots are

$$q = \frac{M + m \pm 2\sqrt{Mm}}{M-m}$$

We only care about $|q| < 1$ since by Assumption 79 the domain consists of square summable points. Therefore, we have

$$\begin{aligned} q &= \frac{M + m - 2\sqrt{Mm}}{M-m} \\ &= \frac{\left(\sqrt{M} - \sqrt{m}\right)^2}{\left(\sqrt{M} - \sqrt{m}\right)\left(\sqrt{M} + \sqrt{m}\right)} \\ &= \frac{1 - \frac{1}{\sqrt{\kappa}}}{1 + \frac{1}{\sqrt{\kappa}}} \end{aligned}$$

We can check this q produces (x_1, x_2) that satisfy $0 = (\nabla F(x))_1$ too.

WLOG, let's assume that we start our descent at $x = 0$. With the provided F, we have that

$$\begin{aligned}||x - x^*||^2 &= \sum_{i=1}^{\infty}(x_i^*)^2 \\ &= \sum_{i=1}^{\infty} q^{2i} \\ &= \frac{q^2}{1-q^2}\end{aligned}$$

We claim that for $l \in \mathbb{N}$, we have that

$$\begin{aligned}&\nabla F(x^{(l-1)}) \in \text{span}(e_1, ..., e_l) \text{ and} \\ &x^{(l)} \in \text{span}(e_1, ..., e_l)\end{aligned}$$

To see why, we can proceed by induction. For $l = 1$, we note that

$$\begin{aligned}\nabla F(x) &= \frac{M-m}{4} e_1 \text{ and} \\ x^{(1)} &= x + t\nabla F(x) \\ &\in \text{span}(e_1)\end{aligned}$$

For the inductive step, suppose that the claim holds for l and we wish to show it for $l + 1$. We have that

$$\begin{aligned}\nabla F(x^{(l)}) &\in \text{span}(Tx^{(l)}, x^{(l)}, e_1) \\ &\in \text{span}(e_1, ..., e_{l+1})\end{aligned}$$

where the only mildly tricky part is noting that $Tx^{(l)} \in \text{span}(x^{(l-1)}, x^{(l)}, x^{(l+1)})$. Thus, we also see that $x^{(l+1)} \in \text{span}(e_1, ..., e_{l+1})$. So,

$$\begin{aligned}||x^{(l)} - x^*||^2 &\geq \sum_{i=l+1}^{\infty}(x_i^*)^2 \\ &= \sum_{i=l+1}^{\infty} q^{2i} \\ &= \frac{q^{2(l+1)}}{1-q^2} \\ &= q^{2l}||x - x^*||^2 \qquad (14)\end{aligned}$$

proving the first lower bound requested by the theorem. Also, from Equation (6), plugging in with (confusingly) $x = x^*$ and $y = x^{(l)}$, we get that

$$\begin{aligned}F(y) &\geq F(x) + (\nabla F(x))'(y - x) + \frac{m}{2}(y-x)'(y-x) \\ \implies F(x^{(l)}) - p^* &\geq \frac{m}{2}||x^{(l)} - x^*||^2 \\ &\geq \frac{m}{2}\left(\frac{1 - \frac{1}{\sqrt{\kappa}}}{1 + \frac{1}{\sqrt{\kappa}}}\right)^{2l} ||x - x^*||^2 \qquad (15)\end{aligned}$$

where the last line follows from applying Equation (14), thus, showing the second desired conclusion. $\square$

4.6.1 Nesterov's Accelerated Gradient Method

Next, we wonder whether there's a gradient method satisfying Assumption 79 that attains the complexity lower bound covered in Theorem 81 for problems given by Assumption 80. The answer is yes! Nesterov's Optimal (accelerated) Gradient Method described in Algorithm 4 attains it.

Algorithm 4 Nesterov's Accelerated Gradient Method

Input: : f as in Assumption 80, x, $\epsilon > 0$, q and M as in Theorem 81
Output: : x^*
Start Algorithm:
$k \leftarrow 0$
$y^{(k)} \leftarrow x$
$x^{(k)} \leftarrow x$
while True **do**
  $x^{(k+1)} = y^{(k)} - \frac{1}{M}\nabla f(y^{(k)})$
  $y^{(k+1)} \leftarrow x^{(k+1)} + q\left(x^{(k+1)} - x^{(k)}\right)$
  $k \leftarrow k+1$ **Quit if**: $||\nabla f(x^{(k)})|| \leq \epsilon$
end while
return $x^{(k)}$

For that algorithm, we have that

- Choose $y = x \in \mathbb{R}^m$
- For each iteration later, set $x^{(l+1)} = y^{(l)} - \frac{1}{M}\nabla f(y^{(l)})$ and $y^{(l+1)} = x^{(l+1)} + q(x^{(l+1)} - x^{(l)})$

5 Simplex Method

In this section, we will introduce the simplex method for solving Linear Programs in standard form. It will not be fully comprehensive due to time constraints but will express the key ideas in the algorithm. Recall from Definition 56, that a Linear Program in standard form is given as

$$\inf_{x \in \mathbb{R}^m} c'x \quad \text{s.t. } Ax = b, x \geq 0$$

with $A \in \mathbb{R}^{n \times m}, b \in \mathbb{R}^n$. Any linear program can be put into standard form. In Example 82, we look at an example of how to do that.

Example 82 (Putting an Example Linear Program into Standard Form)**.** Consider the linear program

$$\sup_{x \in \mathbb{R}^m} x_1 - 3x_2 + x_3 \quad \text{s.t. } x_1 + 2x_2 - x_3 = 1, x_1 + x_3 \leq 3, x_1, x_2 \geq 0, x_3 \text{ is free}$$

To put this in standard form, we must deal with a few things:

- We will flip the sign of the objective so that we're doing a minimization.
- We will introduce a slack variable $x_4 \geq 0$ so that $x_1 + x_3 + x_4 = 0$.
- We will write $x_3 = x_5 - x_6$ with $x_5, x_6 \geq 0$ so that x_5 captures the positive part of x_3 and x_6 the negative part.

Then, defining $c := [-1, 3, 0, -1, 1]'$, $x := [x_1, x_2, x_4, x_5, x_6]$, $b := [1, 3]'$, and

$$A := \begin{bmatrix} 1 & 2 & 0 & -1 & 1 \\ 1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

we recover an equivalent linear program in standard form.

Since any linear program can be put in standard form, if we can devise an algorithm to solve linear programs in standard form, we will be all set to solve all linear programs.

Remark 83 (Approaching Equality Constraints of Linear Program)**.** Suppose that $\mathrm{rank}(A) = n$. Construct a matrix B, which consists of n linearly independent columns of A. We can enumerate those columns as $\{a_{j_1}, ..., a_{j_n}\}$ with indices $I = \{a_{j_1}, a_{j_2}, ..., a_{j_n}\}$. Note that this is possible to do under the assumption that A has rank m. Next, solve the system $Bx_B = b$ for $x_B \in \mathbb{R}^n$. Then, for $x \in \mathbb{R}^m$ by settings $x_i = (x_B)_i \mathbb{1}_{\{i \in I\}}$ so that x has $m - n$ zeros. We see that $Ax = Bx_B = b$.

Definition 84 (Variable Definitions Relevant for Linear Programs)**.** Consider the x achieved from the algorithm expressed in Remark 83. We say that

- Such an x is called a *basic solution*.
- If such an x has more than $m - n$ zeros, then x is called a *degenerate basic solution*.
- If such an x additionally satisfies $x \geq 0$, we say that x is a *basic feasible solution*.
- If a variable $y \in \mathbb{R}^m$ satisfies $Ay = b$ and $y \geq 0$, we say that y is *feasible* for the linear program.
- Lastly, any solution to the linear program that is feasible an minimizes the linear program is called an *optimal feasible solution.*

Theorem 85 (Fundamental Theorem of Linear Programming)**.** Take a linear program in standard form with A having full row rank. Then,

(i) If there exists a feasible y for the linear program, then, there exists a basic feasible solution to the linear program.

(ii) If there exists an optimal feasible solution to the linear program, then there exists an optimal solution to that's a basic feasible solution.

Proof.

[(i)]:

Assume that x is feasible meaning that $Ax = b$ and $x \geq 0$. Organize x and the columns of A such that the first $p \leq m$ entries of x contain the nonzero elements. Suppose first that $a_1, ..., a_p$ are linearly independent. Note that that means that $p \leq n$ by our assumption on A. Trivially, then x is a basic feasible solution.

Now suppose that $a_1, ..., a_p$ are linearly dependent. Then, $\exists y \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ such that $\sum_{i=1}^{p} y_i a_i = 0$ and we can further require that $y_i > 0$ for some i WLOG. Then, for $\epsilon > 0$, we have that

$$\sum_{i=1}^{p} (x_i - \epsilon y_i) a_i = b$$

We can choose $\epsilon := \min_{i\in[p]}\{x_i/y_i : y_i > 0\}$ so as to get a new solution $x - \epsilon y$ with $p-1$ nonzero coefficients. We can recurse until we hit the linearly independent case and then conclude.

[(ii)]:

Omitted but very similar in approach to (i).

Definition 86 (Extremal Points of a Convex Set)**.** For a convex set C, we *extremal points* of C are given by

$$\text{extreme}(C) := \{x \in C : \nexists y, z \in C \setminus \{x\}, t \in (0,1) \text{ with } x = ty + (1-t)z\}$$

An open convex set has no extremal points whereas a closed convex set contains extremal points.

Theorem 87 (Extremal Points of Feasible Region of Linear Program)**.** Consider a linear program in standard form. The point $\tilde{x} \in \mathbb{R}^m$ is an extreme point of $K := \{x \in \mathbb{R}^m : Ax = b, x \geq 0\}$ if and only if $\tilde{x}$ is a basic feasible point.

Proof.

[$\Longleftarrow$]:

Suppose that $\tilde{x}$ is a basic feasible point but not an extreme point of K. That means that $\exists y, z \in K \setminus \{\tilde{x}\}$ and $t \in (0,1)$ such that $x = ty + (1-t)z$. Clearly, y and z must have 0s in the same (at least) $m-n$ coordinates that $\tilde{x}$ has zeros due to $y, z \geq 0$ and $x = ty + (1-t)z$. Restrict A to the corresponding n nonzero coordinates of $\tilde{x}$ to form B (padding with additional linearly independent columns so that $B \in \mathbb{R}^{n\times n}$. Let $\tilde{x}_B$ denote the restriction of $\tilde{x}$ onto the selected columns and analogously for y and z. Then, we have that $B\tilde{x}_B = b, By_B = b, Bz_B = b$ implying that $\tilde{x}_B = y_B = z_B$, which is a contradiction.

[$\Longrightarrow$]:

Suppose that $\tilde{x}$ is an extremal point of K and that $\tilde{x}$ is not a basic feasible solution. That means that $\tilde{x}$ has strictly less than $m-n$ zeros meaning that $\tilde{x}$ loads onto strictly more than n columns of A so that the loaded onto columns must be linearly dependent. Let those columns be $\{a_1, ..., a_p\}$ with $p > n$. That means that there exists $y \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ such that $\sum_{i=1}^p a_i y_i = 0$. Extend y so that it's an m vector by filling the missing entries with 0s. Then, we have that for sufficiently small ϵ, $\tilde{x} \pm \epsilon y$ are both feasible and $\tilde{x} = \frac{1}{2}(\tilde{x} + \epsilon y) + \frac{1}{2}(\tilde{x} - \epsilon y)$ so that $\tilde{x}$ is not an extremal point of K. $\square$

Jointly, Theorems 85 and 87 imply that to find an optimal feasible solution it suffices to search the extremal points of $K := \{x \in \mathbb{R}^m : Ax = b, x \geq 0\}$. To solve linear programs, we aim to traverse these vertices and reducing the value of the objective until we reach an optimal point.

5.1 Algebraic Approach to Simplex Method

Inspired by the analysis of the previous section, the BFS algorithm has the following steps:

(1) Find a basic feasible solution.

(2) While not at optimum, move to "adjacent" basic feasible solution such that objective decreases.

5.1.1 Step (2)

First we will explain how to do step (2) for the problem given at the start of Section 5. Take an $x \in \mathbb{R}^m$ that's a basic feasible solution. We can decompose $x = [x_B', x_N']'$ where x_B contains the first n entries, which are potentially nonzero and x_N' contains all zeros. In a corresponding fashion, we can decompose $A = [A_B, A_N]$ with $Ax = b$, $A_B x_B = b$,

$A_B \in \mathbb{R}^{n\times n}$ invertible, and $c = [c_B', c_N']'$. Next, we see that

$$\begin{aligned} Ax = b &\iff A_B^{-1}Ax = A_B^{-1}b \\ &\iff x_B = A_B^{-1}b - A_B^{-1}A_N x_N \\ &\iff c'x = c_B' x_B + c_N' x_N \\ &\iff c'x = c_B' A_B^{-1} b + \left(\underbrace{c_N' - c_B' A_B^{-1} A_N}_{=:r_N'} \right) x_N \end{aligned} \tag{16}$$

We similarly define,

$$\begin{aligned} r_B &:= c_B - A_B' \underbrace{(A_B')^{-1} c_B}_{=:y} \\ r &:= c - A'(A_B')^{-1} c_B \end{aligned}$$

with the observation that $r_B = 0$. The variable r is called the *reduced gradient*.

Theorem 88 (Optimality Condition of Simplex Method)**.** Take the analysis to start Section 5.1. If $r_N \geq 0$ (which is $\iff$ $r \geq 0$) at a basic feasible solution $x \in \mathbb{R}^m$, then (a) x is optimal for the linear program and (b) y is the optimal dual variable when both problems are feasible (ie., strong duality holds).

Proof.

[a]:

Observe Equation (16). For $x \in \mathbb{R}^n$, we've written the objective as a function of the x_N coordinates in such a way that the equality condition $Ax = b$ holds. If $r_N \geq 0$, then no permissible local perturbation of x_N will decrease the objective so that we're at a local optimum. Further, note that the objective decomposition is valid for *any* feasible $z \in \mathbb{R}^m$. Thus, we have that

$$\begin{aligned} c'x &= c_B' A_B^{-1} b \\ &\leq c_B' A_B^{-1} b + \underbrace{r_N' z_N}_{\geq 0} \end{aligned}$$

[b]:

Following Section 3.1, we can write the dual problem as

$$\begin{aligned} &\sup_{y \in \mathbb{R}^n} b'y \\ &\quad A'y \leq c \end{aligned}$$

The dual objective is given by $b'y$ for $y \in \mathbb{R}^n$. By strong duality of linear programs from Corollary 63, we know that the optimal value of the primal problem, when feasible is equal to the optimal value of the dual problem. The objective of the dual problem at $y = (A_B')^{-1} c_B$ is equal to $b'(A_B')^{-1} c_B = c_B A_B^{-1} c_B$, which is exactly the primal objective as seen in Equation (16) since $x_N = 0$. $\square$

From Theorem 88, we now know the optimality condition that will determine termination of the algorithm. We just need to specify what we mean by moving to an "adjacent" basic feasible solution. We pick an arbitrary $j \in N$ with $r_j < 0$. Then, we set

$$\begin{aligned} d &:= A_B^{-1} A_j \\ x_B(\theta) &:= x_B - \theta d, \text{ for } \theta \geq 0 \end{aligned}$$

For the new point to be feasible, we pick $\theta^* = \min_{i \in B, d_i > 0} \frac{(x_B)_i}{d_i}$ and define the new $x_B \leftarrow x_B(\theta^*)$ and $x_N \leftarrow [0, ..., 0, \underbrace{\theta}_{\text{in position } j}, 0, ..., 0]'$. The motivation for this update is that we're picking θ as large as possible for the entering variable such that we keep $x_B(\theta) \geq 0$. Now some comments:

- At θ^*, the inequality constraint $x \geq 0$ is satisfied by construction. Further, we have that at the new $x \in \mathbb{R}^m$:

$$\begin{aligned} Ax &= [A_B A_N][x_B' - \theta d', ..., \theta, 0, ...]' \\ &= b - \theta A_B d + \theta A_j \\ &= b \end{aligned}$$

  so that the equality constraint is still satisfied too.

- It's a good and easy exercise by way of contradiction to check that the columns of the new A_B are linearly independent.

- If there doesn't exist an $i \in B$ with $d_i > 0$, then we can send $\theta \to \infty$, which will send the objective to $-\infty$ since $r_j < 0$, as seen in the decomposition of the objective in Equation (16). This way, we can identify if the objective is unbounded in the running of the algorithm.

- In the update, if $\theta^* = 0$, we're dealing with a degenerate x. We ignore the corresponding updates in these notes but refer the reader to Bland's Rule here.

5.1.2 Step (1)

For step (1), we must identify a basic feasible solution. For that, we consider an auxiliary linear programming problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^m, u \in \mathbb{R}^n} \quad & u' 1_n \\ \text{s.t. } & Ax + u = b, x \geq 0, u \geq 0 \end{aligned}$$

We note that $x = 0_m, u = b_n$ is a basic feasible solution for this problem so we can solve it using step (2) above. In the optimal solution of this auxiliary problem, if we find that $u \neq 0$, then we know that the original problem is infeasible by a simple argument by way of contradiction. Further, if $u = 0$, then we know that the original problem is feasible and since we're solving it by step (2) we know that the optimal solution will be a basic feasible solution.

6 Alternating Direction Method of Multipliers (ADMM) Method

In this secion, we define subgradients, subdifferentials, and several related concepts. Then, we give some important results on those topics and then move towards explaining the ADMM method.

6.1 Some Prerequisite Definitions and Fundamental Results

Here, we define subgradients, subdifferentials, and several related concepts.

Definition 89 (Proper Function)**.** A function $f : \mathbb{R}^m \to \mathbb{R}$ is *proper* if $x \in \mathbb{R}^m$ such that $f(x) < \infty$ and $f(x) > -\infty \; \forall x \in \mathbb{R}^m$. We write that $\text{dom}(f) = \{x \in \mathbb{R}^m : f(x) < \infty\}$, which is nonempty.

In this section, unless specifically stated otherwise, we assume that $f : \mathbb{R}^m \to \mathbb{R} \cup \{\infty\}$ is convex and *proper*.

Definition 90 (Subgradient of Convex Function)**.** The point $y \in \mathbb{R}^m$ is a *subgradient* of convex and proper f at $x \in \text{dom}(f)$ if

$$f(x + z) \geq f(x) + y'z \; \forall z \in \mathbb{R}^m$$

We note that $[y', -1]'$ is normal to the hyperplane in $\mathbb{R}^{m+1}$ passing through $[x', f(x)]'$ and lying beneath the graph of f. Why? Well, we can write the equation above as

$$f(z) \geq f(x) + y'(z-x) \; \forall z \in \mathbb{R}^m$$

which means that the hyperplane with $(z', t)' \in \mathbb{R}^{m+1}$ satisfying

$$\begin{aligned} & t = f(x) + y'(z-x) \\ \iff & [y', -1][z'-x', t-f(x)]' = 0 \end{aligned}$$

goes through $(x, f(x))$ and lies entirely below the graph of f.

Definition 91 (Subdifferential of Convex Function)**.** The set of all subgradients of convex and proper f at $x \in \operatorname{dom}(f)$ is called the *subdifferential* of f at x. We denote it by $\partial f(x)$.

Lemma 92 (Simple Characterization of Differentiability)**.** A convex and proper function f is differentiable at $x \in \operatorname{int}(\operatorname{dom}(f))$ if and only if $\partial f(x) = \{q\}$ for some $q \in \mathbb{R}^m$. Moreover $q = \nabla f(x)$.

Proof.

[$\implies$]:

Take any $x \in \mathbb{R}^m$. By Theorem 38, we have that

$$\begin{aligned} & f(y) \geq f(x) + (\nabla f(x))'(y-x) \; \forall y \in \mathbb{R}^m \\ \implies \; & f(x+z) \geq f(x) + (\nabla f(x))'z \; \forall z \in \mathbb{R}^m \\ \implies \; & \nabla f(x) \in \partial f(x) \end{aligned}$$

Next, take any $g \in \partial f(x)$. We have that for any $h \in \mathbb{R}^n$ and $\lambda > 0$,

$$\begin{aligned} & f(x+\lambda h) - f(x) \geq \lambda g' h \\ \implies & \frac{f(x+\lambda h) - f(x)}{\lambda} \geq g'h \\ \implies & (\nabla f(x) - g)' h \geq 0, \text{ taking } \lim_{\lambda \downarrow 0} \text{ and using first order taylor expansion} \\ \implies & -||\nabla f(x) - g||^2 \geq 0, \text{ picking } h = g - \nabla f(x) \\ \implies & g = \nabla f(x) \end{aligned}$$

[$\impliedby$]:

Omitted.

Lemma 93 (Simple Subdifferential Lemma)**.** For convex and proper $f : \mathbb{R}^m \to \mathbb{R}$, we have that $\partial f(x)$ is a closed, convex, and nonempty set for any $x \in \operatorname{dom}(f)$. For $x \in \operatorname{int}(\operatorname{dom}(f))$, we additionally have that $\partial f(x)$ is bounded.

Proof.

Take any $y \in \operatorname{dom}(f) = \mathbb{R}^n$. To see nonemptiness, consider the epigraph of f, which is given by

$$\operatorname{epi}(f) := \{[x', t]' \in \mathbb{R}^{n+1} : x \in \operatorname{dom}(f), f(x) \leq t\}$$

This set is convex by Theorem 43. Thus, we can apply Theorem 28 to $\operatorname{epi}(f)$ and $(y', f(y))'$, to say that there exists

$a \in \mathbb{R}^{m+1} \setminus \{\mathbf{0}\}$, such that

$$\begin{aligned}
&a'(y', f(y))' \geq a'(x', t)' \; \forall (x', t)' \in \text{epi}(f) \\
\implies & a_{m+1} f(y) \geq a_{m+1} t + a'_{1:m}(x - y) \; \forall (x', t)' \in \text{epi}(f) \\
\implies & f(y) \geq f(x) + \frac{a'_{1:m}}{a_{m+1}}(x - y) \; \forall (x', f(x))' \in \text{epi}(f) \\
\implies & f(x + z) \geq f(x) + \frac{a'_{1:m}}{a_{m+1}} z \; \forall z \in \mathbb{R}^m \text{ s.t. } (y' - z', f(y - z))' \in \text{epi}(f) \\
\implies & f(x + z) \geq f(x) + \frac{a'_{1:m}}{a_{m+1}} z \; \forall z \in \mathbb{R}^m
\end{aligned}$$

This second implication follows from the fact that $a_{m+1} \neq 0$. To see that, notice that if it did equal 0, since $a \neq \mathbf{0}$, we could find $a_i \neq 0$ for $i \in [m]$ and then perturb it in both directions, and remain in the epigraph for suitably large target value. In turn, violate the inequality in one of the cases. The last equality follows from the fact that $(y' - z'), f(y - z))' \in \text{epi}(f)$.

Next, we show convexity. Take any $y, z \in \partial f(x)$ and any $\theta \in [0, 1]$. We have that for any $t \in \mathbb{R}^m$

$$\begin{aligned}
& f(x + t) \geq f(x) + y't \text{ and } f(x + t) \geq f(x) + z't \\
\implies \; & f(x + t) \geq \theta f(x) + (1 - \theta) f(x) + \theta y't + (1 - \theta) z't \\
& = f(x) + [\theta y + (1 - \theta) z]' \, t
\end{aligned}$$

which implies that $\theta y + (1 - \theta) z \in \partial f(x)$ in light of the arbitrariness of $t \in \mathbb{R}^m$.

Next, we show closedness. Take any $(y_n)_{n \in \mathbb{N}} \in \partial f(x)$ with $y_n \to y$. Suppose for the sake of contradiction that $y \notin \partial f(x)$. Then, we have that $\exists z \in \mathbb{R}^m$ such that

$$f(x + z) < f(x) + y'z$$

But then since $y_n \to y$, we must have that $f(x + z) < f(x) + y'_n z$ for sufficiently large n, which is a contradiction.

Finally, we show the last claim. Take any $x \in \text{int}(\text{dom}(f))$. Take any $g \in \partial f(x)$. We know that $\exists \epsilon > 0$, such that $B(x, \epsilon) \subseteq \text{dom}(f)$. We then have that

$$\begin{aligned}
& f(x + z) \geq f(x) + g'z \; \forall x + z \in \text{dom}(f) \\
\implies \; & f\left(x + \frac{\epsilon g}{||g||}\right) \geq f(x) + \epsilon ||g|| \\
\implies \; & ||g|| \leq \max_{y \in B(x, \epsilon)} f(y) - f(x) \\
& < \infty, \text{ by Lemma 44}
\end{aligned}$$

□

Example 94 (Subdifferential Example)**.** Consider $f(x) = \max_{i \in [n]} x_i$. We have that $\partial f(x) = \text{conv}(\{e_i \in \text{CB}(\mathbb{R}^n) : x_i = m\})$ where $\text{CB}(\mathbb{R}^n)$ denotes the set of canonical basis vectors in $\mathbb{R}^n$ and $\text{conv}(\cdot)$ takes the convex hull of its argument and $m := f(x)$.

Proof.

[$y \in \text{conv}(\{e_i \in \text{CB}(\mathbb{R}^n) : x_i = m\}) \implies y \in \partial f(x)$]:

Suppose that $y \in \text{conv}(\{e_i \in \text{CB}(\mathbb{R}^n) : x_i = m\})$ so that we can write $y := \sum_{i=1}^n \theta_i e_i$ where $\theta \in \Delta^{n-1}$ and

$x_i < m \implies \theta_i = 0$. Then, we have that for any $z \in \mathbb{R}^m$

$$\begin{aligned} f(x) + y'z &= m + \left(\sum_{i=1}^n \theta_i e_i\right)' z \\ &= m + \sum_{i=1}^n \theta_i z_i \\ &\leq m + \max_{i\in[n]}\{z_i : \theta_i > 0\}, \text{ since } \theta \in \Delta^{n-1} \\ &= \max_{i\in[n]}\{x_i + z_i : \theta_i > 0\}, \text{ since } \theta_i > 0 \implies x_i = m \\ &\leq \max_{i\in[n]} x_i + z_i \\ &= f(x+z) \end{aligned}$$

[$y \notin \operatorname{conv}(\{e_i \in \operatorname{CB}(\mathbb{R}^n) : x_i = m\}) \implies y \notin \partial f(x)$]:

Alternatively, suppose that $y \notin \operatorname{conv}(\{e_i \in \operatorname{CB}(\mathbb{R}^n) : x_i = m\})$. First, suppose that $y_i \neq 0$ for some $i \in [n]$ where $x_i < m$. Then, we can pick $z = \lambda \operatorname{sign}(y_i) e_i$ for $\lambda > 0$. We get that for sufficiently small λ, $f(x+z) = m$. On the other hand, we get that

$$\begin{aligned} f(x) + y'z &= m + \lambda |y_i| \\ &> m \end{aligned}$$

proving that $y \notin \partial f(x)$. Next, suppose that $y_i = 0 \; \forall i \in [n]$ with $x_i < m$ but $y_i < 0$ for some $i \in [n]$. Then, we can take $z := -e_i$. We get that

$$\begin{aligned} f(x+z) &\leq m \\ &= f(x) \\ &< f(x) - y_i \\ &= f(x) + y'z \end{aligned}$$

proving that $y \notin \partial f(x)$. Next, suppose that $y_i = 0 \; \forall i \in [n]$ with $x_i < m$, $y \geq 0$, and $1_n' y < 1$. Then, pick $z := -1_n$. Then, we have that $f(x+z) = m - 1$ and

$$\begin{aligned} f(x) + y'z &= m - \sum_{i=1}^n y_i \\ &> m - 1 \end{aligned}$$

proving that $y \notin \partial f(x)$. Finally, suppose that $y_i = 0 \; \forall i \in [n]$ with $x_i < m$, $y \geq 0$, and $1_n' y > 1$. Then, pick $z := 1_n$. We get that $f(x+z) = m + 1$ and

$$\begin{aligned} f(x) + y'z &= m + \sum_{i=1}^n y_i \\ &> m + 1 \end{aligned}$$

proving that $y \notin \partial f(x)$. □

Theorem 95 (Fenchel-Young Theorem)**.** For convex and proper f with $\operatorname{dom}(f) = \mathbb{R}^m$ we have that $f(x) + f^*(y) \geq x'y$ with equality if and only if $y \in \partial f(x)$ where f^* is defined in Definition 52.

Proof.

[First result]:

First, in the degenerate case that $f(x) = \infty$, then, we note that $f^*(y) > -\infty$ since f is proper so that the inequality holds. Suppose alternatively that $f(x) < \infty$. We have that

$$\begin{aligned}
f(x) + f^*(y) &= f(x) + \sup_{z \in \mathbb{R}^m} z'y - f(z) \\
&\geq f(x) + x'y - f(x) \\
&= x'y
\end{aligned}$$

[$\implies$]:

Suppose that $f(x) + f^*(y) = x'y$. Then, we have that $x \in \arg\max_{z \in \mathbb{R}^m} z'y - f(z)$. That means that,

$$\begin{aligned}
& x'y - f(x) \geq z'y - f(z)\ \forall z \in \mathbb{R}^m \\
\implies & f(z) \geq f(x) + (z - x)'y\ \forall z \in \mathbb{R}^m \\
\implies & f(x + z) \geq f(x) + z'y\ \forall z \in \mathbb{R}^m \\
\implies & y \in \partial f(x)
\end{aligned}$$

[$\impliedby$]:

Suppose that $y \in \partial f(x)$. Then, we have that

$$\begin{aligned}
f(x + z) \geq f(x) + z'y\ \forall z \in \mathbb{R}^m &\implies f(z) \geq f(x) + (z - x)'y\ \forall z \in \mathbb{R}^m \\
&\implies x'y - f(x) \geq z'y - f(z)\ \forall z \in \mathbb{R}^m \\
&\implies x \in \arg\max_{z \in \mathbb{R}^m} z'y - f(z)
\end{aligned}$$

That in turn implies that

$$\begin{aligned}
f(x) + f^*(y) &= f(x) + \sup_{z \in \mathbb{R}^m} z'y - f(z) \\
&= f(x) + y'x - f(x) \\
&= y'x
\end{aligned}$$

□

Definition 96 (Directional Derivative)**.** The *directional derivative* of convex and proper f at $x \in \mathbb{R}^m$ in direction $d \in \mathbb{R}^m$ is given by

$$f'(x; d) = \lim_{t \downarrow 0} \frac{f(x + td) - f(x)}{t}$$

Proposition 97 (Directional Derivative Proposition)**.** For convex and proper f with $\mathrm{dom}(f) = \mathbb{R}^m$, we have that

$$y \in \partial f(x) \iff y'd \leq f'(x; d)\ \forall d \in \mathbb{R}^m$$

Proof.

[$\implies$]:

Suppose that $y \in \partial f(x)$. Then, we have that

$$\begin{aligned} f(x+tz) &\geq f(x) + \lambda y'z \; \forall (z,t) \in \mathbb{R}^m \times \mathbb{R}_{++} \\ \implies \frac{f(x+tz)-f(x)}{t} &\geq y'z \; \forall (z,t) \in \mathbb{R}^m \times \mathbb{R}_{++} \\ \implies f'(x;z) &\geq y'z \; \forall z \in \mathbb{R}^m, \text{ taking } \lim_{t \downarrow 0} \end{aligned}$$

[$\Longleftarrow$]:

Suppose $y \in \mathbb{R}^m$ is such that

$$y'd \leq f'(x;d) \; \forall d \in \mathbb{R}^m$$

Convexity of f gives that $f(x+td) \leq (1-t)f(x) + tf(x+d) \; \forall d \in \mathbb{R}^m$ and $\forall t \in (0,1]$. That implies that

$$\begin{aligned} \frac{f(x+td)-f(x)}{t} &\leq f(x+d) - f(x) \\ \implies f'(x;d) &\leq f(x+d) - f(x), \text{ taking } \lim_{t \downarrow 0} \\ \implies f(x+d) &\geq f(x) + y'd \end{aligned}$$

which implies $y \in \partial f(x)$ by the arbitrariness of $d \in \mathbb{R}^m$. □

Proposition 98 (Chain Rule for Subdifferentials)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ be convex and proper with $\mathrm{dom}(f) = \mathbb{R}^m$. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Define

$$h(\xi) := f(A\xi + b), \; \xi \in \mathbb{R}^n$$

By Example 48, we know that h is convex. We have that

$$\begin{aligned} \partial h(\xi) &= A' \partial f(A\xi + b) \\ &=: \{A'y : y \in \partial f(A\xi + b)\} \end{aligned}$$

Proof.

Omitted.

Proposition 99 (Optimality Conditions)**.** We have that $0 \in \partial f(x) \iff x$ is a global minimizer of convex and proper f.

Proof.

Recall from Definition 91. We have that

$$y \in \partial f(x) \iff f(x+z) \geq f(x) + y'z \; \forall z \in \mathbb{R}^m$$

From which it's straightforward to conclude that x is a global minimizer of f if and only if $0 \in \partial f(x)$. □

6.1.1 Dual Ascent

Consider the following minimization problem:

$$\inf_{x\in\mathbb{R}^m} f(x) \quad \text{s.t. } Ax = b \tag{17}$$

with f convex and $A \in \mathbb{R}^{n\times m}$ and $b \in \mathbb{R}^n$. The Lagrangian function is:

$$L(x,y) = f(x) + y'(Ax-b) \tag{18}$$

The Lagrangian dual function is

$$g(y) = \inf_{x\in\mathbb{R}^m} L(x,y)$$

The Lagrangian dual problem is

$$\max_{y\in\mathbb{R}^n} g(y)$$

We will additionally assume that there exists primal optimal x^* and dual optimal y^* with strong duality and that g is differentiable. We then have that

$$L(x^*,y) \le L(x^*,y^*) \le L(x,y^*)\ \forall (x,y) \in \mathbb{R}^m \times \mathbb{R}^n$$

In the *dual ascent method*, we solve the dual problem using gradient ascent as prescribed in Algorithm 5. Note that by the Envelope theorem (eg., see here), $\nabla g(y) = Ax^* - b$ where $x^* = \arg\min_{x\in\mathbb{R}^m} L(x,y)$ if f is strictly convex. If f is an affine function of x, then the Envelope theorem doesn't apply so the method fails.

Algorithm 5 Dual Ascent Method

Input: : Optimization problem as in Equation (17) with associated Lagrangian as in Equation (18), y, $\epsilon > 0$, step sequence $(\alpha^{(k)})_{k\in\mathbb{N}}$
Output: : x^*
Start Algorithm:
$k \leftarrow 0$
$y^{(k)} \leftarrow y$
while True **do**
  $x^{(k+1)} \leftarrow \arg\min_{x\in\mathbb{R}^m} L(x, y^{(k)})$
  $y^{(k+1)} \leftarrow y^{(k)} + \alpha^{(k)}(Ax^{(k+1)} - b)$
  $k \leftarrow k+1$ **Quit if**: $||Ax^{(k+1)} - b|| \le \epsilon$
end while
return $x^{(k)}$

Remark 100 (Parallelization in Dual Ascent)**.** Suppose that f is *separable* meaning that $f(x) = \sum_{i=1}^M f_i(x_i)$ where $\{x_i\}_{i\in[M]}$ partition x. We also partition $A = [A_1, A_2, ..., A_M]$. Then, we have that $Ax = \sum_{i=1}^M A_i x_i$. Then, we have that

$$\begin{aligned} L(x,y) &= \sum_{i=1}^M L_i(x_i,y) \\ &= \sum_{i=1}^M \left(f_i(x_i) + y'A_ix_i - \frac{1}{n}y'b \right) \end{aligned}$$

The first step of the dual ascent algorithm changes to (for $i \in [n]$),

$$x_i^{(k+1)} = \operatorname*{arg\,min}_{z_i \in \mathbb{R}^{\dim(x_i)}} L_i(z_i^{(k)}, y^{(k)})$$

meaning that we can do all of the i's in parallel.

Example 101 (Dual Ascent Example)**.** Consider the problem

$$\inf_{x \in \mathbb{R}} x^2 \text{ s.t. 2x = 0}$$

The dual ascent method consists of the updates:

$$\begin{aligned} &L(x,y) = x^2 + y(2x) \\ \implies &\nabla_x L(x,y) = 2x + 2y \\ \implies &x^{(k+1)} = -y^{(k)} \\ \implies &g(y) = -y^2 \\ \implies &\nabla g(y) = -2y \\ \implies &y^{(k+1)} = y^{(k)} + \alpha^{(k)}(-2y^{(k)}) \end{aligned}$$

6.1.2 Augmented Lagrangian Method

The augmented Lagrangian methods (also known as Methods of Multipliers) were developed to bring robustness to the dual ascent method and to yield convergence without assumptions on the strict convexity of f nor its finiteness. We will define the augmented Lagrangian function

$$L_\rho(x,y) = f(x) + y'(Ax - b) + \frac{\rho}{2}||Ax - b||_2^2$$

This is equivalent to the usual Lagrangian for the problem

$$\begin{aligned} \inf_{x \in \mathbb{R}^m} \; & f(x) + \frac{\rho}{2}||Ax - b||^2 \\ \text{s.t. } & Ax = b \end{aligned} \tag{19}$$

Given the constraint, the previous problem has an equivalent objective and optimum as this problem

$$\begin{aligned} \inf_{x \in \mathbb{R}^m} \; & f(x) \\ \text{s.t. } & Ax = b \end{aligned} \tag{20}$$

If we apply the dual ascent of Algorithm 5 to the problem in Equation (19), we get the following updates (swapping ρ for $\alpha^{(k)}$),

$$\begin{aligned} x^{(k+1)} &:= \operatorname*{arg\,min}_{x \in \mathbb{R}^m} L_\rho(x, y^{(k)}) \\ y^{(k+1)} &:= y^{(k)} + \rho(Ax^{(k+1)} - b) \end{aligned}$$

By the convexity of f and the presence of the norm, the problem in Equation (19) is bounded below. Why are we replacing $\alpha^{(k)}$ with a iteration-invariant ρ that appears in the objective? Assume that f is differentiable though that's actually not needed. For (x^*, y^*) to be primal and dual optimal for the problem in Equation (20), by Proposition 64 we need that

$$\begin{aligned} &Ax^* - b = 0, \text{ primal feasibility} \\ &\nabla f(x^*) + (y^*)'A = 0, \text{ dual feasibility (ie., } g(y^*) > -\infty) \end{aligned}$$

By definition, $x^{(k+1)}$ minimizes $L_\rho(x, y^{(k)})$. That means that

$$\begin{aligned} 0 &= \nabla_x L_\rho(x^{(k+1)}, y^{(k)}) \\ &= \nabla f(x^{(k+1)}) + A' y^{(k)} + \rho A' A x^{(k+1)} - \rho A' b \\ &= \nabla f(x^{(k+1)}) + A'(y^{(k)} + \rho(A x^{(k+1)} - b)) \\ &= \nabla f(x^{(k+1)}) + A' y^{(k+1)}, \text{ with second update equation} \end{aligned}$$

The pair $(x^{(k+1)}, y^{(k+1)})$ is *dual feasible* (not primal feasible) in the problem of Equation (20) using ρ as the step-size. That's because $g(y^{(k+1)}) = \inf_{x \in \mathbb{R}^m} f(x) + (y^{(k+1)})'(Ax - b) > -\infty$ because the objective is convex and there exists a point $x^{(k+1)} \in \mathbb{R}^m$ with $\nabla f(x^{(k+1)}) + A' y^{(k+1)} = 0$.

6.2 ADMM

We will consider solving the problem

$$\begin{aligned} &\inf_{(x,z) \in \mathbb{R}^m \times \mathbb{R}^n} f(x) + g(z) \\ &\text{s.t.} Ax + Bz = c \end{aligned} \tag{21}$$

where g is not the dual function. On dimensions, we have that $A \in \mathbb{R}^{p \times m}, B \in \mathbb{R}^{p \times n}, c \in \mathbb{R}^p$. The optimal value is p^*. We assume that f and g are convex, but not necessarily differentiable. As explained in Section 6.1.2, the augmented Lagrangian is given by

$$L_\rho(x, z, y) = f(x) + g(z) + y'(Ax + Bz - c) + \frac{\rho}{2} ||Ax + Bz - c||_2^2 \tag{22}$$

The ADMM method as prescribed in Algorithm 6 does sequential as opposed to joint updates of the primal variables as in the Dual Ascent method of Algorithm 5.

Algorithm 6 ADMM Method

Input: : Optimization problem as in Equation (21) with associated augmented Lagrangian as in Equation (22), y, z, $\epsilon > 0$, ρ
Output: : x^*, z^*
Start Algorithm:
$k \leftarrow 0$
$y^{(k)} \leftarrow y$
$z^{(k)} \leftarrow z$
while True **do**
  $x^{(k+1)} \leftarrow \arg\min_{x \in \mathbb{R}^m} L_\rho(x, z^{(k)}, y^{(k)})$
  $z^{(k+1)} \leftarrow \arg\min_{z \in \mathbb{R}^n} L_\rho(x^{(k+1)}, z, y^{(k)})$
  $y^{(k+1)} \leftarrow y^{(k)} + \rho \left(A x^{(k+1)} + B z^{(k+1)} - c\right)$
  $k \leftarrow k + 1$ **Quit if**: $||Ax^{(k+1)} + Bz^{(k+1)} - c|| \leq \epsilon$
end while
return $x^{(k)}$

For (x^*, z^*, y^*) to be primal and dual optimal for the problem in Equation (20), by a generalization of Proposition 64 inspired by Proposition 99 we need that

$$\begin{aligned} &Ax^* + Bz^* - b = 0, \text{ primal feasibility} \\ &0 \in \partial_x f(x^*) + (y^*)' A = 0, \text{ part of dual feasibility} \\ &0 \in \partial_z g(z^*) + (y^*)' B = 0, \text{ other part of dual feasibility} \end{aligned}$$

Dual feasibility means that

$$\inf_{(x,z) \in \mathbb{R}^m \times \mathbb{R}^n} f(x) + g(z) + (y^*)'(Ax + Bz - c) > -\infty$$

Given the augmented Lagrangian in Equation (22), we have that dual feasibility is equivalent to

$$0 \in \partial_x L_0(x^*, z^*, y^*)$$
$$0 \in \partial_z L_0(x^*, z^*, y^*)$$

Assuming a "saddle point" (x^*, z^*, y^*) exists (ie., the primal and dual optimal values are attained), then we have that

$$L_0(x^*, z^*, y) \leq L_0(x^*, z^*, y^*) \leq L_0(x, z, y^*) \; \forall (x, z, y) \in \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^p$$

6.2.1 ADMM– Scaled Version

The ADMM method of Algorithm 6 can be written in a different form, which is often more convenient.

Let the residual be $r := Ax + Bz - c$ and the scaled dual variable be $u := \frac{1}{\rho} y$. We then have that the non-objective part of the augmented Lagrangian of Equation (22) can be written as

$$\begin{aligned} y'(Ax + Bz - c) + \frac{\rho}{2} ||Ax + Bz = c||_2^2 &= y'r + \frac{\rho}{2} ||r||_2^2 \\ &= \frac{\rho}{2} \left|\left| r + \left(\frac{1}{\rho} y \right) \right|\right|_2^2 - \frac{1}{2\rho} ||y||_2^2 \\ &= \frac{\rho}{2} ||r + u||_2^2 - \frac{\rho}{2} ||u||_2^2 \end{aligned}$$

Given this analysis, we now note that we can write the update steps corresponding to those in Algorithm 6 (removing irrelevant parts of the objectives) as

$$\begin{aligned} x^{(k+1)} &\in \operatorname*{arg\,min}_{x \in \mathbb{R}^m} L_\rho(x, z^{(k)}, y^{(k)}) \\ &= \operatorname*{arg\,min}_{x \in \mathbb{R}^m} f(x) + \frac{\rho}{2} ||Ax + Bz^{(k)} - c + u^{(k)}||_2^2 \\ z^{(k+1)} &\in \operatorname*{arg\,min}_{z \in \mathbb{R}^n} L_\rho(x^{(k+1)}, z, y^{(k)}) \\ &= \operatorname*{arg\,min}_{z \in \mathbb{R}^n} g(z) + \frac{\rho}{2} ||Ax^{(k+1)} + Bz - c + u^{(k)}||_2^2 \\ u^{(k+1)} &= u^{(k)} + Ax^{(k+1)} + Bz^{(k+1)} - c \end{aligned}$$

Defining the iteration specific residual $r^{(k)} := Ax^{(k)} + Bz^{(k)} - c$, we see that

$$u^{(k+1)} = u^{(0)} + \sum_{i=1}^{k} r^{(k)}$$

6.2.2 ADMM– Convergence Theory

In this section without proof, we state convergence results for the problem of Equation (21).

We additionally assume f, g are closed (see Definition 71), convex (see Definition 35), and proper (see Definition 89) and that the primal and dual optimal points are attained at (x^*, z^*, y^*). Then $\forall (x^{(0)}, z^{(0)}, y^{(0)})$, we have primal convergence, which means that

$$r^{(k)} \to 0 \text{ as } k \to \infty$$
$$f(x^{(k)}) + g(x^{(k)}) \to p^*$$

and dual variable convergence

$$y^{(k)} \to y^* \text{ and}$$
$$u^{(k)} \to u^* = \frac{1}{\rho} y^* \text{ as } k \to \infty$$

But $x^{(k)}, z^{(k)}$ may not converge. The proofs of these convergence claims are left to Boyd et al. (2011).

6.3 ADMM- Examples

In this section, we will present various examples and applications of the ADMM algorithm.

Example 102 (Optimization in Closed Convex Set)**.** Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^m} & f(x) \\ \text{s.t. } & x \in C \end{aligned}$$

where C is a closed and convex set. This problem rewrites as

$$\begin{aligned} \min_{(x,z) \in \mathbb{R}^m \times \mathbb{R}^m} & f(x) + g(z) \\ \text{s.t. } & x - z = 0 \end{aligned}$$

where $g(z) = \begin{cases} 0, & \text{if } z \in C \\ \infty, & \text{otherwise} \end{cases}$ The ADMM algorithm in this example has the following update steps

$$\begin{aligned} x^{(k+1)} &\in \arg\min_{x \in \mathbb{R}^m} f(x) + \frac{\rho}{2}||x - z^{(k)} + u^{(k)}||_2^2 \\ z^{(k+1)} &\in \arg\min_{z \in \mathbb{R}^m} g(z) + \frac{\rho}{2}||x^{(k+1)} - z + u^{(k)}||_2^2 \\ &= \Pi_C(x^{(k+1)} + u^{(k)}) \\ u^{(k+1)} &= u^{(k)} + x^{(k+1)} - z^{(k+1)} \end{aligned}$$

where Π_C is the projection operator onto C. Since C is closed and convex, there's a single closest point.

Example 103 (Finding Point in Intersection of Two Closed Convex Sets)**.** We wish to find $\mathbb{R}^m \ni X \in C \cap D$ where C and D are closed and convex sets. This problem rewrites as

$$\begin{aligned} \min_{(x,z) \in \mathbb{R}^m \times \mathbb{R}^m} & f(x) + g(z) \\ \text{s.t. } & x - z = 0 \end{aligned}$$

where $f(x) = \begin{cases} 0, & \text{if } x \in C \\ \infty, & \text{otherwise} \end{cases}$ and $g(z) = \begin{cases} 0, & \text{if } z \in D \\ \infty, & \text{otherwise} \end{cases}$.The ADMM update steps are

$$\begin{aligned} x^{(k+1)} &= \Pi_C(z^{(k)} - u^{(k)}) \\ z^{(k+1)} &= \Pi_D(x^{(k+1)} + u^{(k)}) \\ u^{(k+1)} &= u^{(k)} + x^{(k+1)} - z^{(k+1)} \end{aligned}$$

This ADMM algorithm is much faster than the Von Neumann's alternating projection algorithm. That algorithm has the following update steps

$$\begin{aligned} x^{(k+1)} &= \Pi_C(z^{(k)}) \\ z^{(k+1)} &= \Pi_D(x^{(k+1)}) \end{aligned}$$

Example 104 (Least Absolute Shrinkage and Selection Operator (LASSO))**.** We consider the least squares problem with an L_1 penalty

$$\min_{x\in\mathbb{R}^m} \frac{1}{2}||Ax-b||_2^2 + \lambda||x||_1$$

where $A \in \mathbb{R}^{n\times m}$ and $b \in \mathbb{R}^n$. The parameter $\lambda \in \mathbb{R}_{++}$ governs the tradeoff between the least squares objective and the regularization $||x||_1$, which promotes sparsity. In the context of the ADMM algorithm, this problem rewrites as

$$\begin{aligned} &\min_{(x,z)\in\mathbb{R}^m\times\mathbb{R}^m} f(x)+g(z) \\ &\text{s.t. } x-z=0 \end{aligned}$$

where $f(x) = \frac{1}{2}||Ax-b||_2^2$ *and* $g(z) = \lambda||z||_1$. The ADMM update equations are

$$x^{(k+1)} = \arg\min_{x\in\mathbb{R}^m} \frac{1}{2}||Ax-b||_2^2 + \frac{\rho}{2}||x-z^{(k)}+u^{(k)}||_2^2$$

The whole thing is convex and differentiable so that the optimum is characterized by the first order condition. We get the first order condition

$$\begin{aligned} 0 &= (A'A)x^{(k+1)} - A'b + \rho x^{(k+1)} - \rho(z^{(k)} - u^{(k)}) \\ \implies x^{(k+1)} &= (A'A+\rho I_m)^{-1}(A'b + \rho(z^{(k)} - u^{(k)})) \end{aligned}$$

Since $\rho > 0$, we get that $A'A + \rho I_m \succ 0$ meaning that it's invertible. For the update equation, we get that

$$z^{(k+1)} = \arg\min_{z\in\mathbb{R}^m} \lambda||z||_1 + \frac{\rho}{2}||x^{(k+1)} - z + u^{(k)}||_2^2$$

This separates component wise so that

$$(z^{(k+1)})_i = \arg\min_{z_i\in\mathbb{R}} \lambda|z_i| + \frac{\rho}{2}(z_i - v_i)^2$$

where $v_i := x_i^{(k+1)} + u_i^{(k)}$. This leads to an idea of "soft thresholding". We have the function

$$h(\xi) = \lambda|\xi| + \frac{\rho}{2}(\xi - v)^2$$

and we wish to find ξ to minimize h. For that, we need $0 \in \partial h(\xi)$. We have that

$$\partial h(\xi) = \begin{cases} [-\lambda, \lambda] - \rho v, & \text{if } \xi = 0 \\ \lambda\,\text{sign}(\xi) + \rho\xi - \rho v, & \text{otherwise} \end{cases}$$

Thus, we have that

$$0 \in h(0) \iff |v| \leq \frac{\lambda}{\rho}$$

Otherwise, we need that

$$\frac{\lambda}{\rho}\,\text{sign}(\xi) + \xi - v = 0$$

So if $v > \frac{\lambda}{\rho}$, we set $\xi = v - \frac{\lambda}{\rho} > 0$ and if $v < -\frac{\lambda}{\rho}$, we set $\xi = v + \frac{\lambda}{\rho} < 0$, and we get a minimizer. We will let the function $\xi(v) := S_{\lambda/\rho}(v)$ denote the minimizer of the problem. In summary, we get that the LASSO update becomes

$$(z^{(k+1)})_i = S_{\lambda/\rho}\left(x_i^{(k+1)} + u_i^{(k)}\right), \text{ for } i \in [m]$$

and

$$u^{(k+1)} = u^{(k)} + x^{(k+1)} - z^{(k+1)}$$

7 Nuclear Norm Optimization

In this section, we will look at nuclear norm optimization after better understanding singular value decomposition and matrix norms.

7.1 Prerequisites

Here we discuss singular values and eigenvalues, which are prerequisites to understanding nuclear norm optimization.

Definition 105 (Singular Value Decomposition (SVD) and Variants)**.** The *reduced singular value decomposition* of $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ is $A = U\Sigma V'$ where $U \in \mathbb{R}^{m\times n}$ with $U'U = I_m$, $\Sigma \in \mathbb{R}^{n\times n}$ with $\Sigma = \operatorname{diag}(\sigma_1, ..., \sigma_n)$, and $V \in \mathbb{R}^{n\times n}$ with $V'V = I_n$. It is customary to sort the singular values $\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_n \geq 0$ and let $\sigma(A)$ be the sorted vector of singular values of A. The singular values are nonnegative WLOG as we can adjust the V matrix accordingly and multiply appropriate columns by -1.

We say that columns v_j of V are *right singular vectors* of A and columns u_j of A are left singular vectors of A. When we right-multiply $A = U\Sigma V'$ by V, we get that $AV = \Sigma U$ from which we see that $Av_j = \sigma_j u_j$ for $j \in [n]$ where u_j is the jth column of U and v_j is the jth column of V.

From the reduced singular decomposition of A, we can form the *full singular value decomposition* of A. When $m \geq n$, that consists of adding columns $u_{n+1}, ..., u_m$ to U to form $\hat{U}$ so that $\hat{U} \in \mathbb{R}^{m\times m}$ and is orthonormal. We form $\hat{\Sigma} = \begin{bmatrix} \Sigma \\ 0_{(m-n)\times n} \end{bmatrix}$ and set $\hat{V} = V$. We get the full singular decomposition $A = \hat{U}\hat{\Sigma}\hat{V}'$.

Suppose we know that $\operatorname{rank}(A) = r \leq n$, then we can decompose $A = \tilde{U}\tilde{\Sigma}\tilde{V}'$ with $\tilde{U} \in \mathbb{R}^{m\times r}$, $\tilde{\Sigma} \in \mathbb{R}^{r\times r}$ as a diagonal matrix, and $\tilde{V} \in \mathbb{R}^{n\times r}$. We can do that from the reduced singular value decomposition under the observation that if $\operatorname{rank}(A) = r$, then there are r non-zero singular values in A.

Theorem 106 (Existence of SVD)**.** Given $A \in \mathbb{R}^{m\times n}$, A has an SVD decomposition.

Proof.

Omitted. See here for instance.

We don't prove it here but the SVD of a matrix always exists.

Lemma 107 (Relationship Between Singular Values of A and Eigenvalues of $A'A$)**.** Take $A \in \mathbb{R}^{m\times n}$. We have that $\operatorname{eig}(A'A) = \sigma(A)$.

Proof.

Let $U\Sigma V'$ be the reduced SVD of A. We have that

$$\begin{aligned} A'A &= (U\Sigma V')'(U\Sigma V') \\ &= V\Sigma \underbrace{U'U}_{I_m} \Sigma V' \\ &= V\Sigma^2 V' \end{aligned}$$

The eigenvalues of $A'A$ are the diagonal entries of $A'A$ which are the squares of the singular values of A, which are in Σ. ☐

Lemma 108 (Relationship Between Singular Values and Eigenvalues of Symmetrized Matrix)**.** Take $A \in \mathbb{R}^{m\times n}$. We have that $\pm\sigma(A) \in \text{eig}\,(M)$ where $M := \begin{bmatrix} 0_{m\times m} & A \\ A' & 0_{n\times n} \end{bmatrix}$.

Proof.

Write the SVD $A = U\Sigma V'$. We note that for any $i \in [n]$, we have that

$$Av_i = \sigma_i u_i \text{ and } A'u_i = \sigma_i v_i$$

We form the vectors $z_i^{(+)} = [u_i', v_i']'$ and $z_i^{(-)} = [u_i', -v_i']'$. We note that $Mz_i^{(+)} = \sigma_i z_i^{(+)}$ and $Mz_i^{(-)} = -\sigma_i z_i^{(-)}$ so that $\pm\sigma(A) \in \text{eig}(M)$. ☐

Definition 109 (Dual Matrix Norm)**.** Like in Definition 33, given a matrix norm $||\cdot||$ on $\mathbb{R}^{m\times n}$, we define the dual norm to be

$$||X||_d := \sup\{\langle X, Y\rangle : ||Y|| \leq 1\}$$

where $\langle X, Y\rangle := \text{Tr}(X'Y)$. Since the dual norm is the pointwise supremum over a collection of convex functions, we have that $||\cdot||_d$ is convex for any norm $||\cdot||$.

Definition 110 (Frobenius Norm of Matrix)**.** For $X \in \mathbb{R}^{m\times n}$, we define its *frobenius norm* as

$$\begin{aligned} ||X||_F &:= \left(\sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij}^2\right)^{1/2} \\ &= (\langle X, X\rangle)^{1/2} \\ &= (\text{Tr}(X'X))^{1/2} \\ &= ||\,\text{vec}_R(X)||_2 \end{aligned}$$

Definition 111 (Nuclear Norm of Matrix)**.** For $X \in \mathbb{R}^{m\times n}$, we define its *nuclear norm* as

$$||X||_* := \sum_{i=1}^{n} \sigma_i(A)$$

Definition 112 (Schur Complement)**.** Consider a block matrix

$$M := \begin{bmatrix} P & Q \\ R & S \end{bmatrix}$$

where $P \in \mathbb{R}^{m \times m}$, $Q \in \mathbb{R}^{m \times n}$, $R \in \mathbb{R}^{n \times m}$, and $S \in \mathbb{R}^{n \times n}$. The *Schur complement* of the block S is given by

$$M/S := P - QS^{-1}R$$

Lemma 113 (Schur Complement PSD Lemma)**.** Let $M := \begin{bmatrix} A & B \\ B' & C \end{bmatrix}$ where $A \in \mathbb{R}^{m \times m}$ and $\mathbb{R}^{n \times n} \ni C \succ 0$ are symmetric matrices and $B \in \mathbb{R}^{m \times n}$. Then, we have that $M/C \succeq 0 \iff M \succeq 0$.

Proof.

We have that for $a := [x', y']' \in \mathbb{R}^{m+n}$

$$\begin{aligned} a'Ma &= x'Ax + 2x'By + y'Cy \\ &= x'(A - BC^{-1}B')x + z'Cz, \text{ where } z := y + C^{-1}B'x \\ &= x'(M/C)x + z'Cz \end{aligned} \tag{23}$$

[$\implies$]:

If $M/C \succeq 0$, since $C \succ 0$ by assumption, by the RHS of Equation (23), we have that $M \succ 0$.

[$\impliedby$]:

If $M \succeq 0$, we have that the LHS of Equation (23) is non-negative for any $[x', y']' = a \in \mathbb{R}^{m+n}$. Choose $y = -C^{-1}B'x \implies z = 0$. Then, Equation (23) implies that $x'(M/C)x \geq 0 \; \forall x \in \mathbb{R}^m \implies M/C \succeq 0$. $\square$

Lemma 114 (Spectral Norm Bound Lemma)**.** For $Z \in \mathbb{R}^{m \times n}$ and $t > 0$, we have that

$$\begin{aligned} ||Z||_2 \leq t &\iff t^2 I_m - ZZ' \succeq 0 \\ &\iff t^2 I_n - Z'Z \succeq 0 \\ &\iff \begin{bmatrix} tI_m & Z \\ Z' & tI_n \end{bmatrix} \succeq 0 \end{aligned}$$

Proof.

The first two equivalences follow from Lemma 107 and the third from Lemma 113. $\square$

Theorem 115 (Dual of Spectral Norm is Nuclear Norm)**.** The dual of $|| \cdot ||_2$ is $|| \cdot ||_*$.

Proof.

For notation in this proof, before we've characterized the dual norm of $|| \cdot ||_2$ to be $|| \cdot ||_*$, we will refer to the dual norm as $|| \cdot ||_{2,d}$.

Take any matrix $X \in \mathbb{R}^{m\times n}$. We assume that it has rank $r \leq n \leq m$ and we write the rank SVD decomposition

$$X = U\Sigma V'$$

where $U \in \mathbb{R}^{m\times r}, \Sigma \in \mathbb{R}^{r\times r}$ with only non-zero diagonal entries and $\sigma_{r+1} = ... = \sigma_n = 0$ and $V \in \mathbb{R}^{n\times r}$. By definition, we have that $||X||_* = \mathrm{Tr}(\Sigma)$.

Next, we observe that $||UV'||_2 = ||UI_rV'||_2 = 1$ by the argument in Example 51. We now have that

$$\begin{aligned}
||X||_{2,d} &= \sup_{||Y||_2 \leq 1} \langle X, Y\rangle \\
&\geq \langle X, UV'\rangle, \text{ since } ||UV'||_2 \leq 1 \\
&= \mathrm{Tr}(V\Sigma U'UV') \\
&= \mathrm{Tr}(V\Sigma V'), \text{ since } U'U = I_r \\
&= \mathrm{Tr}(\Sigma), \text{ since } V'V = I_r \text{ and } \mathrm{Tr}(\cdot) \text{ is invariant to circular permutations} \\
&= ||X||_*
\end{aligned}$$

Now, to find $||X||_{2,d}$, we must solve

$$\begin{aligned}
&\sup_{Y\in\mathbb{R}^{m\times n}} \langle X, Y\rangle \\
&\quad \text{s.t. } ||Y||_2 \leq 1 \\
\iff & \\
&\sup_{Y\in\mathbb{R}^{m\times n}} \langle X, Y\rangle \\
&\quad \text{s.t. } \begin{bmatrix} I_m & -Y \\ -Y' & I_n \end{bmatrix} \succeq 0, \text{ by Lemma 114} \\
\iff & \\
&\sup_{Y\in\mathbb{R}^{m\times n}} \sum_{i=1}^m \sum_{j=1}^n X_{ij}Y_{ij} \\
&\quad \text{s.t. } \begin{bmatrix} I_m & 0_{m\times n} \\ 0_{n\times m} & I_n \end{bmatrix} - \sum_{i=1}^m \sum_{j=1}^n Y_{ij} \underbrace{\left(e_ie_{m+j}' + e_{m+j}e_i'\right)}_{=:2E_{ij}} \succeq 0 \\
\iff & \\
&\sup_{Y\in\mathbb{R}^{m\times n}} \sum_{i=1}^m \sum_{j=1}^n X_{ij}Y_{ij} \\
&\quad \text{s.t. } \sum_{i=1}^m \sum_{j=1}^n Y_{ij}E_{ij} \preceq \frac{1}{2} \begin{bmatrix} I_m & 0_{m\times n} \\ 0_{n\times m} & I_n \end{bmatrix}
\end{aligned}$$

where the canonical basic vectors are in dimension $\mathbb{R}^{m\times n}$. Using our work from Section 3.4, we recognize the primal problem associated with this problem to be

$$\begin{aligned}
&\inf_{W\in S_+^{m+n}} \frac{1}{2}\langle I_{m+n}, W\rangle \\
&\quad \text{s.t. } \langle E_{ij}, W\rangle = X_{ij} \ \forall (i,j) \in [m\times n]
\end{aligned}$$

We observe that the problem has the following feasible point

$$\begin{aligned} W &= \begin{bmatrix} U\Sigma U' & U\Sigma V' \\ V\Sigma U' & V\Sigma V' \end{bmatrix} \\ &= \begin{bmatrix} U \\ V \end{bmatrix} \Sigma [U', V'] \\ &\succeq 0 \end{aligned}$$

and the primal objective is

$$\begin{aligned} \frac{1}{2}\langle I_{m+n}, W\rangle &= \frac{1}{2}\left(\mathrm{Tr}(U\Sigma U') + \mathrm{Tr}(V\Sigma V')\right) \\ &= \mathrm{Tr}(\Sigma) \\ &= ||X||_* \end{aligned}$$

Now, any feasible point for the primal problem gives an upper bound for the optimal solution of the dual problem by weak duality as shown in Section 3.4. That means that $||X||_{2,d} \leq ||X||_*$. Given our earlier demonstration that $||X||_{2.d} \geq ||X||_*$, we conclude that $||X||_{2.d} = ||X||_*$. □

Corollary 116 (Nuclear Norm is Convex)**.** We have that $|| \cdot ||_*$ is convex.

Proof.

By Theorem 115, we have that $|| \cdot ||_*$ is the dual norm of $|| \cdot ||_2$. Thus, by the statement in Definition 109, we have that $|| \cdot ||_*$ is convex. □

Definition 117 (Convex Envelope)**.** Let $f : C \to \mathbb{R}$ where C is a convex set and f not necessarily convex. The *convex envelope* of f is the pointwise maximum of all convex functions that lie below f.

7.2 *Matrix Completion*

In this section, we discuss the problem of matrix completion with nuclear norm optimization. The matrix completion problem is that of finding a low-rank matrix that equals provided entries of a given matrix. We then consider conditions under which a relaxed and convex version of the problem, which minimizes the nuclear norm gives identical solutions to the original problem.

Example 118 (Matrix Completion)**.** Suppose we're given a matrix $X \in \mathbb{R}^{m\times n}$ with some entries known $(i,j) \in \Omega$ with corresponding value m_{ij} and others not known. The matrix completion problem is

$$\begin{aligned} \inf_{X\in\mathbb{R}^{m\times n}} \ & \mathrm{rank}(X) \\ \text{s.t. } & X_{ij} = m_{ij} \text{ for } (i,j) \in \Omega \end{aligned}$$

The motivation is that we wish to find the matrix with the smallest number of "features" (corresponding to nonzero singular values) that agrees with all of the observed entries of X. An SDP relaxation is

$$\begin{aligned} \inf_{X\in\mathbb{R}^{m\times n}, W\in\mathbb{S}_+^{m+n}} \ & \frac{1}{2}\mathrm{Tr}(W) \\ \text{s.t. } & \begin{bmatrix} (W)_{1:m,1:m} & X \\ X' & (W)_{m+1:m+n,m+1:m+n} \end{bmatrix} \succeq 0, X_{ij} = m_{ij}\ \forall (i,j) \in \Omega \end{aligned}$$

To see that this is in fact a relaxation, note that any feasible X of the original problem is also feasible in the relaxation.

Also note that for any X, we can take a rank SVD of $X = U\Sigma V'$ and take the feasible point

$$A = \begin{bmatrix} U\Sigma U' & U\Sigma V' \\ V\Sigma U' & V\Sigma V' \end{bmatrix}$$

at which the objective is $||X||_*$ given our work at the end of the proof of Theorem 115. We note that the dual to this problem has value at least $||X||_*$ as shown in the proof of Theorem 115, which lower bounds the valued of this problem by weak duality. Thus, we have that the optimal value of this problem is $||X||_*$ for some X, which matches the observed values $\{m_{ij} : (i,j) \in \Omega\}$. In other words, the optimal value of the problem matches the optimal value of

$$\begin{aligned} &\inf_{X \in \mathbb{R}^{m \times n}} ||X||_* \\ &\text{s.t. } X_{ij} = m_{ij} \; \forall (i,j) \in \Omega \end{aligned}$$

Next, observe that for $||X||_2 \leq 1$, $||X||_* \leq \text{rank}(X)$. Thus, if X is optimal in the original problem, X is feasible in the relaxation and the objective of the relaxation is weakly smaller. In fact, $||\cdot||_*$ is a convex (ie., Corollary 116) lower bound on $\text{rank}(\cdot)$. In fact, $||\cdot||_*$ is the convex envelope on $\{X : ||X||_2 \leq 1\}$. Also, for $||X||_2 \leq M$, we have $r = \text{rank}(X) \geq \frac{||X||_*}{M}$. In fact, in this case, we have that $\frac{||X||_*}{M}$ is the convex envelope for $\text{rank}(\cdot)$ on $\{X : ||X||_2 \leq M\}$.

Now consider

$$X_0 = \arg\min_X \text{rank}(X) \text{ s.t. } A(X) = b$$

where A is a linear operator and

$$X_* = \arg\min_X ||X||_* \text{ s.t. } A(X) = b$$

Suppose $||X_0||_2 = M$. Then, we have that

$$\begin{aligned} \frac{||X_*||_*}{M} &\leq \frac{||X_0||_*}{M}, \text{ by the second optimization problem} \\ &\leq \text{rank}(X_0), \text{ by the convex envelope} \\ &\leq \text{rank}(X_*) \end{aligned}$$

Intuitively, this means that the optimal value X_0 of the problem of interest is sandwiched between two functions of the optimal value of the relaxed problem.

Definition 119 (Additivity and Subadditivity)**.** We say that $f : \mathbb{R}^n \to \mathbb{R}$ is *subadditive* if $f(x+y) \leq f(x) + f(y)$. We say that f is *additive* if $f(x+y) = f(x) + f(y)$.

Definition 120 (Cardinality of Vector)**.** For a vector $x \in \mathbb{R}^n$, we say that cardinality of x is $||x||_0$. That is, the cardinality of x is the number of nonzero entries of x.

It's plain to see that $||\cdot||_0$ and $||\cdot||_1$ are subadditive. It's also plain to see that $||\cdot||_0$ and $||\cdot||_1$ are additive if the considered vectors have *disjoint* support (nonzero values do not overlap).

Lemma 121 (Rank of Matrices is Subadditive)**.** For matrices $A, B \in \mathbb{R}^{m \times n}$, we have that

$$\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$$

Proof.

We have that $\text{col}(A+B) \subseteq \text{col}(A) + \text{col}(B)$. Thus, trivially, we have that $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Lemma 122 (Nuclear Norm is Subadditive)**.** For matrices $A, B \in \mathbb{R}^{m \times n}$, we have that

$$||A+B||_* \leq ||A||_* + ||B||_*$$

Proof.

We have that

$$\begin{aligned}
||A+B||_* &= \sup_Y \{\langle A+B, Y\rangle : ||Y||_2 = 1\} \\
&\leq \sup_Y \{\langle A, Y\rangle : ||Y||_2 = 1\} + \sup_Y \{\langle B, Y\rangle : ||Y||_2 = 1\} \\
&\leq ||A||_* + ||B||_*
\end{aligned}$$

Theorem 123 (Rank is Additive Characterization)**.** For matrices $A, B \in \mathbb{R}^{m \times n}$, we have that

$$\begin{aligned}
&\text{rank}(A+B) = \text{rank}(A) + \text{rank}(B) \\
\iff& \\
&\text{col}(A) \cap \text{col}(B) = \{0\} \text{ and } \text{col}(A') \cap \text{col}(B') = \{0\}
\end{aligned}$$

Proof.

Omitted.

Lemma 124 (Nuclear Norm is Additive Sufficient Condition)**.** For matrices $A, B \in \mathbb{R}^{m \times n}$, we have that

$$\begin{aligned}
&\text{col}(A) \perp \text{col}(B) \text{ (ie., } A'B = 0) \text{ and } \text{col}(A') \perp \text{col}(B') \text{ (ie., } AB' = 0) \\
\implies& ||A+B||_* = ||A||_* + ||B||_*
\end{aligned}$$

Proof.

Consider the rank SVDs $A = U_A \Sigma_A V_A'$ (rank r_1) and $B = U_B \Sigma_B V_B'$ (rank r_2). We then have that

$$\begin{aligned}
\Sigma_A^{-1} U_A' A &= \Sigma_A^{-1} U_A' U_A \Sigma_A V_A' \\
&= V_A'
\end{aligned}$$

and similarly that

$$\Sigma_B^{-1} U_B' B = V_B'$$

Thus, we have that

$$\begin{aligned}
V_A' V_B &= \Sigma_A^{-1} U_A' A B' U_B \Sigma_B^{-1} \\
&= 0, \text{ since by assumption } AB' = 0
\end{aligned}$$

Likewise, we have that $U_A' U_B = 0$. We next observe that

$$A + B = [U_A, U_B] \begin{bmatrix} \Sigma_A & 0 \\ 0 & \Sigma_B \end{bmatrix} [V_A, V_B]' \tag{24}$$

Since $V_A' V_B = 0$, $U_A' U_B = 0$, $U_A' U_A = I$, ... we have that Equation (24) is an SVD of $A + B$. From here, we conclude that $||A + B||_* = ||A||_* + ||B||_*$. □

Definition 125 (Restricted Isometry and Low Rank Recovery (RIP))**.** Let $A : \mathbb{R}^{m\times n} \to \mathbb{R}^p$ be linear with $m \le n$. For every integer r with $1 \le r \le m$, we define the *restricted isometry property* (RIP) constant $\delta_r(A) \in [0, 1]$ as the smallest number in the range such that

$$(1 - \delta_r(A))||X||_F \le ||A(X)||_2 \le (1 + \delta_r(A))||X||_F$$

holds for all matrices $X \in \mathbb{R}^{m\times n}$ of rank less than or equal to r.

We show some examples of RIP numbers:

- If $A(X) = \mathrm{vec}_R(X)$, then we have that $||A(X)||_2 = ||X||_F$ so that $\delta_r(A) = 0$.
- If $A(X) = x_{11}$, for $r = 1$ and $n = 2$, $\nexists \delta_r(A) < 1$. That's because we can take A to be the zero matrix except with a nonzero entry at x_{12}. In that case, we have that $||X||_F > ||A(X)||_2$.

We also note that $\delta_r(A) \le \delta_s(A)$ if $r \le s$. That's because the smallest number that satisfies the RIP property for matrices of rank less than or equal to s also satisfies the RIP property for matrices of rank less than or equal to r since $r \le s$, by assumption.

Theorem 126 (RIP Theorem 1)**.** Let $A : \mathbb{R}^{m\times n} \to \mathbb{R}^p$ be linear with $m \le n$. Suppose that $\delta_{2r}(A) < 1$ for some $r \ge 1$. Let X_0 have rank r and let $b = A(X_0)$. Then X_0 is the only matrix of rank $\le r$ with $A(X) = b$.

Proof.

Suppose not. Let X have rank $\le r$ with $A(X) = b$. Let $Z := X_0 - X \ne 0$. We know that

$$\begin{aligned} \mathrm{rank}(Z) &\le \mathrm{rank}(X_0) + \mathrm{rank}(X) \\ &= 2r \end{aligned}$$

Also, we have that $A(Z) = A(X_0) - A(X) = 0$. Thus, we have that by the RIP property

$$0 = ||A(Z)||_2 \ge (1 - \delta_{2r}(A))||Z||_F > 0$$

since $Z \ne 0$ so we've reached a contradiction. □

Lemma 127 (RIP Lemma)**.** Let $A, B \in \mathbb{R}^{m\times n}$. Then, $\exists B_1, B_2$ such that

(1) $B = B_1 + B_2$

(2) $\mathrm{rank}(B_1) \le 2\,\mathrm{rank}(A)$

(3) $AB_2' = 0, A'B_2 = 0$ so $||A + B_2||_* = ||A||_* + ||B_2||_*$ by Lemma 124

(4) $\mathrm{Tr}(B_1' B_2) = 0$ so $\langle B_1, B_2\rangle = 0$

Proof.

Consider a full singular value decomposition of $A = U \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V'$ where $U \in \mathbb{R}^{m \times m}, \Sigma \in \mathbb{R}^{r \times r}$ where r is the rank of A, and $V \in \mathbb{R}^{n \times n}$. Let $\hat{B} := U' B V$. We partition $\hat{B}$ as

$$\hat{B} := \begin{bmatrix} \hat{B}_{11} & \hat{B}_{12} \\ \hat{B}_{21} & \hat{B}_{22} \end{bmatrix}$$

where $\hat{B}_{11} \in \mathbb{R}^{r \times r}, \hat{B}_{12} \in \mathbb{R}^{r \times (n-r)}, \hat{B}_{21} \in \mathbb{R}^{(m-r) \times r}$, and $\hat{B}_{22} \in \mathbb{R}^{(m-r) \times (n-r)}$. We also define

$$B_1 := U \begin{bmatrix} \hat{B}_{11} & \hat{B}_{12} \\ \hat{B}_{21} & 0 \end{bmatrix} V', B_2 := U \begin{bmatrix} 0 & 0 \\ 0 & \hat{B}_{22} \end{bmatrix} V'$$

Property (1) follows in a very straightforward manner. For property (2), note that

$$\begin{aligned}
\operatorname{rank}(B_1) &= \operatorname{rank}\left(\begin{bmatrix} \hat{B}_{11} & \hat{B}_{12} \\ \hat{B}_{21} & 0 \end{bmatrix}\right), \text{ since } U, V \text{ are orthonormal} \\
&\leq \underbrace{\operatorname{rank}\left(\begin{bmatrix} \hat{B}_{11} & 0 \\ \hat{B}_{21} & 0 \end{bmatrix}\right)}_{\leq r} + \underbrace{\operatorname{rank}\left(\begin{bmatrix} 0 & \hat{B}_{12} \\ 0 & 0 \end{bmatrix}\right)}_{\leq r}, \text{ by Lemma 121} \\
&\leq 2r \\
&= 2\operatorname{rank}(A)
\end{aligned}$$

The remaining two properties are easy to check by direct verification. □

Theorem 128 (RIP Theorem 2)**.** Suppose that $r \geq 1$ is such that $\delta_{5r}(A) < \frac{1}{10}$ for $A \in \mathbb{R}^{m \times n}$. Let X_0 have rank r and let $b = A(X_0)$. By Theorem 126, X_0 is the only solution to $A(X) = b$ with $\operatorname{rank}(X) \leq r$. Also define $X_* := \arg\min_{X \in \mathbb{R}^{m \times n}} \{||X||_* : A(X) = b\}$. Then, we have that $X_* = X_0$.

Proof.

Omitted; uses Lemma 127.

8 Primal-Dual Interior Point Methods

In this section, we will discuss interior point methods, which are the same as barrier point methods.

8.1 Newton's Method for Solving Nonlinear Equations

Before we discuss primal-dual interior point methods, we will discuss Newton's method for solving nonlinear systems of equations. Suppose that we're trying to solve the problem

$$f(x) = \mathbf{0}$$

for $f : \mathbb{R}^n \to \mathbb{R}^n$. The motivation for Newton's method is that we linearize the equation at a current value x_0 to get a "better" guess of the solution.

$$\begin{aligned}
0 &\approx f(x_0) + (Df)(x - x_0) \\
\implies x &= x_0 - (Df)^{-1} f(x_0)
\end{aligned}$$

We then iterate on this scheme until convergence is reached. The procedure is presented in Algorithm 7.

Algorithm 7 Newton's Method for Nonlinear System

Input: : $f, x, \epsilon > 0$
Output: : x^*
Start Algorithm:
while True **do Quit if**: $||f(x)|| \leq \epsilon$
$x \leftarrow x - (Df)^{-1} f(x)$
end while
return x

8.2 Linear Programs

We begin with the primal problem

$$\begin{aligned} &\inf_{x \in \mathbb{R}^m} c'x \\ &\text{s.t. } Ax = b, x \geq 0 \end{aligned} \tag{25}$$

with $A \in \mathbb{R}^{n \times m}$ with $m \geq n$. We define the *barrier function* $B_\mu(x) = c'x - \mu \sum_{i=1}^m \log(x_i)$, for $\mu > 0$, which goes to ∞ as $x \to \mathrm{bd}(\mathbb{R}^m_{++})$.

We consider a related optimization problem

$$\begin{aligned} &\inf_{x \in \mathbb{R}^m} B_\mu(x) \\ &\text{s.t. } Ax = b \end{aligned} \tag{26}$$

so that we approximately bake the inequality constraint (ie., $x \geq 0$) into the barrier function's domain. We see that this is a convex optimization problem so that the optimum is characterized by the following KKT conditions:

$$\begin{aligned} &A'\nu = \nabla B_\mu(x) = c - \mu \,\mathrm{diag}(x)^{-1} 1_m \\ &Ax = b \end{aligned}$$

where $\nu \in \mathbb{R}^n$ is the Lagrangian multiplier. With the aim of using Newton's method for solving nonlinear systems of equations, we can stack the optimality conditions:

$$\begin{aligned} \mathbf{0} &= F_\mu(x, \nu) \\ &= \begin{bmatrix} A'\nu - \nabla B_\mu(x) \\ Ax - b \end{bmatrix} \\ &= \begin{bmatrix} A'\nu + \mu \,\mathrm{diag}(x)^{-1} 1_m - c \\ Ax - b \end{bmatrix} \end{aligned} \tag{27}$$

so that optimality requires solving $F_\mu(x, \nu) = \mathbf{0}$. For Newton's method for nonlinear equations, we must write down the Jacobian

$$DF_\mu(x, \nu) = \begin{bmatrix} -\mu \,\mathrm{diag}(x)^{-2} & A' \\ A & 0 \end{bmatrix}$$

The Newton's method update is

$$\begin{aligned} \begin{bmatrix} x^+ \\ \nu^+ \end{bmatrix} &= \begin{bmatrix} x \\ \nu \end{bmatrix} - (DF_\mu)^{-1} F_\mu(x) \\ &= \begin{bmatrix} x \\ \nu \end{bmatrix} - \begin{bmatrix} -\mu \,\mathrm{diag}(x)^{-2} & A' \\ A & 0 \end{bmatrix}^{-1} \begin{bmatrix} A'\nu + \mu \,\mathrm{diag}(x)^{-1} 1_m - c \\ Ax - b \end{bmatrix} \end{aligned}$$

It's generally a bad method to compute the inverse $(DF_\mu)^{-1}$ so that we make progress in doing the update using block Gaussian elimination on

$$-\begin{bmatrix} -\mu \,\mathrm{diag}(x)^{-2} & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} x^+ - x \\ \nu^+ - \nu \end{bmatrix} = \begin{bmatrix} A'\nu + \mu \,\mathrm{diag}(x)^{-1} 1_m - c \\ Ax - b \end{bmatrix}$$

If we add $\frac{1}{\mu} A \operatorname{diag}(x)^2$ times the first block to the second block, we get the following equivalent system

$$-\begin{bmatrix} -\mu \operatorname{diag}(x)^{-2} & A' \\ 0 & \mu^{-1} A \operatorname{diag}(x)^2 A' \end{bmatrix} \begin{bmatrix} x^+ - x \\ \nu^+ - \nu \end{bmatrix} = \begin{bmatrix} A'\nu + \mu \operatorname{diag}(x)^{-1} 1_m - c \\ Ax - b + \mu^{-1} A \operatorname{diag}(x)^2 A'\nu + A \operatorname{diag}(x) 1_m - \mu^{-1} A \operatorname{diag}(x)^2 c \end{bmatrix}$$

We can then solve the second equation using Cholesky since $A \operatorname{diag}(x)^2 A' \succ 0$ and then use the solution for $(\nu^+ - \nu)$ to compute $(x^+ - x)$ in the first equation. When $n \ll m$, this method is particularly good. If $x^+ \not> 0$, then we must contract $x^+ - x$ to make it smaller to keep x^+ positive.

8.2.1 *Incorporating the Dual Problem*

Above in the problem of Equation (26), we solve the problem for fixed $\mu > 0$, which makes for a different problem than the original problem of Equation (25). To see how we can use the barrier problem to get a solution to the original problem, consider the dual problem:

$$\begin{aligned} &\sup_{\nu \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad b'\nu \\ &\text{s.t. } A'\nu + z = c, z \geq 0 \end{aligned}$$

We can compare the constraint to the barrier method first order condition:

$$\begin{aligned} &c - \mu \operatorname{diag}(x)^{-1} 1_m = A'\nu, \text{ v.s. } A'\nu + z = c \\ \implies &z = \mu \operatorname{diag}(x)^{-1} 1_m \\ \implies &x_i z_i = \mu \ \forall i \in [m] \end{aligned}$$

which is exactly the complementary slackness optimality condition in Proposition 64 when $\mu \downarrow 0$. With the introduction of this new variable z, we can rewrite the optimality conditions of Equation (27) along with this complementary slackness equation,

$$\begin{aligned} \mathbf{0} &= \tilde{F}_\mu(x, \nu, z) \\ &= \begin{bmatrix} A'\nu + z - c \\ Ax - b \\ \operatorname{diag}(x) \operatorname{diag}(z) 1_m - \mu 1_m \end{bmatrix} \end{aligned}$$

This is the *primal-dual formulation* of the nonlinear equations that govern optimality in the barrier problem. We also define *central path equations* as

$$\{(x, y, z) : \tilde{F}_\mu(x, y, z) = 0 \text{ for some } \mu > 0\}$$

To do Newton's method to solve this nonlinear system of equations, we must compute the Jacobian

$$D\tilde{F}_\mu(x, \nu, z) = \begin{bmatrix} 0 & A' & I_m \\ A & 0 & 0 \\ \operatorname{diag}(z) & 0 & \operatorname{diag}(x) \end{bmatrix}$$

which gives us the following update step

$$\begin{aligned} \begin{bmatrix} x^+ \\ \nu^+ \\ z^+ \end{bmatrix} &= \begin{bmatrix} x \\ \nu \\ z \end{bmatrix} - (D\tilde{F}_\mu)^{-1} \tilde{F}_\mu(x, \nu, z) \\ &= \begin{bmatrix} x \\ \nu \\ z \end{bmatrix} - \begin{bmatrix} 0 & A' & I_m \\ A & 0 & 0 \\ \operatorname{diag}(z) & 0 & \operatorname{diag}(x) \end{bmatrix}^{-1} \begin{bmatrix} A'\nu + z - c \\ Ax - b \\ \operatorname{diag}(x) \operatorname{diag}(z) 1_m - \mu 1_m \end{bmatrix} \end{aligned}$$

which can again be solved more efficiently without computing the full inverse by using block matrix elimination and then Cholesky.

8.3 Semidefinite Programs

We begin with the primal problem

$$\begin{aligned}\inf_{X \in S_+^m} & \langle C, X\rangle \\ \text{s.t. } & A(X) = b, X \succeq 0\end{aligned}$$

with $A \in \mathbb{S}_+^m$ and $b \in \mathbb{R}^p$. We form the *barrier function* $B_\mu(X) = \langle C, X\rangle - \mu \log(\det(X))$ for $\mu > 0$, which goes to ∞ as $X \to \text{bd}(\mathbb{S}_+^m)$.

We consider a related optimization problem

$$\begin{aligned}\inf_{X \in S_+^m} & B_\mu(X) \\ \text{s.t. } & A(X) = b\end{aligned}$$

so that we approximately bake the positive semidefinite constraint (ie., $X \succeq 0$) into the barrier function's domain. We see that this a semidefinite program so that the optimum if characterized by the following KKT conditions:

$$\begin{aligned}0 &= C - \nabla B_\mu(X) - A^*(y) \\ &= C - \mu X^{-1} - A^*(y) \\ b &= A(X)\end{aligned}$$

where $y \in \mathbb{R}^p$ is the Lagrangian multiplier on the equality constraint and A^* is the adjoint operator of A (ie., $A^*(z) = \sum_{i=1}^p z_i A_i$). The dual problem is

$$\begin{aligned}\sup_{y \in \mathbb{R}^p, Z \succeq 0} & \quad b'y \\ \text{s.t. } & C - A^*(y) = Z\end{aligned}$$

We can compare the constraint to the barrier method first order condition:

$$\begin{aligned}0 = C - \mu X^{-1} - A^*(y) &\text{ v.s. } C - A^*(y) = \Lambda \\ \implies Z &= \mu X^{-1} \\ \implies XZ &= \mu I_m\end{aligned}$$

which is exactly the complementary slackness optimality condition in Section 3.4.1. We can write the optimality conditions of the primal problem along with this complementary slackness condition as

$$\begin{aligned}0 &= \hat{F}_\mu(X, Z, y) \\ &:= \begin{bmatrix} A^*(y) + Z - C \\ A(X) - b \\ XZ - \mu I_m \end{bmatrix}\end{aligned}$$

Definition 129 (Column Major Order Vectorization)**.** Given a matrix $A \in \mathbb{R}^{m\times n}$ with entry A_{ij} in the ith row and jth column, we define and denote the *column major order vectorization* of A as

$$\text{vec}_C(A) := [A_{11}, A_{21}, ..., A_{m1}, A_{12}, ..., A_{1n}, ..., A_{mn}]'$$

Definition 130 (Kronecker Product)**.** Given a matrix $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, we define the (ordered)*kronecker product* of A with B as

$$A \otimes B := \begin{bmatrix} A_{11}B & ... & A_{1n}B \\ ... & ... & ... \\ A_{m1}B & ... & A_{mn}B \end{bmatrix} \in \mathbb{R}^{mp\times nq}$$

Lemma 131 (Kronecker Product and Vectorization)**.** Given conformal matrices A, B, C, we have that

$$\text{vec}_C(ABC) = (C' \otimes A)\, \text{vec}_C(B)$$

Proof.

Omitted.

The system of equations $0 = \hat{F}_\mu(X, Z, y)$ is difficult to handle in its current form so that we will end up vectorizing it. Another challenge is that in the optimal solution, X, Z must be positive semidefinite matrices but that restriction will be difficult to work with in Newton's method so that we will relax it when vectorizing those matrices. In other words, for the sake of Newton's method, we will treat the vectorizations as unconstrained vectors in $\mathbb{R}^{m^2}$. Ultimately, the goal is to find step directions

$$\begin{bmatrix} \Delta X \\ \Delta Z \\ \Delta y \end{bmatrix} := \begin{bmatrix} X^+ - X \\ Z^+ - Z \\ y^+ - y \end{bmatrix}$$

For the third block equation, we linearize it as

$$\begin{aligned} \mu I_m &= (X + \Delta X)(Z + \Delta Z) \\ &= XZ + X\Delta Z + \Delta X Z + O(||\Delta X||||\Delta Z||) \\ \implies \text{vec}_C(\mu I_m - XZ) &\sim \text{vec}_C(X\Delta Z) + \text{vec}_C(\Delta X Z) \\ &= (I_m \otimes X)\,\text{vec}_C(\Delta Z) + (Z' \otimes I_m)\,\text{vec}_C(\Delta X) \end{aligned}$$

The Newton step update is given by solving the following linear system

$$\begin{aligned} -D_l\, \text{vec}_C(\hat{F}_\mu(X, Z, y)) \begin{bmatrix} \text{vec}_C(\Delta X) \\ \Delta y \\ \text{vec}_C(\Delta Z) \end{bmatrix} &= \text{vec}_C(\hat{F}_\mu(X, Z, y)) \\ \iff -\begin{bmatrix} 0 & \mathbb{A}' & I \otimes I \\ \mathbb{A} & 0 & 0 \\ Z' \otimes I_m & 0 & I_m \otimes X \end{bmatrix} \begin{bmatrix} \text{vec}_C(\Delta X) \\ \Delta y \\ \text{vec}_C(\Delta Z) \end{bmatrix} &= \begin{bmatrix} \text{vec}_C(A^*(y) + Z - C) \\ \text{vec}_C(A(X) - b) \\ \text{vec}_C(XZ - \mu I_m) \end{bmatrix} \end{aligned}$$

where $\mathbb{A} := \begin{bmatrix} \text{vec}_C(A_1)' \\ \dots \\ \text{vec}_C(A_m)' \end{bmatrix}$ and D_l denotes that we're computing the linearized Jacobian for the matrix block third equation. We again should use block Gaussian elimination to simplify the system before solving though the details of the simplification get very hairy and are differed to another source.

As a remark, a problem with this update is that ΔX and ΔZ are not in general symmetric so that X^+ and Z^+ are not necessarily symmetric. What should one do? There are three main approaches.

- The XZ method. They suggest replacing ΔX by its symmetric part $\frac{1}{2}(\Delta X + \Delta X')$ when incrementing for X and analogously for Z.

- The $XZ + ZX$ method. Replace the matrix equation $XZ = \mu I$ by $\frac{1}{2}(XZ + ZX) = \mu I_m$. Then, Newton's method works in S^m and we can contract the step if need be to keep the matrices in S^m_+.

- The Nesterov-Todd method, which is discussed here.

9 Numerical Methods for Nonconvex and Smooth Optimization

There are two general classes of methods for nonconvex and smooth optimization problems (i) line search methods and (ii) trust region methods. We will only discuss the former. In this section, we consider the problem

$$\inf_{x \in \mathcal{D}} f(x) \tag{28}$$

for $x \in \mathbb{R}^m$ with $f \in C^2$.

9.1 Newton's Method Breaks Down

The first question to ask in nonconvex optimization is what to do with Newton's method? Suppose that $\nabla^2 f(x)$ is not positive definite, meaning that the Cholesky decomposition of the Hessian matrix will break down. Then,

$$\Delta x = -(\nabla^2 f(x))^{-1} \nabla f(x)$$

may not be a descent direction so that Algorithm 3 will not work. Thus, generally, in these problems we must use gradient descent, some variant of Newton's method where we take a convex combination of the newton direction and the gradient descent direction, or something else.

9.2 Introducing the Armijo and Wolfe Conditions

In this section, we discuss the *Armijo and Wolfe conditions* and then present a line search procedure to satisfy both.

Definition 132 (Armijo and Wolfe Conditions)**.** Consider a line search $x^+ = x + t\Delta x$ along a descent direction Δx (ie., $\nabla f(x)'\Delta x < 0$). For $0 < c_1 < c_2 < 1$, we define the following two conditions

- The *Armijo condition* is that $f(x + td) \leq f(x) + c_1 t \nabla f(x)' \Delta x$. See Equation (12) for a previous occurance of this term. This condition ensures that t is not too large (assuming the function turns around and goes up again).

- The (weak) *Wolfe condition* is that $\nabla f(x + t\Delta x)'\Delta x \geq c_2 \nabla f(x)'\Delta x$. This condition ensures that t is not too small. If t where 0, then the condition wouldn't be satisfied.[a]

[a] The strong *Wolfe Condition* is $|\nabla f(x + t\Delta x)'\Delta x| \leq c_2 |\nabla f(x)'\Delta x|$. Line search will not work here when optimizing $f(x) = |x|$. That's because $|\nabla f(x + t\Delta x)'\Delta x| > c_2 |\nabla f(x)'\Delta x|$ for any $c_2 \in (0, 1)$.

Algorithm 8 Bracketing Armijo-Wolfe Line Search

Input: : Δx, f, x, $0 < c_1 < c_2 < 1$
Output: : t
Start Algorithm:
done $\leftarrow$ False
$\alpha \leftarrow 0$
$\beta \leftarrow \infty$
$t \leftarrow 1$
while not done **do**
  $x^+ \leftarrow x + t\Delta x$
  if $f(x^+) > f(x) + c_1 t \nabla f(x)' \Delta x$ **then**
    $\beta \leftarrow t$ # the Armijo condition is violated
  else if $\nabla f(x^+)' \Delta x < c_2 \nabla f(x)' \Delta x$ **then**
    return $\alpha \leftarrow t$ # the Wolfe condition is violated
  else
    $\alpha \leftarrow t$ # both conditions are satisifed
    $\beta \leftarrow t$
  end if
  if $\beta < \infty$ **then**
    $t \leftarrow \frac{\alpha+\beta}{2}$
  else
    $t \leftarrow 2t$

return t

Theorem 133 (Bracketing Armijo-Wolfe Line Search Termination Sufficient Condition)**.** Suppose that $f \in C^1(\mathbb{R})$ and is bounded below on $\{x + t\Delta x : t > 0\}$. Then, we have that Algorithm 8 terminates in a valid Armijo-Wolfe step.[a]

Proof.

Omitted.

[a]In fact, one can show a more general result, which is if f is bounded below in the set and locally Lipschitz, then there exists an interval of points with positive Lebesgue measure that satisfies the Armijo and Wolfe conditions.

9.3 Zoutendijk's Theorem

In this section, we present Zoutendijk's Theorem, which conceptually gives us sufficient conditions for when a nonconvex smooth optimization problem will reach a local optimum.

Definition 134 (Globally Lipschitz Function)**.** Consider a function $g : \mathbb{R}^m \to \mathbb{R}^n$. We say that g is *globally Lipschitz* with Lipschitz constant L if

$$||g(x) - g(y)|| \leq L||x - y|| \; \forall x, y \in \mathbb{R}^m$$

Theorem 135 (Zoutendijk's Theorem)**.** Consider an optimization problem like that of Equation (28). Assume further that f is bounded below, $f \in C^1$, and ∇f is globally Lipschitz with Lipschitz constant L. Define any descent algorithm

$$x^{(k+1)} \leftarrow x^{(k)} + t_k \Delta x^{(k)}$$

where $\{t_k\}_{k\in\mathbb{N}}$ satisfy the Armijo and Wolfe conditions and $\{\Delta x^{(k)}\}_{k\in\mathbb{N}}$ are descent directions. We define $\cos(\theta^{(k)}) =$

$\frac{-\nabla f(x^{(k)})'\Delta x}{||\nabla f(x^{(k)})||||\Delta x^{(k)}||}$. Given that it's a descent direction, we must have that $\theta^{(k)} < \pi/2$ for each $k \in \mathbb{N}$.

The claim is that under these assumption, we have that

$$\sum_{k=0}^{\infty}(\cos(\theta^{(k)}))^2||\nabla f(x^{(k)})||^2 < \infty$$

Trivially, if in addition $\cos(\theta^{(k)}) \geq \tau > 0$ for some $\tau > 0$, then $\nabla f(x^{(k)}) \to 0$. Notice this result also implies that if $\Delta x^{(k)} = -\nabla f(x^{(k)})$ as in gradient descent, we do have that $\cos(\theta^{(k)}) = 1\ \forall k \in \mathbb{N}$, so that we have the result.

Proof.

To simplify notation, let $g_k := \nabla f(x^{(k)})$, $f_k := f(x^{(k)})$, and $d_k := \Delta x^{(k)}$ for $k \in \mathbb{N}$. From the *Wolfe* condition, we have that

$$\begin{aligned}
c_2 g_k' d_k &\leq g_{k+1}' d_k \\
\implies \underbrace{(c_2-1)}_{<0}\underbrace{g_k' d_k}_{<0} &\leq (g_{k+1}-g_k)' d_k \\
&\leq ||g_{k+1}-g_k||||d_k||, \text{ by Cauchy-Schwarz} \\
&\leq L||t_k d_k||||d_k||, \text{ by Lipschitz condition} \\
\implies \frac{c_2-1}{L}\cdot\frac{g_k' d_k}{||d_k||^2} &\leq t_k
\end{aligned}$$

Substituting this into the *Armijo* condition, we have that

$$\begin{aligned}
& f_{k+1} \leq f_k + c_1 t_k g_k' d_k \\
\implies & f_{k+1} \leq f_k + c_1\frac{c_2-1}{L}\frac{(g_k' d_k)^2}{||d_k||^2} \\
\implies & f_{k+1} \leq f_k - \frac{c_1(1-c_2)}{L}(\cos(\theta^{(k)}))^2||g_k||^2
\end{aligned}$$

Next, we will recursively apply last inequality over $0 \leq j \leq k$, to get that

$$f_{k+1} \leq f_0 - \frac{c_1(1-c_2)}{L}\sum_{j=0}^{k}(\cos(\theta^{(j)}))^2||g_j||^2$$

Letting $k \to \infty$ and using the fact that f is bounded from below, we get that $\sum_{j=0}^{k}(\cos(\theta^{(j)}))^2||g_j||^2 < \infty$. □

9.4 Quasi-Newton Methods

We continue to ponder problems like that of Equation (28) though this class of methods also works when f is convex and its domain is $\mathbb{R}^m$ (or some subset of it).

Even for convex problems, each step of Newton's method takes $O(m^3)$ work because we must invert a hessian matrix $\nabla^2 f(x)$. In this section, we will consider techniques to approximate a factorization or inverse of $\nabla^2 f(x)$ in $O(m^2)$ time or even $O(m)$ time.

To motivate quasi-Newton methods, we consider that we have current iterate $x^{(k)} \in \mathbb{R}^m$ and we've just concluded an Armijo-Wolfe line-search as in Algorithm 8 with a descent direction d_k and returned step size t_k. As a result, we have that

$$x^{(k+1)} = x^{(k)} + t_k d_k$$

and we define

$$\begin{aligned} g_k &:= \nabla f(x^{(k)}) \\ g_{k+1} &:= \nabla f(x^{(k+1)}) \\ s_k &:= x^{(k+1)} - x^{(k)} \\ y_k &:= g_{k+1} - g_k \end{aligned}$$

From the fundamental theorem of calculus, we have that

$$\begin{aligned} y_k &= \int_0^1 \nabla^2 f(x_k + \tau s_k) s_k d\tau \\ &= \left[\int_0^1 \nabla^2 f(x_k + \tau s_k) d\tau \right] s_k \end{aligned}$$

This motivates choosing our new approximation of $\nabla^2 f(x^{(k+1)})$, which would be used in Newton's method, say B_{k+1}, to satisfy $B_{k+1}s_k = y_k$. Or, if we're approximating $\left(\nabla^2 f(x_{k+1})\right)^{-1}$, by say H_{k+1}, we'd pick it to satisfy $H_{k+1}y_k = s_k$.

9.4.1 BFGS

In BFGS, the algorithm seeks to approximate $\left(\nabla^2 f(x^{(k+1)})\right)^{-1}$ with H_{k+1}. The algorithm puts:

$$\begin{aligned} H_{k+1} &= V_k' H_k V_k + \gamma_k s_k s_k' \\ d_{k+1} &= -H_{k+1} g_{k+1} \end{aligned}$$

with $\gamma_k := \frac{1}{s_k' y_k}$ and $V_k := I_m - \gamma_k y_k s_k'$. To initialize, the algorithm puts $H_0 = I_m$. There are three properties of this update that are easy to check by direct evaluation and one that we check:

(1) If $H_k \succeq 0$, then we have $H_{k+1} \succeq 0$.

(2) If $H_k \succ 0$, then we have $H_{k+1} \succ 0$.

(3) We have that $H_{k+1} y_k = s_k$.

(4) We have that $s_k' y_k > 0$, which is important for justifying property (2), which is important to keep the algorithm along a descent direction. To see this claim, recall by the conclusion of the Armijo-Wolfe line-search we have that $g_{k+1}' d_k \geq c_2 g_k' d_k$. As a result, we have that

$$\begin{aligned} s_k' y_k &= t_k d_k' y_k \\ &= t_k \left(d_k' g_{k+1} - d_k' g_k \right) \\ &\geq t_k (c_2 - 1) g_k' d_k \\ &> 0, \text{ since } g_k' d_k < 0 \text{ and } c_2 - 1 < 0 \end{aligned}$$

Theorem 136 (Powell (1976))**.** Take an optimization problem like that of Equation (28). Suppose that $f \in C^2$ and $\Omega := \{x : f(x) \leq f(x_0)\}$ is convex for any x_0. Also, assume that f is strongly convex as says Definition 70. Then, we have that $\{x^k\}_{k \in \mathbb{N}}$ generated by BFGS with the Armijo-Wolfe line-search satisfies

$$x^{(k)} \to x^*$$

where x^* is the unique local minimizer of f.

Proof.

Omitted but uses Zoutendijk's Theorem.

Theorem 137 (Superlinear Convergence of BFGS)**.** Take an optimization problem like that of Equation (28). Suppose that $f \in C^2$ and $\Omega := \{x : f(x) \leq f(x_0)\}$ is convex for any x_0. Also, assume that f is strongly convex as says Definition 70 and that $\nabla^2 f(x)$ is Lipschitz with factor L on Ω. Then, we have that $\{x^k\}_{k\in\mathbb{N}}$ generated by BFGS with the Armijo-Wolfe line-search satisfies

$$\lim_{k\to\infty} \frac{||x^{(k+1)} - x^*||}{||x^{(k)} - x^*||} = 0$$

where x^* is the unique local minimizer of f.

Proof.

Omitted.

9.4.2 Limited Memory BFGS (L-BFGS)

The issue with BFGS is that storing and updating H_k still takes $O(m^2)$ time.

In the L-BFGS method, we take $n \ll m$. At iteration $k+1$, we have $x^{(k+1)}$, $\nabla f(x^{(k+1)})$ and $\{s_i, y_i : i \in \{k-n+1, ..., k\}\}$ saved from n previous iterates. We choose an initial $H^0_{k+1} := \begin{cases} \frac{s'_{k-1}y_{k-1}}{s'_k y_k} I_m, & \text{if computable} \\ I_m, & \text{otherwise} \end{cases}$. Then, we set

$$\begin{aligned} H_{k+1} = {} & V'_k \cdot ... \cdot V'_{k-m+1} H^0_{k+1} V_{k-m+1} \cdot ... \cdot V_k \\ & + \gamma_k \left(V'_k \cdot ... \cdot V'_{k-m+2}\right) s_{k-m+1} s'_{k-m+1} \left(V_{k-m+2} \cdot ... \cdot V_k\right) \\ & ... \\ & + \gamma_k s_k s'_k \end{aligned}$$

where V_i is defined in Section 9.4.1. That is the same update as full BFGS would have done after $k+1$ iterations except that we're applying it to H^0_k instead of H_{k-m}, which we don't have access to due to limited memory. We never actually compute H_{k+1}, instead we do that series of matrix operations cleverly so that we're only doing rank-one matrix vector multiplications. We then put $d_{k+1} := -H_{k+1}\nabla f(x^{(k+1)})$, again, never fully forming H_{k+1}, so this is all done in $O(m)$ time.

10 Nonsmooth and Nonconvex Optimization

Here we will discuss the topic of nonconvex and nonsmooth optimization.

10.1 Subgradients and Subdifferentials for Nonconvex and Nonsmooth Optimization

We will start by introducing various subgradients and subdifferentials.

Definition 138 (Regular Subgradient and Regular Subdifferentials)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ be continuous. We say that $y \in \mathbb{R}^m$ is a *regular subgradient* of f at x if $\forall (z_r)_{r\in\mathbb{N}}$ with $z_r \to \mathbf{0}$, $z_r \neq \mathbf{0}$ $\forall r \in \mathbb{N}$, we have

$$\lim\inf_{z_r\to\mathbf{0}} \frac{f(x+z_r) - f(x) - y'z_r}{||z_r||} \geq 0$$

We often write this as casually as

$$f(x+z) \geq f(x) + y'z + o(||z||)$$

If y is a regular subgradient of f as x, we write $y \in \hat{\partial} f(x)$ where $\hat{\partial} f(x)$ is the *regular subdifferential* of f at x.

Compare Definition 138 to Definition 90. For a convex function, the subgradient lies below the function whereas for a general function it lies below the function to first order.

Definition 139 (General Subgradient and General Subdifferential)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ be continuous. We say that $y \in \mathbb{R}^m$ is a *general subgradient* of f at x if $\exists (x_r)_{r\in\mathbb{N}}, (y_r)_{r\in\mathbb{N}}$ with $x_r \to x$, $y_r \to y$ and $y_r \in \hat{\partial} f(x_r)\ \forall r \in \mathbb{N}$. In that case, we write that $y \in \partial f(x)$ where $\partial f(x)$ is the *general subdifferential* of f at x.

It's plain to see that $\hat{\partial} f(x) \subseteq \partial f(x)$.

It is easy to see that the regular subdifferential is closed, convex, but not necessarily compact nor nonempty. The general subgradient is not necessarily convex.

Definition 140 (Horizon Subgradient and Horizon Subdifferential)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ be continuous. We say that $y \in \mathbb{R}^m$ is a *horizon subgradient* of f at x if $\exists (x_r)_{r\in\mathbb{N}}, (y_r)_{r\in\mathbb{N}}, (t_r)_{r\in\mathbb{N}} \in \mathbb{R}_+$ with

- $x_r \to x$
- $t_r y_r \to y$
- $t_r \downarrow 0$
- $y_r \in \hat{\partial} f(x_r)\ \forall r \in \mathbb{N}$

We write that $y \in \partial^\infty f(x)$ where $\partial^\infty f(x)$ is the *horizon subdifferential* of f at x.

If $\hat{\partial} f(z)$ is bounded for $z \in B(x, \epsilon)$ for some $\epsilon > 0$, we then get that $\partial^\infty f(x) = \{\mathbf{0}\}$ (since the only possible limit of $t_r y_r$ is 0 since $t_r \downarrow 0$ and y_r is bounded for sufficiently large $r \in \mathbb{N}$).

Proposition 141 (Differentiability and Regular Differential)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ and suppose that f is differentiable at $x \in \mathbb{R}^m$. Then, we have that

$$\hat{\partial} f(x) = \{\nabla f(x)\}$$

Moreover, if f is continuously differentiable at x, then we have that

$$\partial f(x) = \{\nabla f(x)\}$$

Proof.

Suppose that f is differentiable at x. Then, by a first order Taylor expansion, we have that for any $z \in \mathbb{R}^m$

$$\begin{aligned} & f(x+z) = f(x) + \nabla f(x)'z + o(||z||) \\ \implies & f(x+z) \geq f(x) + \nabla f(x)'z + o(||z||) \\ \implies & \nabla f(x)' \in \hat{\partial} f(x), \text{ by definition} \end{aligned}$$

Next, suppose that $v \in \hat{\partial} f(x)$ meaning that

$$\begin{aligned} & f(x+z) \geq f(x) + v'z + o(||z||) \\ \implies & f(x) + (\nabla f(x))'z + o(||z||) \geq f(x) + v'z + o(||z||), \text{ by a first order Taylor expansion} \\ \implies & (\nabla f(x) - v)'z \geq o(||z||) \end{aligned}$$

Letting $z = tw$ for $t > 0$ and $||w|| = 1$, we can divide both sides of the equation by t and take the limit as $t \downarrow 0$ to see that

$$(\nabla f(x) - v)'w \geq 0$$

Since we can repeat the exercise with $z = t(-w)$, to conclude that $(\nabla f(x) - v)'(-w) \geq 0$, we conclude that $v = \nabla f(x)$.

Next, suppose that f is continuously differentiable at x meaning that $y \mapsto \nabla f(y)$ is continuous in a neighborhood of x. Applying the first part of this proposition, we have that $\forall y$ sufficiently close to x, $\hat{\partial} f(y) = \{\nabla f(y)\}$. That means that if $x_r \to x$, we have that $\nabla f(x_r) \to \nabla f(x)$ meaning that $\partial f(x) = \{\nabla f(x)\}$ by definition. ☐

Example 142 (Square-Root of Absolute Value)**.** Consider $f(x) = |x|^{1/2}$. For $x \neq 0$, we have $y \in \hat{\partial} f(x) \implies y = f'(x)$.[a] That is with

$$f'(x) = \begin{cases} \frac{1}{2}x^{-1/2}, & \text{if } x > 0 \\ -\frac{1}{2}|x|^{-1/2}, & \text{if } x < 0 \end{cases}$$

Now we consider the case of $x = 0$. We have that

$$\lim \inf_{z_r \to 0} \frac{|z_r|^{1/2} - 0 - yz_r}{|z_r|} = \infty \pm y \geq 0 \; \forall y \in \mathbb{R}$$

That implies that $\hat{\partial} f(0) = \mathbb{R}$, which implies that $\partial f(0) = \mathbb{R}$.

Now we compute the horizon subgradient at $x = 0$. For any $y \in \mathbb{R}$, take $t_r = \frac{1}{r}$ for $r \in \mathbb{N}$ so that $t_r \to 0$. Define $(y_r)_{r \in \mathbb{N}}$ by $y_r = \frac{y}{t_r}$ for $r \in \mathbb{N}$. We note that trivially $t_r y_r \to y$ and $y_r \in \hat{\partial} f(x_r)$ $\forall r \in \mathbb{N}$. Thus, we have that $\partial^\infty f(x) = \mathbb{R}$.

[a] See Proposition 141

Example 143 (Negative Square-Root of Absolute Value)**.** Consider $f(x) = -|x|^{1/2}$. For $x \neq 0$, we have $y \in \hat{\partial} f(x) \implies y = f'(x)$. That is with

$$f'(x) = \begin{cases} -\frac{1}{2}x^{-1/2}, & \text{if } x > 0 \\ \frac{1}{2}|x|^{-1/2}, & \text{if } x < 0 \end{cases}$$

Now we consider the case of $x = 0$. We see that for any $z_r \to 0$,

$$\lim \inf_{z_r \to 0} \frac{-|z_r|^{1/2} - 0 - yz_r}{|z_r|} = -\infty \pm y < 0 \; \forall y \in \mathbb{R}$$

From that, we conclude that $\hat{\partial} f(x) = \emptyset$. For any $y \in \partial f(0)$, we need $x_r \to 0$ and $y_r = f'(x_r) \to y$, which fails for any $y \in \mathbb{R}$. From there, we conclude that $\partial f(0) = \emptyset$.

But, we claim that $\partial^\infty f(0) = \mathbb{R}$. To see this, for any $y \geq 0$, we set $x_r = -\frac{1}{r} \to 0$, we set $y_r = f'(x_r) = \frac{1}{2}\left(\frac{1}{r}\right)^{-1/2} \to \infty$. Lastly, we set $t_r = \frac{y}{y_r} \to 0$. Thus, we get that $y \in \partial^\infty f(0)$. For any $y \leq 0$, we set $x_r = \frac{1}{r}$ and do the same argument.

Proposition 144 (Convex Function and Relationship between Regular and General Subdifferentials)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$ be a convex function. Then, we have that $\tilde{\partial} f(x) = \hat{\partial} f(x) = \partial f(x)$ and that $\partial^\infty f(x) = \{\mathbf{0}\}$ for all $x \in \text{int}(\text{dom}(f))$.[a] If additionally $f \in C^1$, then we also have that $\partial f(x) = \{\nabla f(x)\}$.

Proof.

Take any $x \in \operatorname{int}(\operatorname{dom}(f))$. The fact that $\hat{\partial} f(x) \subseteq \partial f(x)$ is trivial. To see the opposite inclusion, suppose that $y \in \partial f(x)$. That means $\exists (x_r)_{r\in\mathbb{N}}, (y_r)_{r\in\mathbb{N}}$ with $x_r \to x$, $y_r \to y$ and $y_r \in \hat{\partial} f(x_r)\ \forall r \in \mathbb{N}$. That means that we have for each $r \in \mathbb{N}$

$$f(x_r + z) \geq f(x_r) + y_r' z + o(||z||)$$

Since f is convex, it's continuous (on the interior of its domain by corollary of Lemma 46, which we're assuming to be $\mathbb{R}^m$) so taking $r \to \infty$, we get that

$$f(x + z) \geq f(x) + y' z + o(||z||)$$

which means that $y \in \hat{\partial} f(x)$.

Next, we show that $\hat{\partial} f(x) = \tilde{\partial} f(x)$, which we define to be the subdifferential of f as given in Definition 91. Clearly, we have that $y \in \tilde{\partial} f(x) \implies y \in \hat{\partial} f(x)$, so that it suffices to show the opposite. To see the opposite, suppose that $y \notin \tilde{\partial} f(x)$. Then, $\exists w \in \mathbb{R}^m$ and $\epsilon > 0$ such that

$$f(x + w) = f(x) + w' y - \epsilon$$

Next, by convexity of f, for any $t \in [0, 1]$, we have that

$$\begin{aligned} f(x + tw) &= f((1-t)x + t(x + w)) \\ &\leq (1-t) f(x) + t f(x + w) \end{aligned}$$

That implies that

$$\begin{aligned} f(x + tw) &\leq (1-t) f(x) + t(f(x) + w' y - \epsilon) \\ &= f(x) + t w' y - t\epsilon \end{aligned}$$

That implies that

$$\begin{aligned} \frac{f(x + tw) - f(x) - w' y}{t||w||} &\leq \frac{-t\epsilon}{t||w||} \\ \implies \liminf_{t} \frac{f(x + tw) - f(x) - w' y}{t||w||} &\leq -\epsilon/||w|| \end{aligned}$$

which implies that $y \notin \hat{\partial} f(x)$.

Next, we see that $\partial^\infty f(x) = \{\mathbf{0}\}$. To that end, we know that any convex function is locally Lipschitz at x by Lemma 46. We let the local Lipschitz constant be L. With that in hand, we claim that $\hat{\partial} f(x)$ is bounded in a neighborhood of x, though we will focus on x arbitrarily. To see that, we observe that for $y \in \hat{\partial} f(x)$, we have that

$$\begin{aligned} & f(x + z) - f(x) \geq y' z + o(||z||) \\ \implies & L \geq y' z + o(||z||), \text{ for } z \text{ small enough} \\ \implies & L \geq ||y||^2, \text{ taking } z = t\frac{y}{||y||} \text{ and } t \downarrow 0 \end{aligned}$$

Then, we apply the intuition at the end of Definition 140 to conclude the argument.

To see the final claim, we apply Proposition 141. ☐

[a] We let $\tilde{\partial} f(x)$ be the subdifferential of f at x as defined in Definition 91. We then have that $\hat{\partial} f(x)$ and $\partial f(x)$ inherit the results from Lemma 93.

Example 145 (Regular Subdifferential Example)**.** Let $f_k : \mathbb{R}^m \to \mathbb{R}$ be given by $f_k(x) = x_{[k]}$ for $k \in [m]$. That is, f takes the kth largest element of x. This function is convex if and only if $k = 1$. We claim that

$$\hat{\partial} f_k(x) = \begin{cases} \text{conv}(\{e_i : x_i = f_k(x)\}), & \text{if } k = 1 \text{ or } (k > 1 \text{ and } f_{k-1}(x) > f_k(x)) \\ \emptyset, & \text{otherwise} \end{cases}$$

We additionally claim that

$$\partial f_k(x) = \{y \in \mathbb{R}^m : y \in \text{conv}\{(e_i : x_i = f_k(x))\} \text{ and } |\{i \in [m] : y_i > 0\}| \leq |\{i \in [m] : x_i \geq f_k(x)\}|\}$$

Proof.

We just prove the first claim.

For the case of $k = 1$, see Example 94 and apply Proposition 144. To address the remainder, let $I = \{i \in [m] : x_i = f_k(x)\}$.

Suppose that $k > 1$ and $f_{k-1}(x) > f_k(x)$, then for points sufficiently close to x, $f_k = h$ where $h(w) = \max_{i \in I} w_i$, which is a convex function with $\partial h(x) = \text{conv}(\{e_i : i \in I\})$ by the case of $k = 1$. So we have that $\hat{\partial} f_k(x) = \text{conv}(\{e_i : i \in I\})$.

Suppose now that $k > 1$ and $f_{k-1}(x) = f_k(x)$. In that case, we have that $|I| \geq 2$. Suppose that $\exists y \in \hat{\partial} f(x)$. In that case, we have that $f_k(x + z) \geq f_k(x) + y'z + o(||z||)$ for any $z \to \mathbf{0}$. For any $i \in I$ and for small $\delta > 0$, we have $f_k(x + \delta e_i) = f_k(x)$. So, we see that $y_i \leq 0 \; \forall i \in I$ as $y'(\delta e_i) = \delta y_i$.

On the other hand, taking $z = \delta \sum_{i \in I} e_i$, we see that $f_k\left(x - \delta \sum_{i \in I} e_i\right) = f_k(x) - \delta$. Also, we see that $y'z = -\delta \sum_{i \in I} y_i \leq -\delta$. That implies that $\sum_{i \in I} y_i \geq 1$, which is a contradiction. □

Definition 146 (Recession Cone)**.** The *recession cone* (or horizon cone) of a nonempty, closed, and convex set $S \subseteq \mathbb{R}^m$, denoted S^∞ is

$$S^\infty := \{v \in \mathbb{R}^m : \text{ for some } w \in S, w + tv \in S \; \forall t \in \mathbb{R}_+\}$$

It's plain to see that $\mathbf{0} \in S^\infty$ for any S. It's also plain to see that if S is additionally compact (ie., closed and bounded), then $S^\infty = \{\mathbf{0}\}$.

Example 147 (Recession Cone Example)**.** For $S = [-1, \infty)$, then we have that $S^\infty = \mathbb{R}_+$. For $S = [1, \infty)$, we also get that $S^\infty = \mathbb{R}_+$.

Definition 148 (Clarke Regular)**.** $f : \mathbb{R}^m \to \mathbb{R}$ continuous is *Clarke regular* (ie., subdifferential regular) at $x \in \mathbb{R}^m$ if

(1) $\partial f(x) = \hat{\partial} f(x) \neq \emptyset$

(2) $\partial^\infty f(x) = \left(\hat{\partial} f(x)\right)^\infty$

Definition 149 (Clarke Regularity in Neighborhood)**.** $f : \mathbb{R}^m \to \mathbb{R}$ directionally differentiable and locally Lipschitz is *Clarke regular in a neighborhood* of a point x when its directional derivative $f'(\cdot, d)$ is upper semicontinuous there for every $d \in \mathbb{R}^m$.

Proposition 150 (Convexity Implies Clarke Regularity)**.** If $f : \mathbb{R}^m \to \mathbb{R}$ is continuous and convex, then it is Clarke regular at all $x \in \mathrm{int}(\mathrm{dom}(f))$.

Proof.

Take any $x \in \mathrm{dom}(f)$. The first condition of Clarke regularity is taken care of by Proposition 144. As for the second condition, observe that also by Proposition 144, we have that $\partial^\infty f(x) = \{\mathbf{0}\}$. Moreover, by Lemma 93 and Proposition 144 we see that $\hat{\partial} f(x)$ is compact so that by the observation at the end of Definition 146, we have that $\left(\hat{\partial} f(x)\right)^\infty = \{\mathbf{0}\}$. That satisfies the second condition of Clarke regularity. ☐

Proposition 151 (Differentiability Implies Clarke Regularity)**.** If $f : \mathbb{R}^m \to \mathbb{R}$ is continuously differentiable, then it is Clarke regular at all $x \in \mathrm{int}(\mathrm{dom}(f))$.

Proof.

By Proposition 141, we satisfy the first condition of Clarke regularity. Moreover, since $\hat{\partial} f(x)$ is a singleton for every $x \in \mathrm{int}(\mathrm{dom}(f))$, it's compact so that by the observation at the end of Definition 146, we have that $\left(\hat{\partial} f(x)\right)^\infty = \{\mathbf{0}\}$. That satisfies the second condition of Clarke regularity. ☐

Theorem 152 (Chain Rule)**.** Let $F : \mathbb{R}^m \to \mathbb{R}^n$ be continuously differentiable with Jacobian $F' \in \mathbb{R}^{n\times m}$. Let $g : \mathbb{R}^n \to \mathbb{R}$ be continuous. Suppose also that at $\bar{x} \in \mathbb{R}^m$:

- g is Clarke regular at $F(\bar{x})$.
- $\mathrm{null}(F'(\bar{x})') \cap \partial^\infty g(F(\bar{x})) = \{\mathbf{0}\}$.

Define $f := g \circ F$ so that $f : \mathbb{R}^m \to \mathbb{R}$. Then f is Clarke regular at $\bar{x}$ with

$$\begin{aligned}\partial f(\bar{x}) &= \left(F'(\bar{x})\right)' \partial g(F(\bar{x})) \\ &= \{y \in \mathbb{R}^n : y = F'(\bar{x})' z \ \exists z \in \partial g(F(\bar{x}))\}\end{aligned}$$

and

$$\partial^\infty f(\bar{x}) = \left(F'(\bar{x})\right)' \partial^\infty g(F(\bar{x}))$$

Proof.

Omitted. See Rockafellar and Wets *Variational Analysis* Theorem 10.6.

Example 153 (Chain Rule Example)**.** Let $F(x) := \sin(x)$, $g(x) := |x|$, and $f(x) := |\sin(x)|$. Theorem 152 says

$$\begin{aligned}\partial f(0) &= \cos(0) \cdot [-1, 1] \\ &= [-1, 1]\end{aligned}$$

Definition 154 (Clarke's Subgradient and Subdifferential)**.** Suppose that $f : \mathbb{R}^m \to \mathbb{R}$ is locally Lipschitz at $x \in \mathbb{R}^m$. We say that $y \in \mathbb{R}^m$ is a *Clarke subgradient* of f at x, written $y \in \partial^c f(x)$, if it a convex combination of general subgradients of f at x. Mathematically, $\partial^c f(x) = \mathrm{conv}(\partial f(x))$. We say that $\partial^c f(x)$ is the Clarke's subdifferential at x.

Clearly, then $\partial f(x) \subseteq \partial^c f(x)$.

As a note, there's a more general definition of a Clarke subgradient but in class we just specialized it to this locally Lipschitz case where it coincides with this characterization.

Definition 155 (Clarke Stationary). For $f : \mathbb{R}^m \to \mathbb{R}$, if $0 \in \partial^c f(x)$, we say that x is Clarke stationary.

Example 156 (Clarke's Subgradient Example). If $f(x) = -|x|$, then we have that $\partial f(0) = \{-1, 1\}$ and $\partial^c f(0) = [-1, 1]$.

Proposition 157 (Clarke's Subgradient Characterization). Suppose that $f : \mathbb{R}^m \to \mathbb{R}$ is locally Lipschitz at $x \in \mathrm{dom}(f)$. We have that $\partial^c f(x) = \mathrm{conv}(G(x))$ where

$$G(x) = \{g : \exists x_r \to x, \text{ with } f \text{ differentiable at } x_r\ \forall r \in \mathbb{N} \text{ with } \nabla f(x_r) \to g\}$$

Proof.

Omitted but involves applying Rademacher's Theorem (see here).

Proposition 158 (Clarke Regularity Implies Equality Between Subgradients). If $f : \mathbb{R}^m \to \mathbb{R}$ is Clarke regular and locally Lipschitz at $x \in \mathrm{dom}(f)$, then we have that

$$\hat{\partial} f(x) = \partial f(x) = \partial^c f(x)$$

Proof.

The first equality holds by definition of Clarke regularity. The second follows since $\hat{\partial} f(x)$ is always convex so that $\partial^c f(x) = \mathrm{conv}(\partial f(x)) = \mathrm{conv}(\hat{\partial} f(x)) = \hat{\partial} f(x)$, because the convex hull of a convex set is itself as shown in Lemma 10. □

Theorem 159 (Optimality Conditions for Nonconvex and Nonsmooth Optimization). Suppose that $f : \mathbb{R}^m \to \mathbb{R}$ is continuous at x. We have that $\mathbf{0} \in \hat{\partial} f(x)$ is a necessary condition for x to locally minimize f.

Proof.

Suppose that $\mathbf{0} \notin \hat{\partial} f(x)$. Then, $\exists z_r \to \mathbf{0}$ with

$$\begin{aligned} 0 &> \lim_{z_r \to \mathbf{0}} \frac{f(x + z_r) - f(x) - 0' z_r}{||z_r||} \\ \implies& f(x + z_r) < f(x) \quad \text{for } r \text{ large enough by continuity of } f \end{aligned}$$

Thus, we can find y arbitrarily close to x such that $f(y) < f(x)$ meaning that x doesn't locally minimize f. □

Definition 160 (Sharp Local Minimizer). For $f : \mathbb{R}^m \to \mathbb{R}$, we say that x is a *sharp local minimizer* of f if $\exists \tau > 0$

such that

$$f(x+z) - f(x) \geq \tau ||z||\ \forall z \text{ with } ||z|| \text{ sufficiently small}$$

Theorem 161 (Sharp Local Minimizer Characterization)**.** For $f : \mathbb{R}^m \to \mathbb{R}$ continuous, we have that x is a sharp local minimizer of f if and only if $\mathbf{0} \in \text{int}(\hat{\partial} f(x))$.

Proof.

[$\Longrightarrow$]:

We have that $\forall w \in \mathbb{R}^m$ with $||w|| \leq 1$ and small $||z|| < 1$

$$\begin{aligned} \frac{f(x+z) - f(x) - \tau w'z}{||z||} &\geq \frac{\tau ||z|| - \tau w'z}{||z||}, \text{ by assumption} \\ &\geq 0, \text{ by Cauchy-Schwarz} \\ \Longrightarrow \lim \inf_{z_r \to 0} \frac{f(x+z_r) - f(x) - \tau w' z_r}{||z_r||} &\geq 0 \end{aligned}$$

so that by definition we get that

$$\tau w \in \hat{\partial} f(x) \text{ so } \tau B[0,1] \subseteq \hat{\partial} f(x)$$

which implies that $\mathbf{0} \in \text{int}(\hat{\partial} f(x))$.

[$\Longleftarrow$]:

Suppose $\mathbf{0} \in \text{int}(\hat{\partial} f(x))$. So $\exists \sigma > 0$ with $\sigma B[0,1] \subseteq \hat{\partial} f(x)$ so

$$\lim \inf_{z_r \to 0} \frac{f(x+z_r) - f(x) - \sigma w' z_r}{||z_r||} \geq 0\ \forall w \in B[0,1]$$

In particular, this is true for any sequence $z_r = \delta_r w$ with $||w|| = 1$ and $\delta_r \downarrow 0$. So, we have that

$$\lim \inf_{\delta_r \downarrow 0} \frac{f(x+\delta_r w) - f(x)}{\delta_r} \geq \sigma$$

Now let $\tau \in (0, \sigma)$. Then, for any $||w|| = 1$ and sufficiently large r, we have that

$$\begin{aligned} \frac{f(x+\delta_r w) - f(x)}{\delta_r} &\geq \tau\sigma \\ \Longrightarrow f(x+\delta_r w) - f(x) &\geq ||\delta_r w|| \tau\sigma \end{aligned}$$

In light of the arbitrariness of $||w|| = 1$, we have that x is a sharp local minimizer of f. □

10.2 Nonsmooth and Nonconvex Optimization Introduction

We consider the problem:

$$\inf_{x \in \mathbb{R}^m} f(x) :$$

where f is taken to be nonsmooth, nonconvex, but locally Lipschitz. By Rademacher's theorem, locally Lipschitz generally means differential almost everywhere? Why might gradient descent methods jam? Intuitively, the gradient might be steepest in the direction of nondifferentiability and jump back and forth across nondifferentiable points, and make little progress towards the nearby optimum.[5] Iterates may be stopped by non-differentiable points, which is nonideal.

Theorem 162 (Convex Problem with Nonconvergence)**.** Consider the optimization problem

$$\inf_{x \in \mathbb{R}^2} a|x_1| + x_2$$

with $a \geq 1$. Let $x^{(0)}$ satisfy $x_1^{(0)} \neq 0$ and define $x^{(k+1)} := x^{(k)} - t_k \nabla f(x^{(k)})$ where t_k satisfies the Armijo-Wolfe conditions with parameter $0 < c_1 < c_2 < 1$. If $c_1(a^2+1) > 1$, then $x^{(k)}$ converges to some $\bar{x} \in \mathbb{R}^2$ with $\bar{x}_1 = 0$, even though f is unbounded below.

Proof.

Omitted.

To do better, than gradient descent, there are three classes of methods: (a) bundle methods, (b) gradient sampling methods, and (c) BFGS as discussed in Section 9.4.1. We will discuss the latter two further.

10.2.1 Gradient Sampling Method

Here, we introduce the gradient sampling method and discuss some convergence results.

Algorithm 9 Gradient Sampling Method

Input: : $x_0, f : \mathbb{R}^m \to \mathbb{R}, n \geq m+1, \epsilon, \tau > 0, \mu, \beta \in (0,1)$
Start Algorithm:
$x \leftarrow x_0$
while True **do**
  while True **do**
    $u_1, ..., u_n \overset{iid}{\sim} B_{\mathbb{R}^m}(\mathbf{0}, 1)$
    $G \leftarrow \{\nabla f(x), \nabla f(x + u_1\epsilon), ..., \nabla f(x + u_n\epsilon)\}$
    $g \leftarrow \arg\min_{g \in \text{conv}(G)} ||g||_2$ # convex optimization problem
    if $||g|| \leq \tau$ **then**
      break inner loop
    end if
    $d \leftarrow -g$
    $t \leftarrow$ Backtracking line-search such that $t \in \{1, 1/2, 1/4, ...\}$ and $f(x+td) < f(x) - \beta t ||g||_2$
    $x \leftarrow x + td$ # maneuver if at nondifferentiable point
  end while
  $\epsilon \leftarrow \mu\epsilon$
  $\tau \leftarrow \theta\tau$
end while=0

Lemma 163 (Stabilized Steepest Descent)**.** Let $C \subseteq \mathbb{R}^m$ be a compact and convex set and $||\cdot|| = ||\cdot||_2$. Then, we have that

$$-\operatorname{dist}(\mathbf{0}, C) = \min_{||d|| \leq 1} \max_{g \in C} g'd$$

Proof.

[5] There cooked up examples, see for instances Dem'janov and Malozemov (1970) for when gradient descent converges to non-optimum even for convex problems.

We have that

$$\begin{aligned}
-\operatorname{dist}(\mathbf{0}, C) &= -\min_{g\in C} ||g|| \\
&= -\min_{g\in C} \max_{||d||\leq 1} g'd, \text{ by Cauchy-Schwarz} \\
&= -\max_{||d||\leq 1} \min_{g\in C} g'd, \text{ by Sion's min-max theorem} \\
&= -\max_{||d||\leq 1} \min_{g\in C} g'(-d), \text{ relabeling} \\
&= \min_{||d||\leq 1} \max_{g\in C} g'd, \text{ flipping max and min}
\end{aligned}$$

Theorem 164 (Gradient Sampling Theorem)**.** Suppose that $f : \mathbb{R}^m \to \mathbb{R}$ is locally Lipschitz, continuously differentiable on an open full measure subset of $\mathbb{R}^n$, has bounded level sets and we apply Algorithm 9. Then, with probability one, f is differentiable at the sampled points, the line search always terminates, and if the sequence of iterates $\{x^{(k)}\}_{k\in\mathbb{N}}$ converges to some point $\bar{x}$, then with probability 1

- The inner loop always terminates so the sequences of sampling radii $\{\epsilon^{(n)}\}_{n\in\mathbb{N}}$ and tolerances $\{\tau^{(n)}\}_{n\in\mathbb{N}}$ converge to 0.
- $\bar{x}$ is Clarke stationary for f (ie., $\mathbf{0} \in \partial^c f(\bar{x})$)

Proof.

Omitted.

10.2.2 Quasi-Newton Methods and BFGS for Nonsmooth Optimization

Recall BFGS as described in Section 9.4.1. In practice, it works well for nonsmooth optimization but the theory is lacking.

REFERENCES

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. Available at https://web.stanford.edu/~boyd/cvxbook/.

Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011. doi: 10.1561/2200000016. URL https://web.stanford.edu/~boyd/papers/pdf/admm_distr_stats.pdf.

Michael L. Overton. Csci-ga.2945-004: Convex and nonsmooth optimization, 2026. URL https://cs.nyu.edu/courses/spring26/CSCI-GA.2945-004/. Graduate course on convex and nonsmooth optimization, Spring 2026.